



Project Information Memorandum

Djermaya Solar 32MWp PV Plant in Chad

Version 01 (31st January 2019)



GROUPE DE LA BANQUE AFRICAINE
DE DÉVELOPPEMENT

PROPARCO
GROUPE AGENCE FRANÇAISE DE DÉVELOPPEMENT



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Important Notice

Purpose

This Project Information Memorandum (the “Memorandum”) has been developed by the Sponsors of the Djermaya Solar project specifically for the African Development Bank (AfDB), Proparco (subsidiary of the French Development Agency), and the Multilateral Investment Guarantee Agency (MIGA) (the “Financing Parties”) to evaluate their interest in financing or supporting, through the provision of guarantees or insurance, a 32 MWp solar PV plant located in Chad that will be delivered as an Independent Power Producer (IPP).

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List of Abbreviations

AC	Alternating Current
Aldwych	Aldwych International Limited
AfDB	African Development Bank
BCEAC	Central Bank of Central African States
BOOT	Build Own Operate Transfer
CAPEX	Capital Expenditure
CFADS	Cash Flows Available for Debt Service
CEMAC	Communauté Economique et Monétaire de l'Afrique Central
COD	Commercial Operations Date
DC	Direct Current
DFIs	Development Finance Institutions
DCE	Djermaya CDEN Energy S.a.r.l.
DSA	Debt Sustainability Analysis
DSCR	Debt Service Cover Ratio
DSRA	Debt Service Reserve Account
DSU	Delay in start-up
ECA	Export Credit Agency
EPC	Engineering, Procurement and Construction
ESIA	Environmental and Social Impact Assessment
ESMP	Environmental and Social Management Plan
EU-AITF	EU-Africa Infrastructure Trust Fund
EURIBOR	Euro Interbank Offered Rate
GHI	Global Horizontal Irradiation
GoC	Government of Chad
HIPC	Heavily Indebted Poor Countries
ICIEC	Islamic Corporation for the Insurance of Investment and Export Credit
IDC	Interest During Construction
IMF	International Monetary Fund
IPP	Independent Power Producer
JDA	Joint Development Agreement
MIGA	Multilateral Investment Guarantee Agency
OHADA	Organisation pour l'Harmonisation en Afrique du Droit des Affaires
O&M	Operations and Maintenance
OPEX	Operating Expenses
PCOA	Put and Call Option Agreement
PIDG	Private Infrastructure Development Group
PPA	Power Purchase Agreement
PRG	Partial Risk Guarantee
PRI	Political Risk Insurance
PV	Photovoltaic
Project Developers	Aldwych Africa Developments Limited and Smart Energies International S.A.S.
Project Sponsors	InfraCo Africa Limited and Smart Energies International S.A.S.
SE4ALL	Sustainable Energy For All
SHA	Shareholders Agreement
SNE	Société Nationale d'Electricité du Tchad
SPV	Special Purpose Vehicle
STEE	Société Tchadienne d'Eau et d'Electricité
TAF	Technical Assistance Facility
UN	United Nations
XAF	Currency of the BCEAC region

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I. Transaction Overview

A. Project Description

Djermaya Solar (the “Project”) will be the first utility-scale solar PV plant in Chad. The project site is located 30 km north of N’Djamena on a 100ha piece of land awarded by presidential decree. The project is structured as an Independent Power Producer (IPP) and will be funded primarily through private investment. When completed, it will produce 62.2¹ GWh per year (13% of total generation in Chad²) of renewable energy in a country that is totally dependent on fossil fuels.

The project consists of the design, financing, construction, operation, and transfer of a 32MWp solar PV plant and interconnection infrastructure. The plant includes PV modules mounted on single-axis trackers, inverters, and step-up transformers. The interconnection infrastructure includes an 18 km 33kV aerial double-circuit transmission line, two 33/90kV step-up transformers at the Lamadji sub-station, and a 4MWh battery stabilisation system.

Private investors will own the plant for 25 years of operations and then transfer it to the Société Nationale d’Electricité (SNE). Interconnection infrastructure (including the battery system) will be transferred to the SNE after construction and commissioning.

The Power Purchase Agreement (PPA) signed in August 2018 establishes the right to sell electricity to SNE on a “take-or-pay” basis over a 25-year period at a predetermined tariff denominated in XAF but linked to a fixed XAF/EUR exchange rate. Invoicing and payment will be carried out in XAF. Payment will be backed by a liquidity enhancement mechanism.

Total project cost of EUR 52.9 million will be funded through sponsor capital and also by long-term non-recourse loans to be provided by the African Development Bank (AfDB) and Proparco (French Development Agency), as well as a EUR 6.35 million refundable public grant from the European Union Africa Infrastructure Trust Fund (EU-AITF) covering the cost of integration and interconnection works.

The sponsor group includes:

- InfraCo Africa, a donor development facility designed to assume the risks and costs of early-stage project development in lower income countries in Africa.
- Aldwych Africa, a private company established to develop and deliver early-stage infrastructure projects in Africa.
- Smart Energies, a developer and financier of solar PV projects with a portfolio of 250MW under development in Sub-Saharan Africa.

Construction of the plant is expected to begin in July 2019 with commercial operations to commence 12 months later.

¹ Average of yearly mean (P50) generation over 25 years of operations.

² Calculated using 2016 data. During the course of 2016, 468 GWh were produced on the N’Djamena grid.

Figure 1: Map of Chad



B. Project Context

Access to reliable, affordable energy has been highlighted by the United Nations (UN) as crucial to achieving the social and economic development of Chad, which is one of the world's poorest countries. The high cost of fuel coupled with an unreliable network and low connectivity have created a large obstacle to economic growth. The country's installed power generation capacity is poorly maintained and entirely reliant upon fossil fuels, which translates into high and volatile generation costs (historically in excess of 30 USD cents/KWh).

Regular supply failures and logistical challenges have meant that Chad's available power generation capacity is insufficient to meet demand. The electrification rate in 2012 was around 6.4% of Chad's population according to data published by the World Bank³. Other than its oil resources, the only alternative resource for power generation near the demand centre of N'Djamena is solar irradiation, which is currently not being exploited in any scale.

While the recent sustained low oil price has in part narrowed the gap between cost of generation and retail electricity prices, the predominance of oil-based activities in Chad's economy has resulted in a sharp contraction in the country's short-term budget. As part of their renewed support to Chad, international donors have recommended that the Government of Chad (GoC) implement reforms to diversify the economy away from oil and improve the business climate.

Djermaya Solar aims to provide a long-term sustainable, reliable and competitive source of renewable energy to the N'Djamena isolated grid and will pave the way for the development of future sustainable energy projects and further private investment in the country.

C. Project Rational

The project is being jointly developed by UK-based Aldwych Africa Developments Limited (on behalf of InfraCo Africa Limited) and France-based Smart Energies International S.A.S.

Djermaya Solar is aiming to finance the 32 MWp photovoltaic power plant (Phase 1 of a 60 MW program) on the basis of non-recourse project finance. This Information Memorandum refers to the first phase only. The Project goals are to:

- Establish the first commercial solar PV plant in Chad by private investment;
- Enable Chad to diversify the national energy mix and decrease dependency on expensive and polluting fossil fuels, saving SNE and the GoC an estimated EUR 5.5 million per annum (based on an estimated average price of diesel of EUR 0.85/L).
- Provide Chad with approximately 1.5⁴ TWh of clean, 100% renewable power over the lifetime of this project Phase 1;
- Enhance local employment through construction (up to 250 jobs) and permanent employment (up to 12 direct jobs) once operational;

³ Reference: <http://data.worldbank.org/indicator>. Up to date verifiable data on electrification rates is hard to come by. World Bank sources are in conflict with over data bases particularly the International Energy Association (IEA) database which suggests that in 2015 the electrification rate was 3.8% (13.9% in urban areas and 0.8% in rural areas).

⁴ Mean (P50) generation over 25 years of operations.

- Build institutional capacity within government/public authorities through the provision of legal support, and pave the way for development of future power infrastructure.

The project has been under discussion since 2013 and is currently at an advanced stage of development. The project site has been secured, the Environmental and Social Impact Assessment (ESIA) for the plant has been submitted to authorities, the electricity tariff has been agreed, and the Power Purchase Agreement (PPA) signed. With Government advisors in place, there is now a clear route to the finalisation of government support agreements which are expected to include *inter alia*:

- An Framework Agreement (*Convention d'établissement*) setting out the fiscal environment;
- Credit enhancements including a payment liquidity support mechanism backed by Partial Risk Guarantee (PRG) and a Put-Call Option Agreement (PCOA).

D. Project Funding

The estimated capital cost of the Project up to Commercial Operations Date is EUR 52.9 million. Financing is being sourced through a combination of long-term senior loans, sponsor equity, and public grants. The EU-ATIF has already approved a EUR 6.35 million repayable grant to finance interconnection and integration equipment and systems. The grant will be junior to all tranches of debt and equity.

Figure 2 sets out the breakdown of budgeted project costs and the sources of funding as currently contemplated (N.B. the complete sources and uses table is included in Figure 62 in Section XIII).

Figure 2: Sources and Uses

Sources of Funds	EUR m		Uses of Funds	EUR m	
Senior Debt	37.0	70.0%	Construction Costs	39.0	73.8%
Equity	9.5	18.0%	Development Costs	8.7	16.5%
Refundable Grant	6.4	12.0%	Closing and Financing Costs	5.1	9.7%
Total Sources	52.9	100.0%	Total Uses	52.9	100.0%

Source: Financial model January 2019

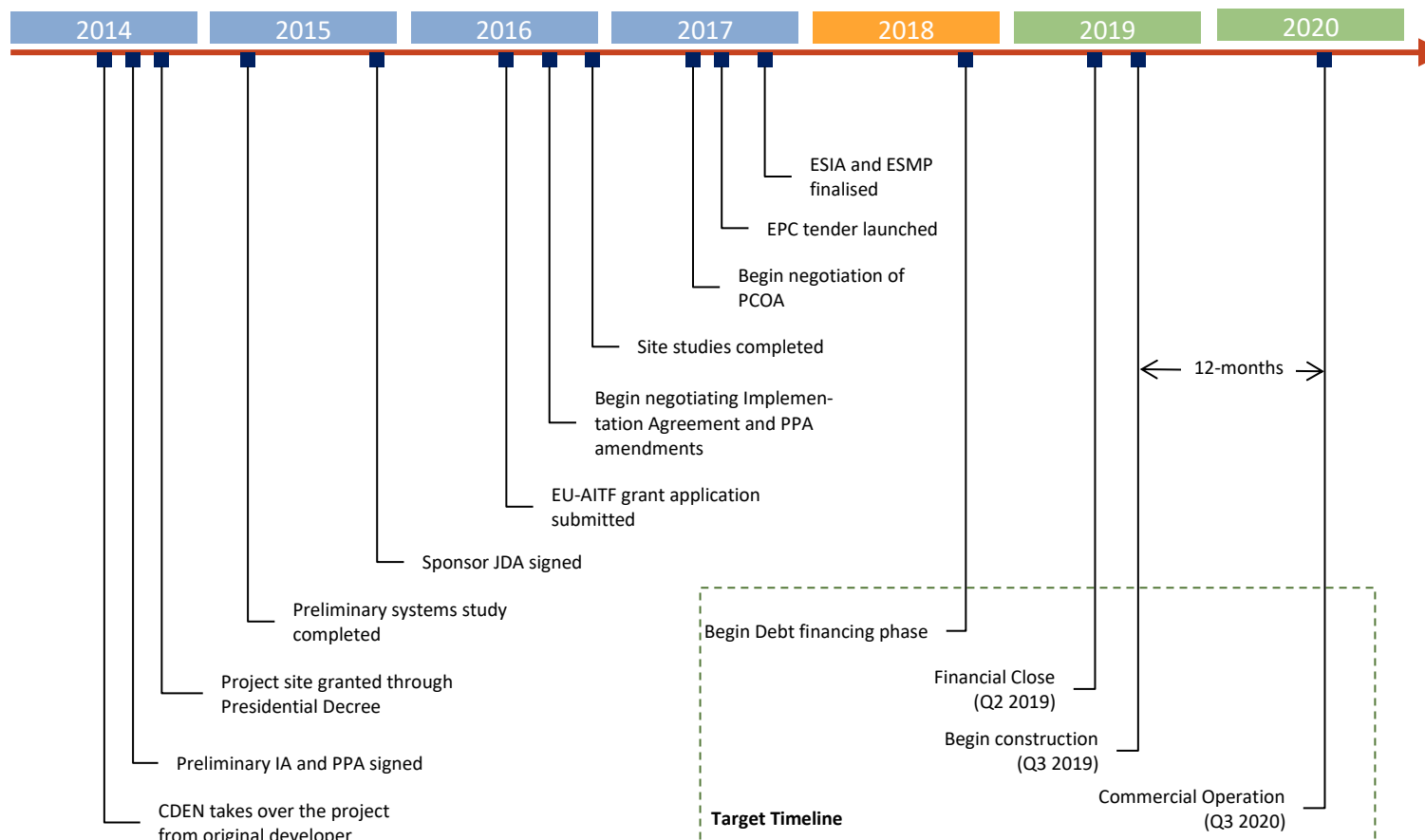
E. Project Timeline

The diagram provided in Figure 3 on the following page traces the project timeline starting in 2014 – when CDEN, a French renewable energy developer, acquired the project rights from the original developer – all the way to target financial close in the second quarter of 2019 and beginning of commercial operations in third quarter 2020.

With respect to short-term target dates in the context of the Lender due diligence process, please note the following:

- On-site due diligence mission W11 (2nd week of March)
- Credit Committee approval W14 (1st week of April)
- Board approval W18 (last week of April)
- Financial close W26 (last week of June)
- First disbursement W31 (first week of August)

Figure 3: Project Timeline



F. Positive Impacts

Djermaya Solar benefits from the following key strengths:

- *Clear rationale* – There is an obvious need for the Project in light of the current expensive and “dirty” generation sources in Chad, the long term demand growth, and the competitiveness of the tariff.
- *Experienced Sponsors and partnerships with “best-in-class” stakeholders* – Djermaya Solar has developed close partnerships with world-class suppliers, contractors, service providers, advisors and co-developers all of whom have deep experience in their respective areas of specialisation.
- *Support from the Chadian Government* – GoC identified this project as a priority and created a cross-sector task force comprising all the main governmental institutions of the power sector to provide the required support to build the foundations for the successful development, construction and operation of the plant.
- *Support from the PIDG* – Technical Assistance Facility (TAF) has provided USD 842,200 of non-refundable grants to fund legal advisory for the GoC and for the Environmental and Social Impact Assessment (ESIA).
- *Support from the EU-AITF* – The fund has provided a EUR 6.35 million grant that will cover all of the interconnector costs.
- *A bankable contractual structure:*
 - A 25-year Power Purchase Agreement;
 - Government support through an Framework Agreement, a Liquidity Support Facility, and a Put and Call Option Agreement;
 - A fully wrapped, lump sum, turn-key Engineering Procurement and Construction (EPC) contract with usual warranties;
 - Operation and Maintenance (O&M) contract with customary performance guarantees;
 - A Shareholder Agreement and LMA standard financing agreements;
 - A market standard security package including first ranking security over project contracts, assets, shares and project accounts etc.
- *Agreement reached on a competitive and sustainable tariff* – Detailed in Annex 14 of the signed Power Purchase Agreement.
- *Environmental and Social Impact Assessment* – Undertaken by an experienced international firm following IFC performance standards, noting that the final report was submitted to the authorities in December 2018.
- *Compliance with best practice procurement processes* – The EPC procurement process was undertaken using best market practices and in full compliance with AfDB procurement guidelines.
- *Emphasis on capacity building initiatives* – Throughout the development cycle emphasis has been placed on building capacity at the local level, exemplified by the securing of a grant to fund independent, international advisory services for the GoC in support of its negotiations of the PPA and other key agreements.

II. Political, Economic, and Social Environment

A. Introduction

Chad is a large, landlocked, semi-arid country located south of the Sahara in Central Africa. The country is one of world's poorest and most corrupt, with most people eking out a subsistence living. The ethnically diverse population of around 14.9 million lives mostly in the more humid south, and speaks Arabic, French, and some hundreds of other tribal languages.

Control lies firmly in the hands of the President Idriss Déby, who has been in power since 1990. Since 2003, the country has benefitted from oil exports flowing via the Chad-Cameroon pipeline. The country has taken a firm stand against Boko Haram in neighbouring Nigeria, and serves as a regional military base for France in its fight against cross-border Islamist terrorism in the Sahel region.

Figure 4: French Military in Sahel



Source: Operation Barkhane

B. Political Environment

Politics in Chad takes place within a presidential framework, where the president is both head of state and head of the government. The current President Idriss Déby has faced a series of rebellions and coup attempts since overthrowing Hissene Habre in 1990, but has nevertheless managed to maintain a tight grip on power.

Déby and his Patriotic Salvation Movement (MPS) won Chad's first multi-party elections in 1996 and was re-elected in 2001. After abolishing term limits in 2005, he was re-elected in 2006, 2011, and most recently in April 2016. Chad's political opposition has frequently alleged electoral irregularities during polls, but their lack of unity and organisation, coupled with the MPS' repressive stance against public dissent, have hindered efforts to mount a credible challenge. The opposition's call for the international community to denounce the regime's authoritarian tendencies has also largely gone unheeded due to a large extent to Déby's prominent role in working with the West to combat terrorism.

On 30th April 2018 the MPS-dominated parliament approved a new constitution which further expands Déby's powers. The new constitution eliminates the post of prime minister and also re-imposes the two-term limit scrapped in the 2005 referendum (although it will not be applied retroactively). It also extends presidential terms from five to six years, meaning Déby could potentially stay in office until 2033.

In recent years, the main threat to the regime's stability has come from deteriorating public finances and associated social unrest. Chad has been grappling with an economic crisis since late 2015 caused by declining international oil prices. Insecurity along its borders with Nigeria, Niger, and Cameroon, as well as with Libya and Sudan, has also negatively impacted the country's trade. The high cost of military operations in neighbouring countries further weakens the country's finances. Authorities have been forced to make sweeping cuts in response to mounting deficits.

In the area of governance, Chad scored 20 out of 100 on Transparency International's 2017 Corruption Perception Index, which resulted in the country ranking 165 out of 180 countries. As indicated by Marie Chêne of Transparency International:

"Chad's recent history has been characterised by insecurity, endemic corruption, human rights abuses and a deeply entrenched patronage system which permeates all sectors of society... oil revenues have been misused by government to strengthen its armed forces and reward its cronies, which contributes to the undermining of the country's governance system... The government has taken legal and institutional steps to fight corruption and launch anti-corruption campaigns in recent years, resulting in the dismissal and arrest of several high level officials. However, these initiatives are perceived by many external observers as politically motivated and there are no clear indications of a firm commitment to effectively eradicate corruption⁵."

C. Economic and Social Environment

Chad's GDP was 12.8% of the total GDP of the Central African economic and monetary union in 2017 and it remains one of the poorest countries of the world. Two-thirds of the 14.9 million inhabitants live below the poverty line. The population is predominantly rural (78%).

With the discovery and subsequent extraction of oil commenced in 2003, Chad's economy became heavily dependent on petroleum. The collapse in oil prices, the rainfall deficits, and the deterioration of the security situation have severely affected the economy of Chad. The country remains in deep recession, and according to World Bank forecasts poverty is expected to rise to 39.8% by 2019. The World Bank also notes that the current economic downturn is reflected in "cuts in public expenditure, low foreign direct investment, and a loss of income caused by the disruption of cross-border trade with Nigeria in livestock."

Despite severe fiscal adjustments, the overall fiscal deficit increased from 4.4% of non-oil GDP in 2014 to 4.9% in 2016. This was financed through the issuance of treasury bonds in the regional debt market, IMF disbursements, and budgetary support from donors. This increase in public debt has led to hard currency rationing. Chad's official reserves continue to fall, representing barely a month's worth of imports as of 2017.

Chad needs an economic diversification policy based on sectors that are less vulnerable to oil shocks. The livestock and food crops (about 32% of the GDP) and services (32%) are the other major economic sectors. The construction of a second cement plant has been launched, and the telecommunication companies continue to grow. The following table summarizes the main economic indicators:

Figure 5: Main Economic Indicators

	Units	2013	2014	2015	2016	2017
Population	Million	13.1	13.6	14.0	14.5	14.9
GDP (current prices)	EUR billion	9.7	10.5	9.9	9.1	8.7
GDP (constant prices)	EUR billion	7.8	8.4	8.5	8.0	7.7
Inflation	Percent	0.2%	1.7%	6.8%	-1.1%	-0.9%
GDP/cap (current prices)	EUR	742	772	704	631	586
Exports (current prices)	EUR billion	3.3	3.6	3.0	2.4	3.0
Imports (current prices)	EUR billion	3.8	4.5	3.6	3.4	3.5
External Debt (current prices)	EUR billion	2.2	2.8	2.7	2.7	2.8
Reserves (current prices)	EUR billion	0.9	0.9	1.0	0.3	0.0
Transparency	CPI Index	163/177	154/174	147/168	159/176	165/180

Source: IMF, World Bank

⁵ Source: <https://knowledgehub.transparency.org/helpdesk/overview-of-corruption-and-anti-corruption-in-chad>.

D. IMF Involvement

1. Debt Sustainability

Chad's external public debt was about 20% of GDP over the decade after the start of oil production in 2003. The external public debt to GDP ratio increased in the 2013 – 2014 period following two non-concessional oil sales advance operations with Glencore Energy. In late 2015, a rescheduling agreement with Glencore consolidated the two oil sales advance operations and extended their maturities. This agreement improved Chad's debt service profile over the medium term.

Chad has seen significant improvement in the economic management of the country in the past years, culminating in a Highly Indebted Poor Country (HIPC) completion point being reached in April 2015, which resulted in a USD 756 million debt relief for Chad (including relief from the International Development Association, the AfDB, the IMF and the Paris Club). The debt relief was in part aimed at enabling Chad to allocate resources for the promotion of economic growth. At the time, the present value of debt-to-exports ratio was projected to fall from 55.1% at end-2013 to 31.3% at end-2015.

The stock of public domestic debt has increased in recent years, reaching 9% of GDP at end-2014. The stock of domestic debt rose to 16% of GDP in 2015, in part a result of the authorities stepping up their domestic debt issuance program in the CEMAC market. As of end-2015, the government accrued a sizable amount of domestic payment arrears. The balance of treasury arrears increased from XAF 52 billion at end-December 2014 to around XAF 200 billion (3.1% of GDP) in 2015.

In March and April 2016, the International Monetary Fund (IMF) conducted bilateral discussions with the Chadian government in order to collect economic and financial information, as well as to discuss with officials the country's economic developments and policies. The mission outcomes confirmed that the government's short term budget remained under extreme pressure. This was followed by a Debt Sustainable Analysis (DSA) in August 2016 which concluded: "Chad remains at a high risk of debt distress, with vulnerabilities concentrated in the short term. With oil prices lower than in the previous DSA, some external debt ratios have deteriorated. The high risk of debt distress is explained by the large debt service payments falling due in the short to medium term linked to the non-concessional oil sales advances from Glencore".

An Extended Credit Facility (ECF) arrangement was approved by the IMF Board in June 2017 for an amount of USD 312.1 million (160% of Chad's quota). The ECF-supported program aims to help Chad restore macroeconomic stability and lay the foundation for robust and inclusive growth. In December 2018 the IMF completed the third review of Chad's economic and financial program supported by the ECF and this enabled a disbursement of a further USD 48.6 million, bringing total disbursements to about USD 194.5 million.

Under the supervision of the IMF, authorities in Chad are implementing a comprehensive fiscal strategy to foster non-oil growth and reduce banking sector vulnerabilities. The main pillars of the strategy include: (i) better managing the wage bill, (ii) paying down domestic debt and clearing domestic arrears, (iii) increasing domestic revenue mobilization, (iv) improving public financial management, and (v) addressing structural weaknesses in some of the domestic banks.

2. IMF and the Djermaya Project

Within the context of the IMF DSA and ECF programme, the GoC was subject to certain constraints aimed at limiting additional public sector debt. Sponsors and Government legal counsel were aware of the potential impact of such constraints on the financing structure being proposed for the Djermaya project, and in 2018 initiated a dialogue with the IMF both at the country level as well as with the IMF mission to Chad based in Washington D.C.

One of the initial concerns was an apparent IMF requirement for the Djermaya capital structure to have a rather high amount of concessional financing. The other concerns related to the potential off-balance sheet liabilities associated with two key contracts forming part of the credit enhancement of the project financing structure, specifically the PRG and the PCOA.

In January 2019 the AfDB team (working in close cooperation with the Sponsors) received confirmation from the IMF that the proposed financing structure did not fall within any of the constraints imposed by the ECF programme. As follows, two citations from an IMF senior economist:

“We understood that based on your evaluation of the financial arrangements of the project, and in particular the Put and Call Option Agreement (PCOA) and the Partial Risk Guarantee (PRG) backed liquidity support structure described in the documentation submitted you and the IMF team, the Government is not prevented from entering into such financial agreements with the Djermaya solar project company (CDEN) owing to any constraints imposed by the ECF program, given that such constraints do not apply to such financial arrangements of the Djermaya project.”

“I can confirm that constraints in the ECF program related to the zero limit on non-concessional external borrowing would not apply to the Put and Call Option Agreement (PCOA) and the Partial Risk Guarantee (PRG) backed liquidity support structure described in the documentation submitted to the IMF team.”

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III. Electricity Sector

A. Sector Overview

The Ministry of Energy is the government authority in charge of the electricity sector. Whilst legislation provides for the creation of a regulator, no regulatory body current exists.

The *Société nationale d'électricité* (SNE) is the national electricity utility. It was created in 2010 following the separation of the *Société Tchadienne d'Eau et d'Electricité* (STEE) which was a combined water and electricity utility. SNE has a service contract with the Tunisian utility STEG, which provides technical advisory support for the operation of the system.

Electrification remains extremely low at around 6.4% of the population of 14.9 million people. Generation is exclusively based on diesel and heavy fuel oil (HFO) which are supplied from the Djermaya refinery.

B. SNE

1. Five Year Plan

Approximately 90% of SNE activities are in and around Ndjamen, where a 90kV loop and link to the refinery at Djermaya via a 66kV transmission line were completed in 2015. SNE operates a number of power generation stations, manages the nascent transmission grid, and distributes and sells power to end consumers.

In 2015 SNE carried out its first five year plan (*Plan de développement des infrastructures électriques 2016 – 2020*). The aim of this strategic planning effort was to provide more visibility with respect to growth in demand and to develop a coherent investment strategy aimed at meeting this growth. SNE took 2015 demand and then inflated it using three scenarios: Low growth (5%) medium growth (7%), and high growth (10%). The results are provided in Figure 6.

The medium growth figures were taken as the base case, and under this scenario SNE estimated that the total amount of required new investment from 2016 to 20120 was XAF 127 billion (EUR 236 million). This was broken down into: Generation XAF 127 billion (EUR 190 million); transportation and distribution XAF 20.7 billion (EUR 31 million); other investments at locations other than N'Djamena XAF 10.2 billion (EUR 15 million).

As a point of interest, there were five solar PV projects included in SNE's 5-year plan for a potential total solar PV peak installed capacity of 26MW as shown in Figure 7. SNE realised even at that time that choices would have to be made.

Figure 6: Forecast Demand (MW)

Year	Low	Medium	High
2015 (Ref. Year)	66.0	66.0	66.0
2016	82.5	83.8	85.8
2017	102.0	104.8	109.5
2018	118.5	123.6	132.0
2019	133.7	141.5	154.5
2020	146.0	157.0	176.0

Source: SNE

Figure 7: Solar PV Included in 5-year Plan

Sponsor	MWp	Expected COD
Newsolar	60	2017
Citelum (Djermaya)	60	2017
Sunnvest	60	2017
Total	50	2017
Huawei	30	2018
Total	260	

Source: SNE

2. Operational and Financial Performance

Comprehensive data in respect of operational and technical performance of SNE is difficult to secure. Sponsors have had access to a report prepared by the African Development Bank (AfDB) in May 2018. According to this report, SNE technical losses were estimated at 36% and non-technical losses at 25%. In other words, 61% of the electricity produced is not generating revenues. The report also provided an estimated cost of electricity produced of 40 US cents/kWh (35 Euro cents /kWh).

In order to make power more affordable and to combat fraud and non-payment, the Government reduced tariffs by 37% in September 2014 to XAF 125 /kWh (17 Euro cents/kWh) for most customer classes, with a social tariff of XAF 85/kWh (13 Euro cents/kWh) for those consumers using less than 150kWh in a month. Excluding the social tariff, this means that SNE is selling electricity for less than half of what it takes to produce.

The impact of this business model means that SNE must rely on State subsidies to remain solvent. Abbreviated financials are provided in Figure 8, and as can be seen, State subsidies have historically represented around 50% of total revenues.

Figure 8: SNE Abbreviated Profit & Loss Statement

EUR millions	2012	2013	2014	2015	2016
Power Sales	41.3	33.3	33.8	49.2	67.8
Other Income	0.4	0.6	1.8	2.3	2.5
State Subsidies	55.6	35.9	33.5	51.9	48.1
Total Revenues	97.3	69.8	69.0	103.4	118.4
Cost Base	98.6	75.3	71.2	114.5	118.6
Operating Profit / Loss	-1.3	-5.4	-2.2	-11.2	-0.2
Net Profit / Loss	-2.1	-6.0	-2.8	-2.1	-1.7
Subsidies (% Total Revenues)	57%	51%	48%	50%	41%

Source: African Development Bank

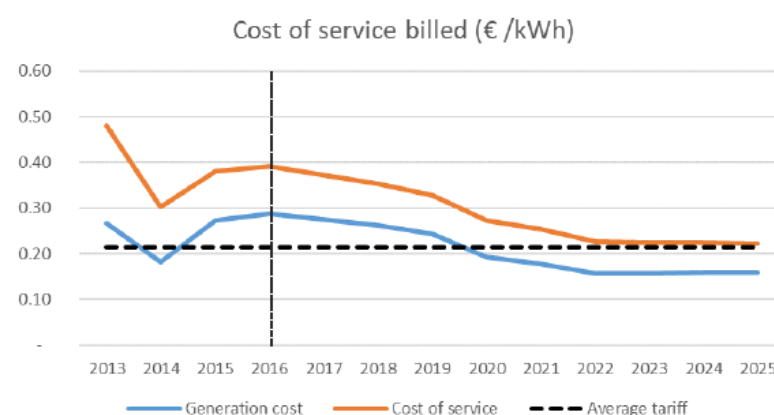
3. Turn-around Plan

SNE management has developed a comprehensive turnaround plan aimed at improving the operational and financial performance of the utility and reaching a positive operating margin by 2020. This plan covers the 2017 – 2021 period and was presented in September 2017 to the country's financial partners (AFD, World Bank, AfDB, Islamic Development Bank, and the European Union) during a roundtable in Paris. The AFD has undertaken a technical and institutional audit, and both the World Bank and AfDB will likely co-finance part of the programme.

The recovery plan has a three-pronged approach which includes: (i) Diversifying the generation mix with HFO and PV generation; (ii) reducing technical and non-technical losses to 20% levels; (iii) rollout prepaid metering (payable via mobile money) to increase billing efficiency.

As can be seen in the graph presented in Figure 9, should the implementation of the recovery plan be successful, the cost of billed service should reach the average end-user tariff by 2020.

Figure 9: Improving Cost of Service Billed



Source: African Development Bank

C. The Electricity Network

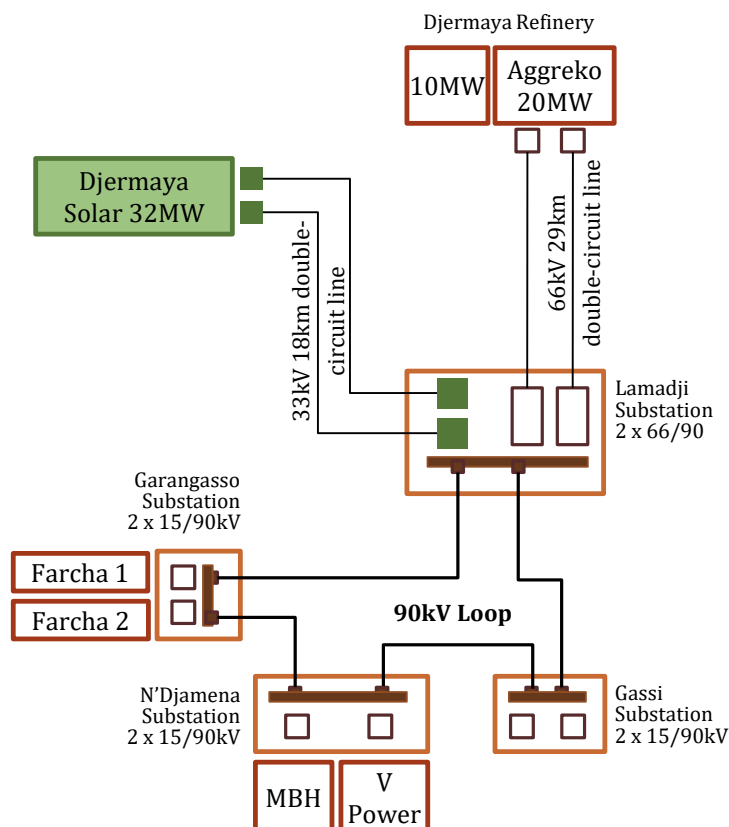
1. N'Djamena Grid

In Chad the “Network” refers to the isolated grid at N'Djamena. This consists of a 90kV transmission backbone built by the Chinese that forms a ring around the capital city, with 66kV spur connecting this ring with the refinery at Djermaya.

With reference to Figure 10 below, four main substations form part of the 90kV loop: Lamadji, Garangasso, N'Djamena, and Gassi.

The Djermaya solar PV plant will connect via a new 33kV aerial transmission line to the existing Lamadji substation where two new 25 MVA transformers will be installed.

Figure 10: N'Djamena Electricity Network



2. Installed Capacity

Excluding the captive power plants of the upstream oil and gas operators, and other large industrial sites such as the cement factor to the north of N'Djamena, the total installed capacity of the grid is 208MW, of which only 88MW is currently operating.

With reference to Figure 11 on the following page, it can be seen that some of the generating units are permanently out of service (due to major repairs), whilst others are just temporarily inoperative due to maintenance activities.

SNE owns and operates the four MBH 5.3MW units, of which only one is operating at a reduced capacity of 3.5MW. SNE also owns and operates two plants: Farcha 1, having three Wärtsilä 7.3MW generators of which only one is currently operating; Farcha 2, having seven 8.6MW generators all of which are current operational and which meet base load and spinning reserve requirements.

There are three third-party suppliers, namely: V-Power (17MW), Aggreko (20MW), and the refinery (20MW). Currently SNE is not buying electricity from either Aggreko or the refinery due to lack of funds (N.B. which is why they are showing as “not operational” in the table in Figure 11).

Figure 11: N’Djamena Generation Capacity

SNE Plants					Dec-18	Forecast 2020
Type	Fuel	N°	Nominal Capacity (MW)	Current Status	Operational Capacity (MW)	Operational Capacity (MW)
1 MBH	Diesel		5.3	Permanently Out of Service	0.0	0.0
2 MBH	Diesel		5.3	Undergoing Maintenance	0.0	4.0
3 MBH	Diesel		5.3	Operational	3.5	4.0
4 MBH	Diesel		5.3	Undergoing Maintenance	0.0	4.0
1 V Power	Diesel	28 x 0.6	17.0	Operational	17.0	17.0
Sub-total			38.2		20.5	29.0

Djermaya Refinery					Dec-18	Forecast 2020
Type	Fuel	N°	Nominal Capacity (MW)	Current Status	Operational Capacity (MW)	Operational Capacity (MW)
-- Refinery	Diesel and HFO		20.0	Cabling Problem	0.0	13.0
1 Aggreko	Diesel and HFO	13 x 0.8	10.0	Not Operational	0.0	5.0
1 Aggreko	Diesel and HFO	13 x 0.8	10.0	Not Operational	0.0	5.0
Sub-total			40.0		0.0	23.0

Farcha 1					Dec-18	Forecast 2020
Type	Fuel	N°	Nominal Capacity (MW)	Current Status	Operational Capacity (MW)	Operational Capacity (MW)
1 Wärtsilä	Diesel		7.3	Permanently Out of Service	0.0	0.0
2 Wärtsilä	Diesel		7.3	Undergoing Maintenance	0.0	7.3
3 Wärtsilä	Diesel		7.3	Operational	7.3	7.3
Sub-total			21.9		7.3	14.6

Farcha 2					Dec-18	Forecast 2020
Type	Fuel	N°	Nominal Capacity (MW)	Current Status	Operational Capacity (MW)	Operational Capacity (MW)
1 Wärtsilä	Diesel		8.6	Operational	8.6	8.6
2 Wärtsilä	Diesel		8.6	Operational	8.6	8.6
3 Wärtsilä	Diesel		8.6	Operational	8.6	8.6
4 Wärtsilä	Diesel		8.6	Operational	8.6	8.6
5 Wärtsilä	Diesel		8.6	Operational	8.6	8.6
6 Wärtsilä	Diesel		8.6	Operational	8.6	8.6
7 Wärtsilä	Diesel		8.6	Operational	8.6	8.6
Sous-total			60.2		60.2	60.2

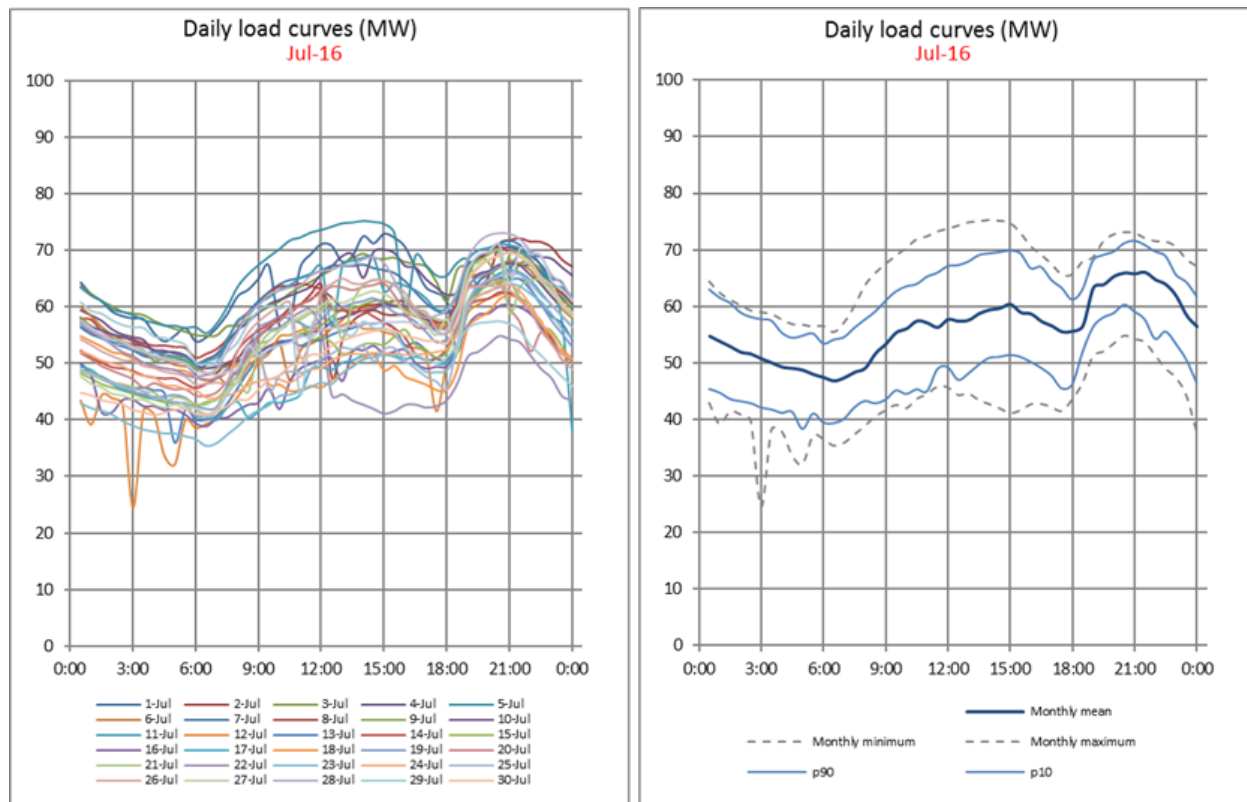
Mobile Power Gas Turbines					Dec-18	Forecast 2020
Type	Fuel	N°	Nominal Capacity (MW)	Current Status	Operational Capacity (MW)	Operational Capacity (MW)
1 Not Available	Diesel		23.8	Not Yet Commissioned	0.0	23.8
2 Not Available	Diesel		23.8	Not Yet Commissioned	0.0	0.0
Sub-total			47.6		0.0	0.0
Total			207.9		88.0	126.8

3. Electricity Demand

The need for air conditioning during the hotter months has a significant influence on the demand for electricity. In the N'Djamena area, temperatures start to increase in February and reach a maximum in April and May. Whilst monsoon rains during the summer months tend to bring down the temperatures, the greater humidity makes the weather unbearably hot. When the monsoon period is over, the temperatures again rise in October and November. What this means is that during the hotter months from April through November, the mid-day peak demand is quite close to the evening peak. In fact, as can be seen in Figure 12 below, in July it can even be slightly in excess of the evening peak.

The daily demand curves for the month of July show peaks from 12:00 to 15:00 and from 19:00 to 22:00. Both peaks reach approximately 70MW. With reference to the right-hand graph in Figure 13 on the following page, it can be seen that during the cooler month of December, the mid-day peak is less pronounced and reaches a maximum of around 55MW, and the peak from 18:00 to 22:00 reaches a maximum of around 65MW.

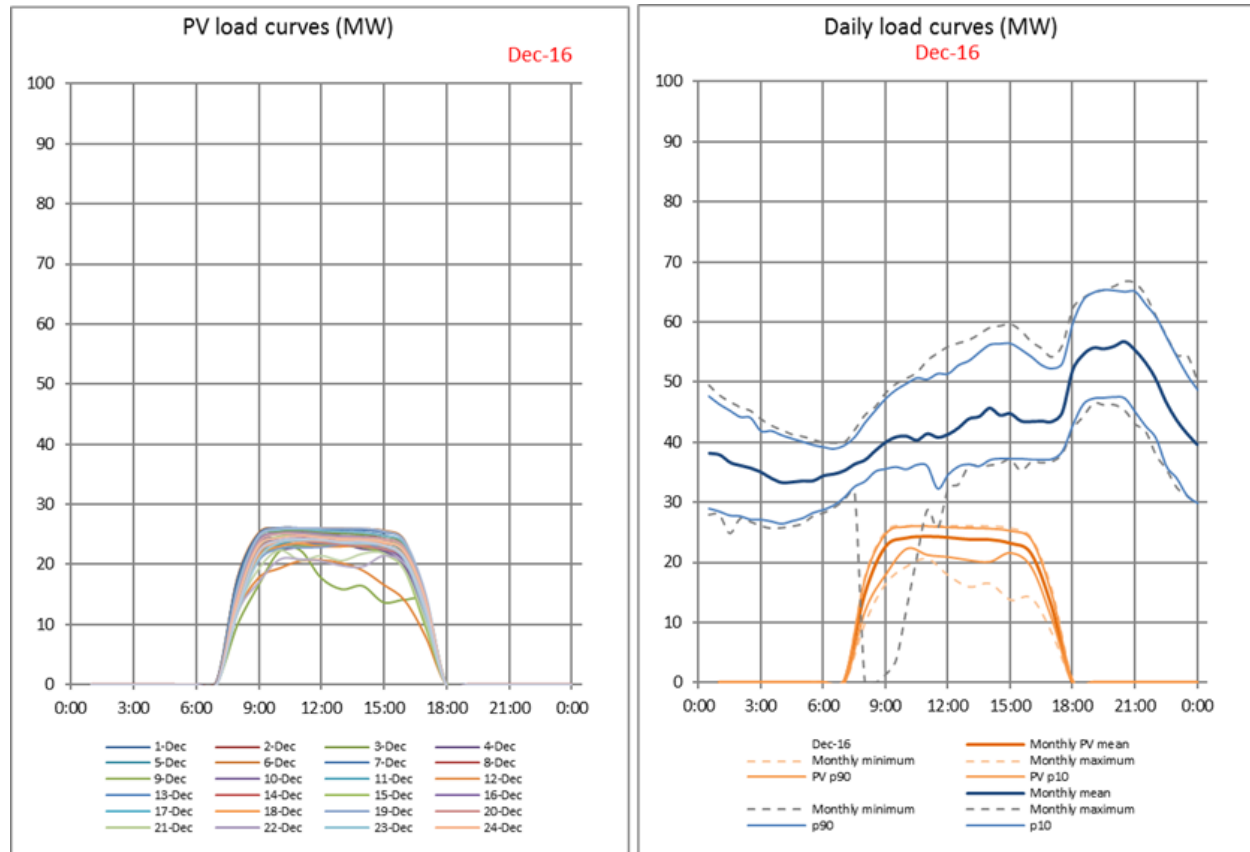
Figure 12: Daily Load Curves (July 2016)



There several important observations can be made from the above demand curves. Firstly, because solar PV only produces power during the day, the addition of solar energy to the grid will not necessarily result in the addition of new customers (otherwise they would need to be cut off during the evening). Secondly, the real benefit of solar PV to such a grid is to off-set thermal generation, thus lowering the cost of generation for SNE as a result of lower fuel expenditures. Thirdly, the addition of a “peak shifting” battery storage – even if it was attractive from a capex perspective – would not make sense because the displacement of thermal generation during the day is equal in value to the displacement of thermal generation in the evening.

The left-hand graph in Figure 13 shows the solar PV generation curve, with the characteristic flat peak between 09:00 and 16:00 associated with single-axis tracker technology. The addition of around 25MW of solar power can be used to off-set expensive thermal generation even during the cooler winter months.

Figure 13: Daily Load Curves (December 2016)



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IV. Regulatory Environment

A. Regional Legal Framework

Chad is a member of various regional organisations including the Organisation for the harmonisation of business law in Africa (OHADA), the Central African Economic and Monetary Community (CEMAC) and the inter-African conference of insurance markets (CIMA).

As a member state of these international organisations, any regulation adopted by them are directly applicable in Chad and will prevail over the laws of Chad in the event of any conflict with national legislation.

- **OHADA** – Chad is a signatory to the OHADA treaty. As such, all OHADA Uniform Acts (which govern many aspects of business law) are directly applicable in Chad, without the need for any transposition, and they prevail over conflicting national legislation. The main goals of OHADA are (i) to harmonise business law with a view to securing legal certainty and encouraging investment, and (ii) to implement a legal and judicial environment favourable to economic activities, along with the progressive unification of legislation. In substance, the nine OHADA uniform acts relate to (i) general commercial law, (ii) commercial companies and economic interest groups, (iii) organising security interests, (iv) organising simplified recovery procedures and measures of execution, (v) organizing collective proceedings for clearing debts, (vi) arbitration, (vii) organising and harmonizing undertakings accountings systems, (viii) contracts for the carriage of goods by road, and (ix) cooperative companies.
- **CEMAC** – The Central African Economic and Monetary Community (*Communauté Economique et Monétaire de l'Afrique Centrale*) is a regional organisation founded in 1994 and composed of six states: Chad, Gabon, Equatorial Guinea, Central African Republic, the Republic of Congo, and Cameroon. The purpose of CEMAC is (i) to create a common market based on the free movement of persons, goods, capital and services, (ii) to ensure the stability of the common currency, (iii) to secure the environment for economic activities and general business, and (iv) to harmonise national sectorial policies. CEMAC produces regulations and directives relating to various areas, including tax, customs, investment, foreign exchange control, banking, etc.
- **CIMAC** – The purpose of the CIMA is to (i) strengthening and consolidating a close co-operation in the field of insurance, so that markets are able to cover, by appropriate coverages that meet African realities and taking into account their contributing potentials, the risks of the agricultural and rural sector, and (ii) creating conditions for a healthy and balanced development of insurance companies.

B. Investment Framework

Investments in Chad are governed by law N°006/PR/2008 of 3rd January 2008 enacting the *Charte d'Investissement* (the "Investment Charter"). The Investment Charter was adopted in order to implement provisions of the Regulation n° 17/99/CEMAC-20-CM-03 dated 17 December 1999 enacting the CEMAC investment charter (the "CEMAC Investment Charter"). The Investment Charter governs both national and foreign investments. The Investment Charter has not been modified since its enactment.

The purpose of the Investment Charter is to encourage the creation and development of activities towards the transfer of relevant technology, the reorganisation (rehabilitation) and extension of existing companies, and the mobilisation of national savings and foreign capital. A list of "productive investments" governed by provisions of the Investment Charter is provided in article 3 of the Investment Charter, which expressly refers to "water and energy production supply activities" and "activities promoting others sources of solar and wind energy". Consequently, the Project is governed by the Investment Charter and the Project shall benefit from the guarantees and tax and customs advantages granted by the Investment Charter.

C. Energy Framework

The electricity regulatory framework in Chad is based on the Electricity Law 14 which was passed in 1999. It is important to note that the implementation decrees provided for under the Electricity Law have not yet been issued, which is a risk that needs to be contractually addressed.

The Electricity Law has created the legal framework cornerstone allowing private power generation. Under this Law, the generation, transmission and supply of power qualify as a commercial and industrial public service (*service public industrial et commercial*) which is the exclusive competency of the State; however, third parties are allowed to be declared as independent power generators or the State is entitled to delegate its power to third parties.

The Electricity Law provides for the creation of an Electric Power Regulator that has not yet been created.

The State has regulatory powers and imposes technical requirements on the power facilities. Specific tax provisions are applicable for renewables projects.

D. Property Legal Framework

1. Land Control

The land legal regime in the Republic of Chad is complex and organized through three laws dated 1967⁶ and their respective implementation decrees⁷.

On 19th September 2014, a site of 100 hectares was attributed to Djermaya CDEN Energy through a Presidential Decree voted on the Council of Ministers. The site was registered with the cadastral plan and the land registry on 21st March 2016.

⁶ Law No. 23 dated 22 July 1967 on the status of the State property; law No. 23 dated 22 July 1967 on the land ownership and customary rights; law No. 23 dated 22 July 1967 on the land ownership limits.

⁷ Decree No. 186/PR dated 1 August 1967 on the land ownership and customary rights; decree No. 187/PR dated 1 August 1967 on the land ownership limits; decree No. 188/PR dated 1 August 1967 implementing the law on the status of the State property.

A land title is normally only granted once the purpose for which the land is granted has materialised. In the case of Djermaya Solar, this means that the power plant would need to have been constructed. In order for the consortium to obtain a land title prior to start of construction, an exemption would need to be requested. A process has started to request an official derogation to obtain the land title prior to the start of construction.

2. Land Use Planning

The construction legal framework in the Republic of Chad has long been governed by Ordinance No.17/PR of 9th July 1970 on regulation of the construction permits. This text was finally repealed by two laws of 2010: the Construction Law and the Town Planning Law. According to the Construction Law, any construction in the Republic of Chad requires a construction permit. The procedure regarding the construction permit is regulated by the Construction Permit Decree issued in 2012.

On March 2014, the Ministry of Energy attributed to Djermaya a preliminary Construction Authorisation. The Construction permit file will be updated and will be submitted to the competent authority to secure the ultimate Construction Permit.

3. Environment

The legislation for environment in Chad is based on the law N°14/PR/98 of 17th August 1998 defining the general principles of the environmental protection. This law is the pillar of the national policy for protecting the environment in Chad. The law is assorted of a decree (N°904/PR/PM/MERH/2009) ruling pollutions and nuisances to the environment. This decree applies to installations categorised as *Installations Classées pour la Protection de l'Environnement* (ICPE). Solar PV farms are classified as ICPE and are falling under the rules listed in the decree N°904/PR/PM/MERH/2009.

The decree N°6300/PR/PM/MEERH/2010, ruling the ESIA in Chad, classifies projects into 3 categories:

- *Category 'A'*: projects with various and significant impacts on the environment, requiring detailed investigations; projects requiring a full Environmental and Social Impact Assessment.
- *Category 'B'*: projects with limited impacts on the environment, requiring a simple notice of impact on the environment.
- *Category 'C'*: projects with very limited impacts on the environment not requiring any notice of impact or ESIA.

Djermaya Solar is classified as a Category 'A' Project. The Terms of References of the ESIA were prepared according to the Ministerial Order N°039/PR/PM/MERH/SG/DGE/DEELCPN/2012, describing the guidelines to conduct an ESIA in Chad. The ESIA is also designed to fulfil IFC and AfDB's guidelines and meet minimal requirements of the EU

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V. Contractual Structure

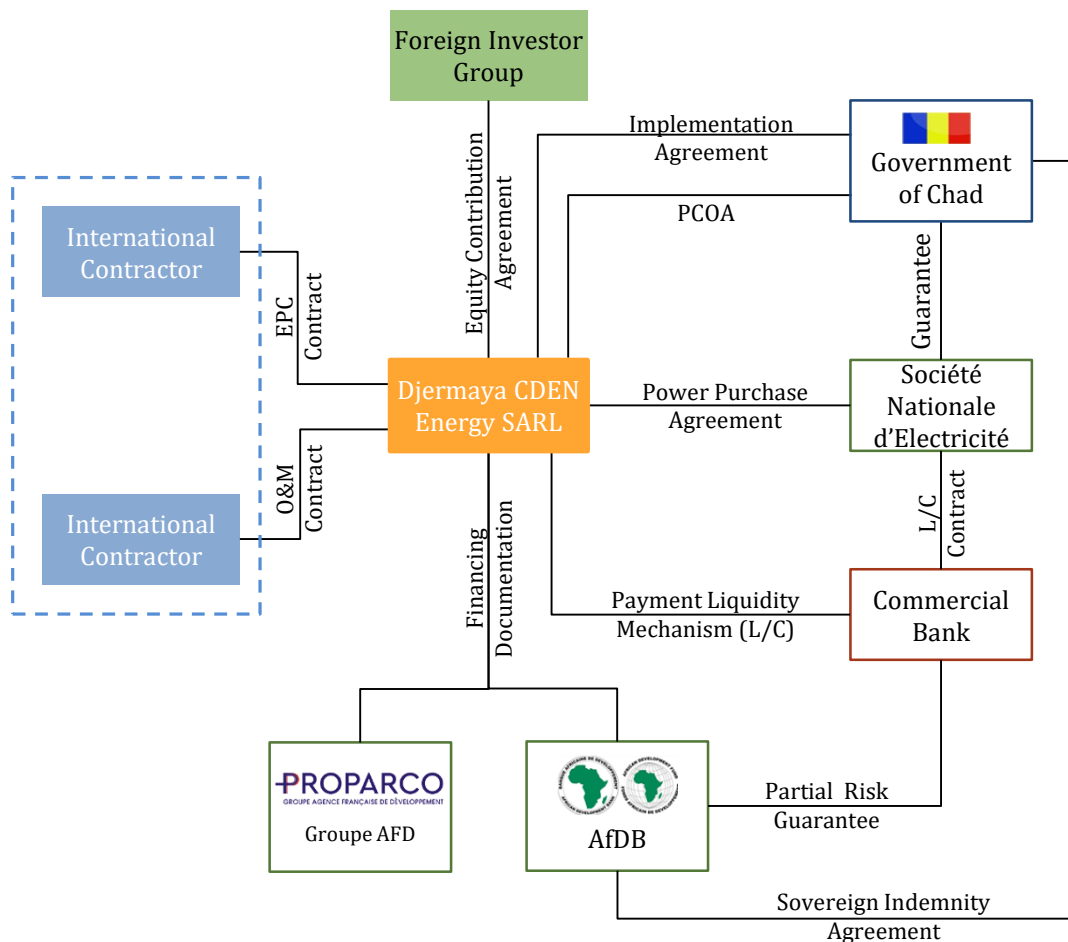
A. Contractual Overview

The contractual structure for the Djermaya project is a classic configuration used in almost all private power and infrastructure projects around the world where the off-taker is not in a strong financial situation.

With reference to Figure 14 below, and starting from the top and moving clockwise, Djermaya CDEN Energy S.a.r.l. (the “Project Company” or the “SPV”) is owned by a consortium of international investors. This will be more thoroughly presented in Section XI of this Memorandum, noting briefly however that the direct owner is Djermaya Holdings Limited (DHL), registered in London. The current shareholders of DHL are InfraCo Africa Limited (UK), Smart Energies International (France), and JCM Clean Power Development Fund (Canada). This shareholding will change from now until financial close, with JCM exiting and Anergi Asset Company (Mauritius) an affiliate of Aldwych International Limited (UK), itself the parent of Aldwych Africa Developments Limited (UK) (AADL) most likely taking a sizeable equity stake.

The Project Company is and will be held by an investor group interested in making a long-term equity commitment in a renewable power asset in Chad. They are bound by a Shareholder’s Agreement, and will likely sign an Equity Contribution Agreement in view of contributing 100% equity upfront at the time of financial close.

Figure 14: Djermaya Contractual Structure



Sponsors are in advanced stages of the application for on a Framework Agreement (“*Convention d’établissement*”) that will be entered into by the Project Company and the Ministry of Commerce. More information about this Agreement will be set out in the next sub-Section B.

Sponsors signed a Power Purchase Agreement (“*Contrat de fourniture et d’achat d’électricité*”, or PPA) with the GoC and SNE on 9th August 2018. Most of the key annexes (including the calculation of the tariff) are already included in the PPA, noting however that a few annexes still need to be finalised but that these can be inserted (and existing annexes can be modified) through a simple process that does not involve renegotiating the PPA as a whole. The PPA is presented in detail in sub-Section C.

The contractual structure includes a payment guarantee mechanism (also referred to as the “liquidity mechanism”) that will involve five stakeholders: The Project Company, the GoC, the SNE, a commercial bank (the “Issuing Bank”) that has not yet been selected, and the public sector of the African Development Bank (AfDB). This structure includes a commercial Letter of Credit (LC), and also a Partial Risk Guarantee (PRG) from the AfDB. Because it is a key credit enhancement, this mechanism will be explained in detail in Section XII “Project Bankability”.

The second credit enhancing feature of this contractual structure is the Put-Call Option Agreement (“*Promesse de vente et d’achat*”, or PCOA) entered into between the Project Company and GoC. This contract provides protection to senior lenders to the Project (Lenders) and investors in the event of termination of the PPA. With respect to Lenders, it always provides for full reimbursement of outstanding loans. With respect to investors, it provides under certain scenarios for repayment of invested capital. The PCOA is also presented in detail in Section XII.

The “Financing Documents” will include *inter alia* the Common Terms Agreement, the Facility Agreements, the Inter-creditor Agreement, the Accounts Agreement, the Hedging Documents, the Direct Agreements, and the Security Documents.

Lastly, the Engineering, Procurement, and Construction (EPC) contract and the Operations and Maintenance (O&M) agreement will most likely be signed with a large Indian power manufacturer – Bharat Heavy Electricals Limited (BHEL) – with whom Sponsors are in exclusive negotiations. Section VIII of the Memorandum reviews the tender process used by Sponsors to select BHEL, provides a corporate overview of BHEL, and then presents both of the contracts in significant detail.

B. Framework Agreement

Framework Agreement sets out the framework within which the project will be undertaken, and addresses key issues with respect to the relationship between the GoC and the Project Company.

The Framework Agreement covers such matters as local sourcing of materials and resources, environmental and social standards, subcontracting conditions, confirms the allocation of government land for the project, indicates that the project company will enter into a power purchase agreement (PPA) with the national electricity company, and confirms GOC undertakings in terms of fiscal matters, legal stability, non-discrimination, and repatriation of funds.

The Framework Agreement will also set out special tax arrangements and exemptions which might, for example, go beyond what is granted under the Investment Charter (Law n° 006/PR/2008). Together, these tax arrangements are a key element in reaching a competitive electricity tariff.

C. Power Purchase Agreement (PPA)

1. General Contractual Information

All information related to the negotiation and execution of the *Contrat de Fourniture et d'Achat d'Énergie* (the “Power Purchase Agreement”, or “PPA”) as well as the information contained within is to remain confidential.

The parties to the Power Purchase Agreement signed on August 9th 2018, are:

- Djermaya CDEN Energy S.a.r.l. (the “Seller”), a company registered under the laws of Chad;
- Société Nationale d'Électricité (the “Buyer”), a company registered under the laws of Chad;
- The Republic of Chad (the “Government”).

The PPA is entered into in accordance with Law N° 014/PR/99 dated 26th May 1999 relating to the generation, transport, and distribution of electricity, which provides for an exemption to the outsourcing of public services and which authorises the generation and sale of electricity between parties per mutual agreement.

The PPA defines Installed Capacity in “MWp” (mega watts peak DC) and differentiates between the installed capacity for the Djermaya project (60MWp) and the installed capacity for Phase 1 (32MWp) and Phase 2 (28MWp). It is important to highlight that the 32MWp installed capacity for Phase 1 is a minimum value⁸.

The PPA will come into force as soon as all condition precedents (CPs) have been satisfied (the “Effective Date”) and will remain in force for 25 years starting from the Effective Date. If all of the CPs are not satisfied before August 9th 2020 parties to the have a right to send notification to render the PPA will become null and void. The main CPs are:

- Granting of a construction permit
- Granting of the environmental authorization
- Entry into force of the Put Call Option Agreement (PCOA)
- Entry into force of the Payment Guarantee
- Completion of all annexes to the PPA

At expiration of the PPA, the Seller will transfer (for a symbolic price of XAF 1.0) the plant to the Buyer along with all the permits and authorizations required to operate and maintain the plant.

⁸ Installed capacity of Solar PV projects is frequently presented as both DC peak capacity (“nominal” or “nameplate” capacity) and AC power that is injected to the grid, for example: 60 MWp (50 Mwac). This is a classic situation where (i) the PPA provides a maximum power output, and (ii) the project sponsor decides to “oversize” the plant in order to maximise annual power produced. The amount of “oversizing” is a tradeoff between increased capex vs. a truncated peak output compensated by more hours at maximum capacity. To be clear, this situation does not apply to Djermaya where the PPA specifies a minimum Phase 1 capacity.

2. Site Attribution and Access, Permitting, and Metering

The PPA acknowledges that the Government, as per Decree N° 1174/PR/PM/MATUH/2014 dated 22nd September 2014, attributed to the Seller free of charge a 100 Ha site where the solar plant will be built. From the Effective Date and until the expiration date of the PPA, the Buyer and the Government will be entitled to access this site up to 12 visits per year to make sure that the construction and operation of the plant are compliant with the conditions set in the PPA.

The Seller will have the obligation to develop and build the plant and the interconnection equipment (i.e. the switch yard on the project site, the 33kV transmission line, and the Lamadji substation upgrade including transformers) in a manner that ensures that the plant and the connection equipment will last at least 25 years.

Although The PPA refers to 1 x 40MVA transformer at Lamadji, the design has changed and now includes 2 x 25MVA transformers, which results in better redundancy. This configuration has been accepted by SNE (ref. meetings between Sponsors, SNE, and BHEL on 26th and 27th November 2018 and minutes that were subsequently signed by SNE). The PPA is currently silent on the battery storage system but the schedule describing the interconnection equipment is intended to be updated in due course. All interconnection equipment (transmission line, transformers at Lamadji, etc.) will be financed and built by the Seller but transferred to the Buyer at Commercial Operation Date (COD).

The Seller will have to install, operate and maintain the metering equipment which will be located on the Sellers side of the plant 33kV bus bar. A verification meter will be installed on the Buyer's side of the same bus bar.

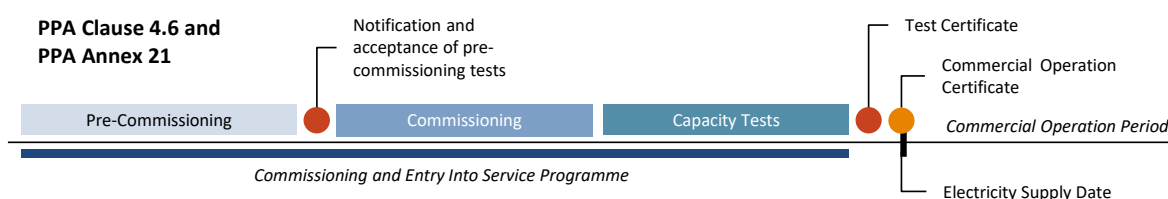
The Buyer and the Government will through best efforts facilitate the procuring of all required permits and authorizations, and to facilitate the construction and the commissioning of the plant.

3. Testing and Commissioning

Testing and commissioning is set out in Clause 4.6 as well as in annex 21. The Seller will bear all the costs related to the testing phase and will appoint an independent engineer who will witness and validate the Pre-Commissioning Tests. If those tests are conclusive, the independent engineer will deliver an Independent Engineer's Certificate.

Within 45 days after the delivery of this certificate, the Commissioning Tests will have to be completed. Those tests can be renewed as many times as necessary. Although the PPA allows for the tests to be completed in several stages, and the plant connected progressively (at least 3MWp per stage), it is likely that the plant will be tested and commissioned at one time. The diagram in Figure 15 below sets out the timeline for the various tests, notifications, and certification.

Figure 15: PPA Timeline



4. Commercial Operations

The Seller will start selling energy to the Buyer as soon as the Seller will notify the Buyer that the Commissioning Tests have been successfully completed.

If the Electricity Supply Date does not occur within 12 months after the Effective Date (the “Target electricity Supply Date”) due to Buyer Default, Government Default, or Political Force Majeure, the Buyer will to pay for deemed energy until Commencement Date or PPA termination.

If the Electricity Supply Date does not occur by the Target Electricity Supply Date due to Seller Default, the Seller will have to pay penalties to the Buyer. Those penalties will be equal to XAF 2,000,000 per day and will be paid until Commencement Date or PPA termination, with a cap equal to 1% of capex.

If the plant does not achieve the Electricity Supply Date within 2 years after the Effective Date, and unless the parties have agreed to extend this date, the parties will be entitled to terminate the PPA.

5. Operation and Maintenance

The Seller will oversee the plant maintenance during the entire Operating Period and the Buyer will do its best efforts to facilitate the operation and maintenance of the plant.

If the Seller decides to use subcontractors, the identity of the subcontractors shall be disclosed to the Buyer and the Government prior to their selection.

At least 45 days before each Contract Year, the Seller will share with the Buyer the operation and maintenance plan, detailing all the actions to be implemented within the next 12 months.

The Seller will also share with the Buyer a monthly table forecasting the daily production of the plant. For grid management considerations, the Buyer will be allowed to disconnect the plant a maximum of 20 hours per year. The Buyer will do its best efforts to avoid disconnecting the plant between 10am and 4pm. The hours above this 20-hour threshold will be compensated to the Seller following the take-or-pay principle.

6. Price and Invoicing

It is important to highlight that the PPA is governed by a take-or-pay principle and that the tariff is paid in XAF but indexed to the EUR.

The Energy Effectively Delivered and the Deemed Energy will be bought by the Buyer as per the following price:

$$P_e = (1 + t_a)^{(ACE)} * P_{ref} * I_{ref}$$

Where:

P_e	energy price in XAF for a given invoicing period
t_a	yearly indexation rate, equivalent to Chad cumulated inflation rate from production start
ACE	number of completed Contract Years before the invoicing period
P_{ref}	reference price for a given invoicing period
I_{ref}	XAF/EUR reference index at invoicing date

Starting from the Effective Date, the reference price is XAF 79/kWh. The reference price will then be updated according to the following table:

Figure 16: Tariff Level

Revision date of the reference price	Reference price
1 st day of the 6 th contract year	XAF 78 / kWh
1 st day of the 11 th contract year	XAF 76 / kWh
1 st day of the 16 th contract year	XAF 70 / kWh
1 st day of the 21 st contract year	XAF 62 / kWh

The Seller will invoice the Buyer in XAF on a monthly basis and the Buyer will pay the Seller within 30 days after having received the invoice. In case of payment delays the applicable interest rate is equal to Euribor 6 months plus 400 basis points.

7. Payment Guarantee

Prior to Financial Close, the Buyer or the Government shall implement an on-demand payment guarantee⁹ covering a 12-month period, and this guarantee shall remain in effect during the entire Operating Period. The inability to maintain a 12-month payment guarantee constitutes an Event of Default.

The Seller may call the payment guarantee 5 days after notification of non-payment is sent to the Buyer. The Buyer and/or the Government must replenish the full amount of drawn funds to the issuer of the guarantee within 10 days. If the payment guarantee is not replenished within 6 months of being called this will constitute an Event of Default.

Until the Electricity Supply Date, the guarantee will be equal to EUR 7,931,000. At the Electricity Supply Date and at the beginning of each Contract Year, the guarantee will be adjusted according to the following formula:

$$M_a = M_u * C_i * (1 - d_m)^{(a-1)} * \frac{P_{ref}}{P_{ref0}} * (1 + t_a)^a$$

Where:

- a is the current contract year
- M_a is the amount of the guarantee, in euros, for the contract year a
- M_u is the amount of the guarantee expressed in euros per MWp et equal to EUR 247,844 per MWp
- C_i is the installed capacity
- d_m is the yearly average degradation rate of the plant (0.4% per year)
- P_{ref} is the price of energy
- P_{ref0} is the price of energy at the 1st day of the 1st contract year
- T_a is the yearly indexation rate

The parties acknowledge that the payment guarantee will be implemented with the support of AfDB through a Partial Risk Guarantee (PRG)¹⁰.

⁹ Although not specified in the PPA, this will be in the form of a revolving letter of credit. The letter of credit needs to be issued by a commercial bank having a minimum international credit rating of BBB+.

¹⁰ It is important to note that PRG will only remain in force during the senior debt amortisation period, or in other words, the Seller will not benefit from PRG after the senior debt is fully amortised until the end of the PPA.

The Seller will bear the costs related to this guarantee comprising a starting fixed fee of 1% and an annual fee of 0.75% of the guaranteed amounts. All the disputes related to invoicing will be solved by an independent expert.

8. Events of default

The following events can be on considered as a Seller Default:

- A false declaration or guarantee that will harm the Buyer interests, and the inability to find a solution to this Default within 45 days
- A non-compliance by the Seller of its obligations under this PPA, and the inability to find a solution to this Default within 45 days

The following events can be considered as a Buyer Default:

- Payment default not covered by the payment guarantee that is not resolved within 20 days;
- A false declaration or guarantee that will harm the Seller interests and the inability to find a solution to this Default within 45 days;
- Failure to recharge the payment guarantee within 6 months after its first use;
- A non-compliance by the Buyer of its obligations under this PPA and the inability to find a solution to this Default within 45 days;
- Government default.

The following events can be on considered as a Government Default:

- A false declaration or guarantee that will harm the Seller interests and the inability to find a solution to this Default within 45 days;
- Failure to recharge the payment guarantee within 6 months after its first use;
- Failure to the renew the payment guarantee for a 12-month period;
- Expropriation event;
- A non-compliance by the Government of its obligations under this PPA and the inability to find a solution to this Default within 45 days;
- Payment default not covered by the payment guarantee that is not resolved within 20 days.

All the above-mentioned events can lead to the termination of the PPA.

9. Force Majeure

The contract provides for a standard definition of Natural Force Majeure and specially mentions as examples the following events:

- Natural events such as drought, lightning, hurricanes, volcanic eruption...
- Plane crashes, train accidents, major interruption of transportation services...
- Plague of disease outbreaks

The contract provides for a standard definition of Political Force Majeure and specially mentions as examples the following events:

- War, invasion, armed conflict, terrorism...
- Radioactive contamination
- General strike

- Discovery of archaeological findings, ammunitions, or mines
- Boycott or embargo imposed by a foreign power preventing the delivery of equipment

A party affected by a Force Majeure can suspend its obligations under the PPA. If a Force Majeure event has effects lasting at least 180 days over a 12-month period, the parties agree to renegotiate the terms and conditions of the PPA.

10. Termination

The Buyer will be entitled to terminate the contract if:

- A Seller Default occurs and no remedy can be found;
- The Commencement Date does not occur within 2 years after the Effective Date;
- In case of Force Majeure.

The Seller will be entitled to terminate the contract if:

- A Buyer Default occurs, and no remedy can be found;
- A Government Default occurs, and no remedy can be found;
- The Commencement Date does not occur within 2 years after the Effective Date;
- In case of Force Majeure;
- In case of modification of the tax and legal conditions applicable to the PPA.

The consequences of a termination are detailed in the PCOA.

11. Transfer and change of control

The parties cannot sell, transfer or pledge their obligations under this PPA except for the following events:

- The Seller can transfer its obligations if such a transfer is required by the applicable law or by the Lenders;
- The shareholders of the Seller can sell the plant to an interested party if the following conditions are met:
 - The interested party has the technical and financial capacities to respect all the Seller obligations;
 - The interested party had over the past 3 years and has no legal dispute with the Government;
 - The interested has no activity that could harm the strategic interests of the Government.

12. Applicable Law and Dispute Resolution

This PPA is governed by the law of Chad. The Government has renounced its right of sovereign immunity. All communications shall be in French.

Both parties should work together to resolve disputes out of court and within a period of 30 days following notification. Should the dispute not be resolved, it will be referred to the International Chamber of Commerce for mediation. If it still remains unresolved 45 days after a mediator has been designated, (or longer if mutually agreed), it can be referred to the International Chamber of Commerce for arbitration.

13. Key PPA Terms in French and English

Figure 17: PPA Terms

Année Contractuelle	Contract Year
Arrêt	Outage
Bouclage Financier	Financial Close
Capacité Installée	Installed Capacity
Cas de Défaut	Event of Default
Cas de Défaut de l'Acheteur	Buyer Default
Cas de Défaut de l'Etat	Government Default
Cas de Défaut du Vendeur	Seller Default
Certificat de Mise en Exploitation Commercial	Commercial Operating Certificate
Certificat d'Essais	Test Certificate
Certificat de l'Ingénieur Indépendant	Independent Engineer's Certificate
Certificat d'Essais	Test Certificate
Contrat de Fourniture et d'Achat d'Electricité	Power Purchase Agreement (PPA)
Convention de Promesses de Vente et d'Achat	Put Call Option Agreement (PCOA)
Convention d'Etablissement	Framework Agreement
Convention PRG	Partial Risk Guarantee (PRG)
Date Butoire de Fourniture d'Energie	Longstop Date
Date Cible de Fourniture d'Energie	Electricity Supply Target Date
Date d'Effet	Effective Date
Disponibilité	Actual Availability
Date de Fourniture d'Energie	Electricity Supply Date
Date de Signature	Signing Date
Date Effective	Effective Date / Taking Over Date
Electricité Nette	Net Energy
Electricité Effectivement Livrée (EEL)	Energy Effectively Delivered
Energie Non Enlevée (ENE)	Deemed Energy / Energy Not Off-taken
Essais	Tests
Tests de Conformité	Pre-Commissioning Tests
Essais de Mise en Service	Commissioning Tests
Levée des Conditions Suspensives	Satisfaction of Conditions Precedent
Garantie de Paiement	Payment Guarantee
Indisponible	Plant Unavailability
Mise en Exploitation Commerciale	Taking Over / Commencement Date
Mise en Service de la Centrale	Commissioning
Période d'Exploitation	Operating Period
Test de Ratio de Performance	Capacity Test

D. Government Involvement

Section V has provided an overview of all key project contracts and then examined the two main project contracts that underpin the entire project: the Framework Agreement and the Power Purchase Agreement. Before ending this section, it would be appropriate to firstly describe how the negotiations are being undertaken with respect to the Framework Agreement in terms of process, and secondly, to highlight that the GoC has been well advised throughout the PPA negotiations and has therefore been able to engage on a level footing with Sponsors.

1. Creation of a Cross Sector Task Force

The Government of Chad identified Djermaya as a priority for the country, and as a sign of support, in May 2015 the Minister of Energy created a cross sector task force (the “Task Force”) comprising:

- SNE (the national utility)
- ADER (the renewable energy agency created in Chad to assist in the coordination of all project aspects and in obtaining authorisations)
- Ministry of Energy (the authority in charge of SNE and the electricity sector)
- Ministry of Infrastructure (an authority that has general oversight over large infrastructure programs)
- Ministry of Finance and Budget

Initially the Task Force met on a bi-monthly basis with the aim of: (i) building awareness of the project; (ii) negotiating the power purchase price; and (iii) selecting a law firm to assist the Government of Chad through detailed negotiations of amendments to project documentation. The task force was successful on all three accounts, culminating in the signature of a letter of agreement by the Ministry of Energy for a revised tariff of XAF 83 /kWh in October 2015 and the selection and appointment of legal firms Kalliopé and Lex Africa in August 2016.

Key stakeholders of the Task Force and their representatives constituted the counterparty to all negotiations on the PPA aided by the counsel.

More recently an inter-ministerial committee (the “Committee”) headed by a representative of the Ministry of Commerce has been tasked with the evaluation of the Framework Agreement application. The Committee makes recommendations to the Minister of Commerce who then validates these and raises them at the Council of Ministers.

The most recent meeting with the Committee was on 23rd October 2018. However, the negotiations of the Framework Agreement and on other key contracts including PRG and PCOA was temporarily placed on hold at the end of 2018 given the uncertainty surrounding potential IMF restrictions that could impede the Government from entering into such contracts. Section II already described recent positive developments coming from the IMF which should now allow the process to pursue its normal course.

2. Legal Assistance

As the Government of Chad had little experience in negotiating the documentation required for an IPP, and to ensure that their interests were protected and that they were fairly represented in negotiations with the Project, the consortium applied for a Technical Assistance Facility (TAF) grant from PIDG to fund strategic and legal advice for the government parties. A grant for an amount of USD 523,000 was approved.

The scope of the legal advisory services has been to:

- Review and mark-up existing signed documentation against changes proposed by the Project Sponsors (PPA, Framework Agreement, PCOA and Liquidity Support, Presidential Decree, Direct Agreements and other key documents);
- Represent the GOC and SNE in subsequent negotiations of these documents;
- Amend, review and finalise transaction documents;
- Facilitate the review and sign-off of the documentation by the relevant government parties at a national level.

The selection process undertaken in 2016 culminated in the appointment of a consortium of international and local legal firms consisting of Kalliope and LexAfric:

- Kalliope is a legal firm based out of Paris. There are few legal firms with experience in the power sector in Chad. Kalliope stands out in that regard as they were involved in the creation of the Chadian legislation on renewable energy. Further information is available at <http://www.kalliope-law.com>
- LexAfric is a regional tax and legal counsel based out of Chad. The firm was created in 2011 and brings together a number of lawyers having gained experience in the renowned international firms. Further information is available at <http://www.lexafric.com/index.php/fr/le-cabinet.html>

Currently Kalliope and Lex Afrique are meeting regularly with government representatives in Chad. With respect to funding, as of January 2019 there remains USD 344,000 available which should be sufficient to pay for legal expenses incurred through to financial close.

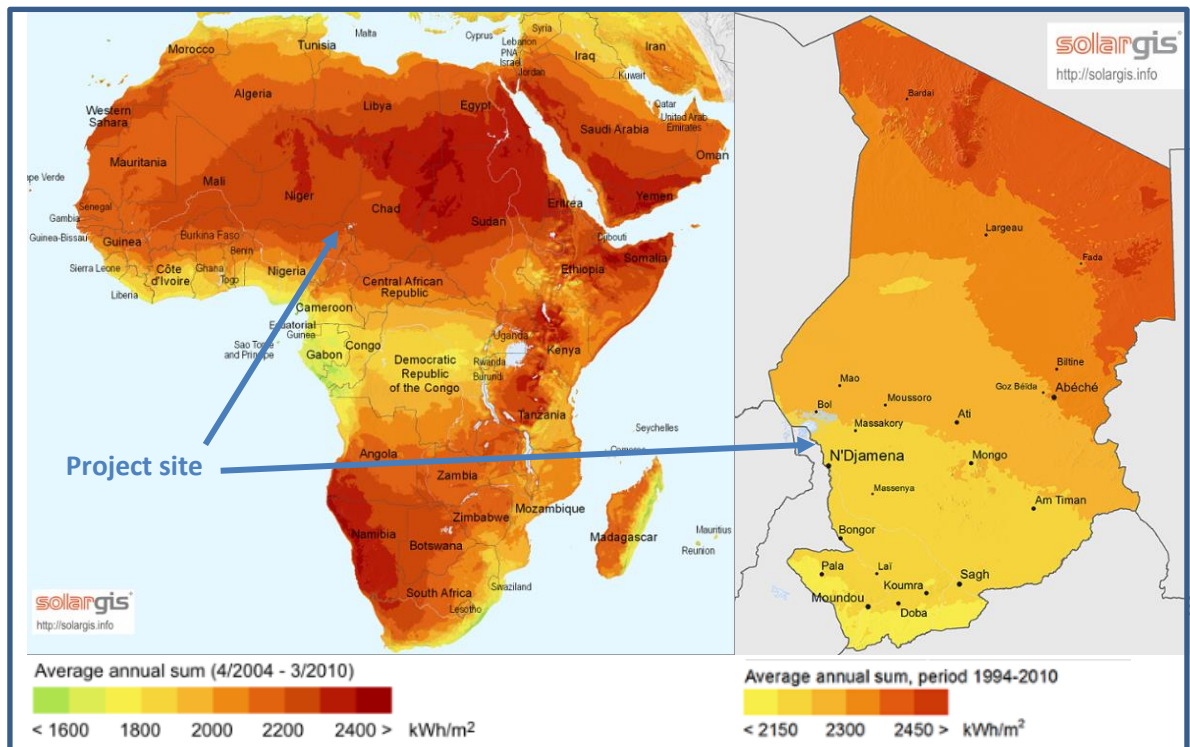
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VI. Solar Resource

A. Initial Assessment of Solar Resource

The project site is located in an area of consistent solar resource. The global horizontal irradiation (GHI) is estimated at 2,191 kWh/m² based on information from three leading satellite datasets: Meteonorm 7.1, SolarGIS, and NREL CSR.

Figure 18: Solar Resource in Africa



Source: SolarGIS

Using the preliminary design, the Project Sponsors' technical advisor determined that there was a 50% probability that each 1 kWp of available capacity would generate in excess of 2,178 kWh annually ("P50"). It should also be highlighted that the seasonality of irradiation is limited and the monthly specific yields are within 12% of the annual average.

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Figure 19: Monthly Yield

Month	GHI (Kwh/m ²)	Average temperature (° C)	Specific yield (kWh/kWp)	Variance to average (%)
January	178	23.4	189	-4
February	170	27.1	173	-5
March	195	30.9	190	4
April	196	34.1	189	4
May	204	33.5	198	9
June	184	30.9	179	-2
July	167	28.3	160	-12
August	177	26.5	172	-5
September	167	27.9	164	-10
October	190	29.3	193	6
November	182	27.6	190	5
December	175	24.4	187	3
Total	2,191		2,178	

Source: OST based on MeteoNorm7.1

B. Independent Assessment of Resource

1. Request for Proposals

In December 2019 Sponsors launched a tender in view of carrying out an independent solar resource assessment. DNV GL (DNV), an international company headquartered in Norway, was selected. DNV has a strong business in renewable energy and is a leading independent advisor within this sector. Services include wind turbine type certification, design consultancy, energy yield assessments, site assessments, permitting, solar plants and turbine design, wind and solar forecasting, etc.

The assessment study will be completed no later than the third week of February. A description of the requested services is provided in sub- section 2 as follows.

2. Scope of Work

DNV will review a variety of solar resource datasets having at least ten years of data, compare these data sources, discuss their uncertainties, and select datasets that are most likely to represent the long-term resource at the site location. DNV should consider:

- Satellite-derived data or data interpolated from ground-based measurements having been appropriately used.
- Site adaption (calibration) of satellite data has been used, where appropriate, to reduce the uncertainty in locations remote from a meteorological station.
- Algorithms have been used to convert global horizontal irradiation to irradiation on the tilted plane of the modules.

Solar resource estimates retained for analysis can come from one dataset, or alternatively be based on a method of weighting of values from several datasets. The choice of resource data will be justified and presented in a logical manner preferably with supporting information provided in tabular format.

DNV will review the following documents and highlight any major concerns and/or issues:

- Hourly generation profiles obtained or synthetically generated.
- Plant design information including capacity, tilt and shading angles, orientation, number of modules per string, total number of modules and inverters.
- Module, inverter and transformer datasheets.
- Topographical studies.
- EPC and O&M contracts.

In view of calculating performance ratio, Sponsors have provide DNV with a topographic survey as well as a preliminary design, including amongst other items spacing between the rows of panels, type and number of inverters, transformers, etc.,. The Consultant will calculate the cumulative losses within the plant design and its key components:

- Generate a 3D shading model using software that integrates horizon and near-shading obstacles.
- Integrate key module characteristics including degradation, low light performance, tolerance, and temperature coefficient.
- Integrate key inverter characteristics including maximum power point tracking capability and efficiency profiles.
- Calculate DC and AC cable losses.
- Calculate auxiliary losses.
- Estimate soiling losses based on precipitation profile, environmental conditions, and cleaning schedule.
- Determine availability losses considering plant reliability.
- Calculate plant PR based on overall losses.

DNV will produce specific yield (kWh/kWp) and plant yield (MWh) figures for the first year of operations based on assumed plant performance losses, and then analyse the evolution of plant performance during the lifetime of the project. Uncertainties estimates should take into consideration climate variability (inter-annual variation), solar resource measurement uncertainty, and all of the other variables affecting plant performance including optical, module, electrical, and degradation uncertainties.

Output will include both yearly and monthly yield figures for the mean expected (P50) irradiation and the corresponding yield results, and also includes uncertainties considered for the calculation of the P75, P90, and P99 exceedance probabilities on an annual basis. The analysis will also include yield under a “Worst Case Scenario” using parameters from the EPC and O&M contracts such as PR guarantee, plant availability, and degradation which will be provided ahead of the commencement of the works.

Uncertainty analysis will include long-term irradiation variability (expressed in standard error of irradiation data), uncertainty in yield modelling loss assumptions, and the combined uncertainties can be calculated for various time periods (e.g. 10 years, 20 years). This in turn will be used to calculate specific yield probability adjustment factors beyond year one.

VII. Project Scope and Design

A. Project Overview

1. Project Site and Security

The project is located approximately 30km north of N'Djamena on a 100Ha plot is adjacent to the main highway and approximately 7km southwest of the main oil refinery. The land is flat and devoid of any agricultural activity, with nearest settlements found within a distance of 1km to 4km from the periphery of the plot.

An abandoned pipeline currently runs through the site, which will be disposed of during construction. At the western boundary of the site is a pond which forms from the gathering of runoff rainwater during the wet season which dries out during the dry season.

The main highway is in good state of repair and does not require any particular reinforcement. It is a well-trodden and secure artery of the country's transport network. The road will be used for the conveyance of equipment and for the commute of the workforce which is anticipated to be based in either Djermaya or N'Djamena itself.

Project site preparation will include clearing and grubbing of land, excavation of topsoil, proper disposal of trees and shrubbery, the construction of a gravel road access to the adjacent highway, roads within the site, fencing and an intruder alarm system to ensure safety and security of equipment and personnel.

A geotechnical study was carried out by Terrasol Setec in April 2016. The main conclusion of this study is that the site is suitable for the construction of the PV plant. For construction ease, the study recommends that heavy earthworks should be completed outside of the rainy season. The clay soils within the first meters are suitable for the pile foundations associated with the PV construction. The study notes that for heavy foundations for large structures it is recommended that lime is utilized to add strength to the soils.

Figure 20: Project Location

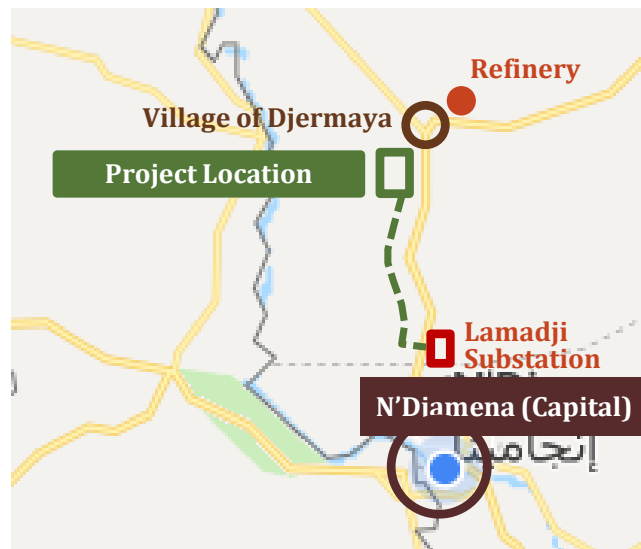


Figure 21: 100ha Plot



GPS Coordinates

NW:	N 12°22'10.14"	E 15° 2'3.0"
NE:	N 12°22'7.89"	E 15° 2'28.53"
SE:	N 12°21'25.89"	E 15° 2'24.69"
SW:	N 12°21'28.09"	E 05° 1'59.22"

A galvanised welded steel wire fence of at least 2.2 metres in height will be installed around the perimeter of the 100ha site. Razor wires (in straight lines or coils) will be mounted on top of the fence if allowed. Fence posts will be installed into the ground with concrete. Although it is not planned for construction and operations personnel to remain at the site overnight, the project scope includes the installation and fitting out of two containers to include bedrooms, toilet and showering facilities, and a meeting room.

The plant security system shall include CCTV coverage (including motion detection and infrared for night-time and poor weather conditions) of the entire perimeter fence and key internal areas including (but not limited to) the site offices, the substation, and the main gate. The system will be designed to continue functioning in the event of grid outage, such that these systems are operational for at least 4 hours.

2. Plant Capacity

Phase 1 of the Djermaya Solar project will have an installed capacity of 32 MWp, which means a peak DC “nominal” or “nameplate” capacity. This capacity is obtained by multiplying the nameplate capacity of each individual module (expected to be 310 watts) by the total number of modules used in the plant.

Module nameplate capacity is verified under standard testing conditions (STC), which includes *inter alia* a module surface temperature of 25 °C. When modules are installed in locations where temperatures are greater than standard, a loss of performance can be expected. For example, the modules that will be used for the Djermaya project will lose an estimated 0.5% for every 1 °C above standard temperature. This means that during the summer in Chad when the module surface temperature could reach 65 °C, this would result in a loss of 20% of efficiency.

The other potential limitation that needs to be considered when calculating the actual AC power injected into the grid is rated capacity of individual equipment. In the case of Djermaya, the limiting piece of equipment will be the inverters, which are rated at 27.5 MVA. So in summary, even though the peak capacity of the plant is 32 MW, the actual capacity in AC will be 27.5 MW. Depending on ambient temperatures, the cause of the reduction in capacity may be a result of reduced module efficiency (i.e. at high ambient temperatures), or else due to reaching the rated capacity of the inverters (i.e. at lower ambient temperatures).

B. Plant Equipment

1. Solar PV Modules

There are two types of crystalline solar PV panel technologies:

- Monocrystalline panels are the most efficient particularly in a hot environment, however they are the more expensive.
- Polycrystalline panels are slightly less efficient but are also less expensive to produce because the manufacturing process is simpler and uses less silicon. The temperature coefficient for polycrystalline technology is higher meaning that efficiency drops off quicker as temperature increases compared to monocrystalline technology. Due to the lower efficiency a slightly larger surface area is needed to produce the same electrical output.

The Djermaya project will use polycrystalline silicon cells having a minimum module efficiency of not less than 14% at standard testing conditions (STC). The modules will be rated at 310 watts and will come with a linear power warranty guaranteeing a minimum of 90% of initial peak power after 10 years and 80% at year 25.

2. Module Mounting

Sunlight has two components: the "direct beam" (carrying about 90% of the solar energy) and the "diffuse sunlight" that carries the remainder. The amount of solar energy collected by PV modules from the direct beam is the amount of light intercepted by the panel which is a function of the cosine of the angle between the incoming light and the panel. When the sun is at the azimuth the cosine is 1 and the energy output will be at the maximum level. Modules can be mounted on fixed structures or on trackers:

- Fixed-mounting system requires modules to be tilted at an angle that is the midpoint of the north-south (seasonal) movement of the sun, which equates to degrees latitude of the project site. A solar panel mounted in a fixed orientation does not track the daily east-west movement of the sun and as a result will lose over 75% of the energy in the morning and evening periods.
- Single-axis trackers allow panels to be rotated in the east-west direction which helps recapture early morning and late afternoon losses. Whilst single-axis trackers do not compensate for north-south (seasonal) movements of the sun, they nevertheless offer significant improvement in total energy output. Compared to fixed mounting systems, use of single-axis trackers will boost annual output by c. 17%.

For the Djermaya project, a single-axis tracking system will be used. The mounting structures will be made from aluminium or galvanised steel.

3. Inverters

With respect to inverters, there are several types of configurations for solar PV installations. Configuration refers to how modules (which produce DC output) will be connected to inverters (which convert DC to AC output). As shown in the diagrams below, there can be one inverter per module as in Figure 22. Solar PV modules can be connected in series to create a string and then connected to an inverter as in Figure 23. There can be multiple strings connected to one inverter as in Figure 24. Lastly, strings can connect in to a central inverter as in Figure 25.

Figure 22: Micro Inverters

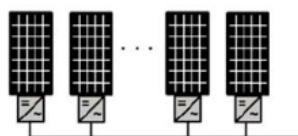


Figure 23: String Inverter

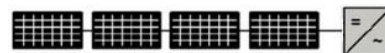


Figure 24: Multi-string Inverter

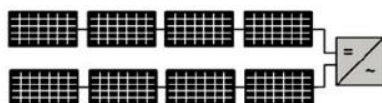
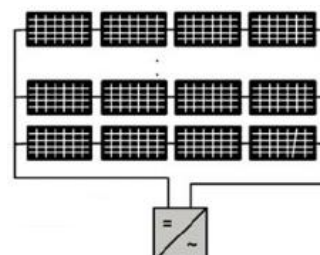


Figure 25: Central Inverter



- String inverters are less expensive in terms of cost per MW compared to central inverters. In this plant configuration, if one inverter faults the output loss is small compared to that of a central inverter and the impact in terms of lost revenue is less. String inverters are held in spares and faulty units swapped out. From an instrumentation and implementation perspective, string inverters are more complex. String inverter communications technology uses serial communications rather than Ethernet resulting in slower data transfer speeds, higher cost of cables, and a greater number of data points to monitor. Over the project life-cycle a string inverter system will be more expensive.
- Central inverters are engineered for reliability more than initial acquisition cost. The system is initially more expensive ultimately but requires less maintenance. Central inverters have improved diagnostics and provide warnings before faulting thus allowing the operator to schedule maintenance before the fault actually occurs.

The Djermaya project will be based on a central inverter design, or more specifically, in a configuration where several central inverters will be coupled using a master/slave arrangement. Inverters will be installed in pre-fabricated lockable substations or in weather proof outdoor installations. Inverter nominal AC capacity will be not less than 30 MVA at 25°C, and not less than 28 MVA at 40°C.

4. Transformers and Plant Substation

At the Djermaya plant there will be eleven (11) 2.7MVA step-up transformers that will inject AC into the 33kV bus bar. Total transformer capacity will therefore be 29.7MVA. The plant substation will have two 33kV bus bars: one which will remain in the ownership of the project, and the other which will be transferred to the off-taker (Société Nationale d'Electricité – SNE). The back-up battery system will connect to the SNE 33kV bus. The SNE 33kV Substation at the PV Plant will be located in a separate room or building from the PV plant substation. The SNE Substation will be handed over to SNE to own and operate.

5. SCADA

Supervisory, Control and Data Acquisition (SCADA) will be offered on a Software as a Service (SaaS) basis. The software and associated data will be centrally hosted on a server which can be accessed by users using a client via a web browser. All historical data will be available during the operational lifetime of the Plant.

The SCADA interface will show the real time parameters and information related to each component and allow easy access all the levels of the system, visualise data and graphs, select time periods, select multiple data sets, create customised tables and graphs, perform basic operations with data and graphs.

6. Weather Stations

There will be three weather stations each of which will have three pyranometers (two for measuring incident irradiation on the plane of module arrays, and one measuring the incident irradiation on the horizontal plane). Data recorded will include:

- Solar irradiance (horizontal and plane-of-array)
- Ambient temperature and module temperature
- Air pressure and humidity
- Wind speed and direction
- Rain level
- Soiling level

C. Interconnection Equipment

1. Battery Storage System

A 4MWh battery back-up system will be included in the overall project design. This system would form part of the interconnection equipment and ownership would be transferred to SNE at commissioning. With respect to physical location of the back-up system, it will be located adjacent to the PV plant and connected to the SNE side of the 33kV bus bar. A single-line diagram of the connection can be found in Annex A.

The battery energy storage system (BESS) is intended to absorb short-term power variations in Solar PV plant output by fast charging or discharging the battery and thus generating a smoother generation curve that can be more easily absorbed by the grid. The smoothing action will be based on an algorithm based on the direct measurement of the last 30 minute average PV plant output power.

- *Frequency Regulation:* Power support will be based on power vs. frequency droop characteristic for system frequency outside of the predefined frequency dead band (49.5 to 50.5 Hz). Operation in this mode will be initiated by detection of low or high grid frequency while the BESS is in any other mode. After normalisation of grid frequency to normal operating range, the BESS shall return to the mode in which it was operating at the start of frequency regulation mode.
- *Fire Protection System:* In line with NFPA or international norms regulations and CBIP guidelines.
- *VAR Support:* BESS will provide VAR support to the grid for voltage regulation purpose and will be able to provide reactive power/power factor set point to the EMS controller/PCS through SCADA HMI (manual intervention). BESS SCADA shall also have the capability to receive the reactive power set point from Energy Management Centre and provide the same to EMS controller/PCS for necessary action.
- *Anti-Islanding Mode:* BESS shall have anti-islanding protection as per IEC 62116 or equivalent international standard.

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Main components of the BESS system are set out in the table shown in Figure 26 below.

Figure 26: Main BESS Parameters

No.	Parameter	Value
1	Rated Discharge AC useful Capacity (measured at PCC)	4.0 MWh
2	Rated charging AC active power (measured at PCC)	[4.0] MW ¹¹
3	Rated discharging AC active power (measured at PCC)	[4.0] MW
5	BESS Round trip AC/AC Efficiency at PCC	0.82
6	Guaranteed Minimum service life	Minimum 10 years
7	Charging rate ¹²	[1]C
	Discharging Rate	[1]C
8	Power factor (Measured at PCC)	Four quadrant capability is required. Operating power factor shall be 0.95 lead or lag
9	Response time:	Response time of BESS system shall not be more than 1 (one) second.
10	Positive and Negative Ramp Rate	Positive and negative ramp rate to support smooth injection of Solar PV plant output power into grid.
11	BESS design operating ambient temperature	0-45 Deg C Ambient
12	Point of Common Coupling (PCC)	BESS feeder at solar plant 33KV or 11kv Switchgear end.

2. Transmission Line

The 18 km aerial double-circuit transmission line will run from the plant southbound parallel to the national highway until reaching a position adjacent to the Lamadji substation, where it will be routed from west to east over the highway and then buried for the final routing to the perimeter of the substation.

The actual routing of the circuits through or around the village of Pombelilé has not yet been decided. A number of existing structures including homes and shops infringe on the minimum regulatory distance to the centreline of the highway. Sponsors are carrying out an assessment of environmental and social constraints and the outcome of this assessment will be integrated into the decision process. It may be necessary to route the circuits as cable through underground conduits for up to a 1.7km segment¹³.

¹¹ The rated discharge and charge values as well as the charging and discharging rate are subject to change following receipt of the CAPSIM system stability report.

¹² C-rate is a measure of the rate at which a battery is discharged relative to its maximum capacity. A 1C rate means that the discharge current will discharge the entire battery in 1 hour. For a battery with a capacity of 100 Amp-hrs, this equates to a discharge current of 100 Amps.

¹³ Cables trenches shall be back filled with appropriate materials to achieve desired cable ratings and accommodate the proximity effects of neighbouring cables. Cables shall be identified with sub surface markers and shall be secured against digging and accidental interference.

The aerial conductor will be 362mm² UPAS AAAC¹⁴ with a 32MVA capacity and will be equipped with an shield wire to provide lightning protection for both lines, and this shield wire will contain fibre optic cables (OPGW). The continuity of the fibre optic cables shall be preserved through underground sections and shall be terminated to provide a complete communications channel between Lamadji and the PV Plant substations.

Lamadji substation: The 2 x 25MVA 33/90 step-up transformers¹⁵ will be located on the north-western corner of the substation. The 33kV side of the transformers will be connected to a 33kV sectionalised bus and switchboard with transformer and feeder cubicles. The 90kV side will feed into the existing 90kV bus-bar (with each of the new transformers having a cable connection to new bays at the remote ends of the 90kV bus-bar. The transformers will be connected to separate sections of the 90kV bus for security.).

E. Grid Integration

1. Analysis of Systems Stability

In 2017 Sponsors commissioned a system integration study carried out by CAPSIM which confirm that 30MWAC of power could reliably be evacuated from the site into the system based on current and future forecast demand profiles. This study was updated in 2018 with similar conclusions.

Following changes to the project scope and design, including the construction of a new 33kV transmission line, the addition of 2 x 25MVA transformers at Lamadji (instead of 1 x 40MVA), and the addition of 4MWh battery backup, a new systems study was necessary. In December 2018 Sponsors sent terms of reference to CAPSIM, which are outlined in the in the next section. These terms of reference were accepted and CAPSIM is currently undertaking the study and expects to have results in mid-February.

2. Scope of Work

The Revised Study will determine the impacts of connecting the Djermaya Solar plant to the N'Djamena transmission network. It should be carried out assuming the plant is online in July 2020, with further sensitivities carried out for year 2025.

The methodology of the study will be based on the SNE network review of the Ndjamen area as a whole or in part and will make recommendations on additional infrastructure and/or operational strategies that may be required.

As concerns voltage performance criteria, it is recommended to use a 5% voltage drop limit for normal circuit operation at 33kV which is reasonably compatible with a 10% standby limit.

¹⁴ Assume an impedance of 0.108 + j0.32 ohms per km per circuit for the overhead line. The resistance is corrected for AC operation at an average operating temperature of 65°C. Also assume approximately 1km of 800mm² 3 x single core Aluminium / XLPE cables per circuit.

¹⁵ The transformers have YNyn0 phasing (rather than the more usual YNyn0(d) arrangement) resulting in high zero-sequence impedance affecting earth faults.

With respect to expected outputs, the Revised Study must:

- Confirm the connection solution as set out in section 3 of Annex A, consisting of a new dedicated 33kV double-circuit transmission line, 2 x 25MVA transformers, and a 4MWh battery back-up system.
- Verify that the PV plant will not result in instability of the network and that the network is stable more generally during production and during certain transient conditions.
- Determine if the plant can continuously evacuate its maximum production potential without problem, taking into consideration any known network expansions and PV plant degradation.
- Consider n and n-1 conditions relating to the 33kV transmission line and the 33/90kV transformers at Lamadji substation.
- Verify and update the configuration (available generation and transmission) of the SNE network as necessary to reflect the current state of the system (N.B. Annex D provides a summary of installed capacity based on Sponsor discussions with SNE in November 2018).
- Carry out the same basic studies as the previous CAPSIM reports and cover as a minimum the following aspects:

Static behaviour

- Steady state power flow
- Voltage profile
- Reactive power flows
- Voltage support: Analysis of “must run” generation whilst solar PV plant is operating to provide voltage support / reactive power to maintain power factors of unity, and 0.95 (lead and lag) at the PV plant, and whether this might lead to any constraint of the PV plant
- kW and kWh losses on the 33kV transmission line

Dynamic behaviour

- Static stability
- Transient stability

Thermal Operations

- Examine the impact of operating thermal generators at a minimum output of 50% (thus maybe limiting the amount of available spinning reserve) and also limiting the amount of spinning reserve to 6MW.
- Determine any load shedding required to maintain system stability.

Short circuit analysis

Harmonic analysis

VIII. Contractor Procurement

This section provides information on the procurement process that was used to select Bharat Heavy Electricals Limited (BHEL) as EPC and O&M contractor with whom Sponsors are in exclusive negotiations. The section then provides a full corporate description of BHEL, including abbreviated financials. The section concludes with the trip report prepared by two members of the Sponsor team that visited BHEL manufacturing facilities in India in December 2018.

A. Procurement Process

1. Introduction

The purpose of the tender procedure was to select a suitable Engineering Procurement and Construction (EPC) and Operation and Maintenance (O&M) contractor for the Djermaya Solar project (the “Project”) in Chad. The tendering process was extensive as well as complex. This can be explained in part by the fact that the Project, which had been under development since May 2014 and will become the first PPP in Chad, had broken new ground in almost every aspect of the development process and involved a wide range of multidisciplinary experts and stakeholders.

The tendering procedures were undertaken in line with international best practices and in full compliance with AfDB procurement rules.

Djermaya Solar used open competition on an international scale. This included a prequalification stage followed by a two-stage bidding process. Tender documents were published on publically accessible internet sites. Bid evaluation and award procedures were based on the lowest resulting tariff from qualified bids, i.e. the lowest resulting tariff from bidders evaluated as having met the qualification requirements as well as the terms and conditions of the bid process.

2. First Prequalification (2016)

The first step in the tender process consisted of a request for Expressions of Interest (Eoi), the key steps of which are outlined as follows.

Starting date: 4th May 2016

Submission deadline: 18th May 2016

Publication of documents:

- Journal N’Djamena Hebdo number 1645
- JCM official website (Procurement Notice - Djermaya Solar Newspaper)

List of tender documents:

- Request For Expression of Interest
- Expression of Interest (Eoi)
- Form 1 – Financial Data
- Form 2 – References
- Form 3 – Time frame
- NDA

Figure 27 below provides the list of 16 contractors that responded to the request for EoI. Considering the nature of the project and in particular the geographical considerations, this number confirmed the success of this first step of the procurement process. However, given the relatively early stage of development of the Project, the Sponsors decided that it was still too soon to provide the full technical requirement that would be needed for the first round of the bid. For this reason the tender was placed on hold and subsequently reinitiated in 2017.

Figure 27: Responses to the first Prequalification Process (2016)

N°	Company Name	Country	Documentation Request	Submission of EOI
1.	Ossi-Yeto-Mitrelli-El-Mor	Angola/Italy/Israel	X	X
2.	CIAT	Chad	X	X
3.	CSCEC	China	X	X
4.	GPEC (Enercap)	China	X	X
5.	Clemessy/RMT/Eiffage	France	X	X
6.	Vinci	France	X	X
7.	ET Solar	Germany-China	X	X
8.	Angelique -Warree	India	X	X
9.	Bharat Heavy Electricals Ltd.	India	X	X
10.	Sterling & Wilson	India	X	X
11.	Enerray	Italy	X	X
12.	CCC	Lebenon	X	X
13.	Desert Technologies-TBEA	Saudi-China	X	X
14.	Arabian Benco /EPS	Saudi/UAE	X	X
15.	AEE	Spain	X	X
16.	Prodiel	Spain	X	X

Note: Sorted by country and company name.

3. Second Prequalification (2017)

After finalising the Project feasibility studies in 2017 the project Sponsors relaunched the EPC tendering process.

Starting date: 21st June 2017

Submission deadline: 5th July 2017

Publication of documents:

- UN website
- PIDG website

List of tender documents:

- Request For Expression of Interest
- Expression of Interest (EoI)
- Form 1 – Financial Data
- Form 2 – References
- Form 3 – Time frame
- NDA

Figure 28 below provides the list of 23 contractors that responded to the request for EoI. Twelve of the submission packages were considered to be complete, and 8 of these were shortlisted to participate in the first round of the tender process¹⁶.

Figure 28: Responses to the Second Prequalification Process (2017)

N°	Company Name	Country	Date Received	Status	Pre-selection	Qualified
1.	Ossi-Yeto-Mitrelli-El-Mor	Angola-Italy-Israel	17/07	Complete		
2.	CIAT	Chad	Did not submit	N/A		
3.	CSCEC	China	Did not submit	N/A		
4.	GPEC (Enercap)	China	17/07	Complete	X	X
5.	Hexing Electrical Co	China	Did not submit	N/A		
6.	Wärtsilä	Finland				X
7.	ENGIE Ineo energy & Systems	France	26/07	Complete	X	X
8.	Clemessy/RMT/Eiffage	France	17/07	Complete	X	X
9.	Vinci	France	11/08	Complete	X	X
10.	Fichtner	Germany	Did not submit	N/A		
11.	ET Solar	Germany-China	Did not submit	N/A		
12.	Angelique -Warree	India	17/07	Complete	X	
13.	Bharat Heavy Electricals Ltd.	India	18/07	Complete	X	X
14.	Sterling & Wilson	India	17/07	Complete	X	X
15.	Enerray	Italy	17/07	Complete		
16.	CCC	Lebanon	12/07	Complete		
17.	Arabian Bemco /EPS	Saudi-UAE	21/07	Complete		
18.	Desert Technologies-TBEA	Saudi-China	Did not submit	N/A		
19.	AEE	Spain	25/07	Complete		
20.	Gamma solutions	Spain	19/07	Complete		
21.	Prodiel	Spain	19/07	Incomplete		
22.	Solarig	Spain	Did not submit	N/A		
23.	NRECA International	US	Did not submit	N/A		

Note: Sorted by country and company name.

It should be noted that Angelique-Warree ended up being disqualified as well as this company was included in the World Bank Group blacklist (Sanctions Board Decision No.102).

Therefore only 7 companies were qualified to participate in the first round of tendering process.

¹⁶ The other firms were not considered to be acceptable, which in most cases was due to smaller balance sheets.

4. First Round of the Tendering Process (Request for Quotes)

Tendering documents were sent to the 7 companies listed in the following table which were then supposed to submit technical and financial offers:

Figure 29: Recipients of Tender Documents

N°	Company Name	Country
1.	GPEC (Enercap)	China
2.	Wärtsilä	Finland
3.	ENGIE Ineo energy & Systems	France
4.	RMT Industrie (Eiffage)	France
5.	Vinci	France
6.	Bharat Heavy Electricals Ltd.	India
7.	Sterling & Wilson	India

Note: Sorted by country and company name.

The original submission deadline was extended by 28 calendar days at the unanimous request of bidders.

Starting date: 22nd February 2018

Questions deadline: 27th February 2018

Original submission deadline: 20th March 2018

Extension: 17th April 2018

Questions deadline from Sponsors: 31st May 2018

Answers deadline from Bidders: 7th June 2018

List of tender documents:

- Djermaya Solar Tender document (RFQ)
- Hydrological study
- Geotechnical study
- ESIA draft
- Topography survey
- Site description
- PPA Schedules
- Taxes scope EPC O&M
- Sample Bank Reference Letter
- EPC and O&M contracts drafts
- EPC and O&M contracts schedules draft
- Chad Security Report
- Direct Agreement with Lenders

On 17th April 2018 Tender Management received proposals from the following bidders: BHEL (16:44 CET) and Engie Ineo Energy & Systems (17:58 CET).

Tender Management was contacted by Wärtsilä and RMT after 18:00 CET. Both companies explained that they had faced technical (IT) problems relating to the submission of their offers and asked for permission to submit offers on 18th April 2018 prior to 12:00 CET. Hugues Guinoiseau, a member of Tender Management, gave consent to the bidders. The proposal from RMT was received on 18th April 2018 at 10:20 CET. Wärtsilä sent part of administrative documents and yield study on 19th April 2018 at 9:54 am CET. Other Wärtsilä documents (commercial/technical proposals) were sent on 20th April at 14:32 CET. The proposal from Wärtsilä was subsequently disqualified because of an unjustifiably late submission.

Of the 4 submitted offers, one was disqualified (Wärtsilä) meaning that just three firms were qualified: Bharat Heavy Electricals Ltd. (BHEL), RMT, and Engie Ineo.

Figure 30: First Round Tender

N°	Company Name	Country	Offer Received	Qualified offers
1.	GPEC (Enercap)	China		
2.	Wärtsilä	Finland	X	
3.	ENGIE Ineo energy & Systems	France	X	X
4.	RMT	France	X	X
5.	Vinci	France		
6.	Bharat Heavy Electricals Ltd.	India	X	X
7.	Sterling and Wilson	India		

Note: Sorted by country and company name.

After receiving the offers, the Sponsors prepared clarification questions that were sent to the three qualified bidders.

In order to continue participating in the tender, RMT was awaiting the Chad Security Report to obtain confirmation from their internal security department to participate in the tender. Subsequent to submitting their bid, and after consultation with their internal risk assessment committee, RMT decided to withdraw their bid submission citing security concerns.

5. Second Round of the Tendering Process (Best and Final Offer)

Provided as follows are the names of the 2 companies that provided their answers on qualification questions on time and were included in the second and final round of the tender.

Figure 31: Second Round Tender

N°	Company Name	Country	Offer Received	Selected
1.	ENGIE Ineo energy & Systems	France	X	X
2.	Bharat Heavy Electricals Ltd.	India	X	X

Discussions of offers and responses to clarification questions with both companies continued during meetings organised by Sponsors in Paris.

Meetings date: 5th July 2018 with Engie-Ineo, 27th July 2018 with BHEL

Best and Final Offer submission deadline: 27th August 2018

Subsequent to these meetings Tender Management asked the contractors to submit their Best and Final Offers (BAFO). With respect to the deadline, both companies submitted BAFOs which were determined to be compliant on 27th August 2018. Details of the BAFOs are given in Figure 32 below.

The lowest tariff resulted from the BHEL proposal. Negotiations with BHEL started in Paris on 4th and 5th October using the EPC and O&M Contract mark-ups submitted with the BAFO. Draft EPC and O&M contracts were initialled with BHEL in London on 31st October 2018. Sponsors signed a letter of intent on 1st November 2018 setting out a period of exclusivity through 31st December 2018. This period was subsequently extended to 31st March 2019.

Figure 32: BAFO Commercial Offers

In EUR cents per Wp	ENGIE	BHEL
A- Transport & logistics: transport to port	4	5
B- Transport & logistics: Cameroun fees & duties included in transport from port to site cost	11	12
C- Equipment: PV module	21	29
D- Equipment: trackers / module mounting structure	15	10
E- Equipment: electrical equipment & auxiliaries (SCADA, metering)	5	2
F- Equipment: inverters & transformer station	8	2
G- Equipment: plant switchgear / switchyard	1	1
H- Equipment: spare parts	1	1
I- Equipment: other BoP equipment	2	4
J- Civil works: foundations/ Piers Type	-	6
K- Civil works: access & internal roads	5	0
L- Civil works: site clearing & other	5	3
M- Other works: electrical works	3	3
N- Other works: mechanical works	3	5
O- Other works: testing & start-up	1	-
P- Other works: general costs (e.g. water sourcing, utilities, H&S)	3	0
Q- Other works: studies, design, engineering & project management	4	-
R- Grid: interconnection / transmission line	7	9
S- Grid: substation upgrade	11	0
T- Other costs (VAT, duties, levies, WHT, custom duties and other taxes)	5	2
TOTAL	114	95

B. Bharat Heavy Electricals Limited

1. Corporate Overview

Bharat Heavy Electricals Limited – BHEL (Crisil AA+)¹⁷ – was founded in 1964 by the Government of India in order to push the power generation installed capacity in the country to 100 GW by the turn of the century. BHEL is one of the largest engineering and manufacturing companies in India. As of 2018 the company employed 37 540 employees and is the second largest employer in the Indian capital goods industry.

BHEL manufactures products and systems for a wide range of activities such as power generation, electrical transmission systems, renewable energy and energy storage, transportation, water management, oil & gas, and defence. Its industrial base includes 17 manufacturing units, 8 service centres, 15 regional marketing centres, and more than 150 project sites across India and abroad.

With respect to power, BHEL has an installed base of 183 GW of power generating equipment in India and abroad and currently has more than 6,000 MW of power projects being executed. BHEL is present in 83 countries and has contributed to the development of 11GW power generation capacity in overseas markets.

BHEL shares are listed on the Bombay Stock Exchange and the National Stock Exchange of India with 63% of the shares held by the State and the remaining 37% by international institutional investors. The share price was at a high of INR 182.65 in April 2017 but has experienced a continuous decline since then and is currently trading at INR 68.20 (30/11/2018). This results in a market capitalisation of EUR 3.2 billion, of which EUR 1.2 billion is free-float.

2. Business Overview

BHEL provides a wide range of solutions to customers in a number of industry sectors:

- **Power generation:** BHEL manufactures a full range of power plant equipment such as turbines, generators, and boilers that are used in coal, gas, nuclear, and hydro power projects.
- **Captive power projects:** BHEL delivers captive power plants using steam and gas turbine technologies to large industrial customers.
- **Transmission:** BHEL manufactures transformers, switchgear, and controls, and also offers on an EPC basis turnkey transmission project execution from concept to commissioning including substation and converter stations.
- **Renewable Energy:** BHEL manufactures PV cells, modules, and the entire range of equipment required for solar PV plants and also offers EPC solutions from concept to commissioning of solar PV plants and grid interactive systems (including battery storage systems).
- **Energy Storage Solutions:** This division was created in 2017 and focuses on lithium-ion battery packs and electric vehicles.

¹⁷ Crisil is an S&P group company. The BHEL rating is a domestic rating. For reference, India sovereign rating is Baa2 / BBB- / BBB-.

- **Transportation:** BHEL manufactures electric locomotives and diesel-electric locomotives, traction drive systems and controls, and traction transformers.
- **Water management solutions:** BHEL offers complete water management solutions for power plants, industries, and municipal applications including pre-treatment, desalination, and sewage treatment plants.
- **Industrial equipment:** BHEL offers drilling technology, well head equipment, compressors, and electrical machines for the petroleum industry.
- **Defence & aerospace:** BHEL supplies strategic equipment for the Indian Defence Forces.

BHEL is also present internationally. Examples of projects executed outside of India include:

- **Gas turbines plants:** 3 x 159MW gas turbines contracted from Siemens with one of the turbines operating in Italy and the other two assigned to projects in Vietnam.
- **Steam Power Plants:** Turnkey execution of the 2 x 120 MW Tripoli West Power Station (extension) in Libya.
- **Sub-Stations:** Execution of a 42MW turnkey power station project at Jizan, Saudi Arabia.
- **Hydro Power Plant:** Execution of 2 x 30 MW each in Nepal.

In the 2017-18 financial year BHEL registered new orders worth EUR 5.2 billion, bringing the total order book to EUR 15 billion.

3. Financial Overview

BHEL financial year runs from 1st April to 31st March. For the purposes of this report INR/EUR exchange rate of 79.42 was used.

BHEL has over the last two financial years focused on profitability and growth. In order to achieve this, the company has been paying close attention to order execution coupled with strict cost control and efficient use of resources through consolidation.

FY 2017-18 financial statements show revenues of EUR 3.7 billion. EBITDA margin of 8.9% is up significantly from the 5.9% achieved in FY 2016-17, and net profit of EUR 65.7 million demonstrates a turnaround compared to the EUR 98.5 loss in FY 2015-16.

Good operating performance can also be seen from the turnover per employee ratio, which has risen from EUR 84.0k in FY 2015-16 to 99.7k in FY 2017-18.

BHEL has almost no long-term debt and is well capitalised as demonstrated by a gearing ratio (liabilities/equity) of 0.97x.

Abbreviated financials are included on the following page.

Figure 33: BHEL Abbreviated Financials
Abbreviated Income Statement

<i>All values Euro millions</i>		2015-16	2016-17	2017-18
A	Turnover	3,359.0	3,838.2	3,746.1
B	EBITDA	-11.7	227.9	333.4
C	EBIT	-102.9	120.9	234.3
D	Net Interest Expense	45.3	44.2	32.1
	Net Profit	-98.5	53.7	65.7

Abbreviated Cash Flow Statement

	Gross Cash Flow	222.0	243.7	536.4
	Change in Working Capital	-174.9	-173.2	-411.9
	Net (Operating) Cash Flow	47.1	70.6	124.5
	Cash Flow from Investing Activities	-2.9	71.2	121.0
	Cash Flow from Financing Activities	15.4	-58.9	-84.0
	Net Increase / Decrease in Cash	34.6	-59.5	161.5
	Cash Beginning of Year	n/a	246.6	187.1
	Cash End Year	n/a	187.1	348.6

Abbreviated Balance Sheet

E	Current Assets	5,683.4	5,401.9	5,438.3
	<i>of which Cash</i>	247.5	187.1	348.6
	Fixed Assets	2,540.9	2,322.1	2562.3
	Total Assets	8,224.3	7,724.0	8,000.7
F	Current Liabilities	2,611.4	2,536.1	2,796.6
G	Long-term Liabilities	1,544.2	1,110.1	1,133.4
H	<i>of which Long-Term Debt</i>	15.9	11.3	7.2
I	Shareholders' Equity	4,068.7	4,077.8	4,070.7
J	Total Liabilities & Shareholders' Equity	8,224.3	7,724.0	8,000.7
K	Number of Employees	42,198	39,821	37,540

Key Financial Ratios

A / K	Turnover per Employee	84,006	96,387	99,791
B / A	EBITA Margin	0.4%	5.9%	8.9%
C / A	EBIT Margin	-4.1%	1.9%	4.1%
E / F	Current Ratio	2.18	2.13	1.94
D / C	Interest Cover Ratio	n/a	1.67	4.76
H / B	Leverage Ratio	1.06	0.05	0.02
(F + G) / J	Gearing Ratio	1.02	0.89	0.97

Note: Exchange rate used: INR/EUR = 79.4242

4. Renewable Energy Strategy

BHEL has been placing increased focus on renewable energy expertise. Starting in 1983, BHEL was the first manufacturer of solar photovoltaic cells and modules in India. Today the company manufactures not only PV cells and modules, but also an entire range of associated equipment including inverters, transformers, power conditioning units, and passive trackers. BHEL also provides end-to-end solutions, including conceptualization, design, engineering, manufacturing, erection, testing, commissioning and O&M of solar PV plants.

BHEL manages a portfolio of 530MW of solar PV, 359MW of which are based in India. This includes ground mounted projects and also rooftop and floating solar installations. During the latest financial year the company commissioned 170MW of solar PV, and the total order book stood at 162MW which included both ground mounted and rooftop solar.

Figure 34 below includes examples of ground mounted solar PV projects:

Figure 34: BHEL Ground Mounted Solar PV Projects

Project Location	Country	Capacity (MW)	Phase
Andra Pradesh	India	50	Commissioned
Odisha	India	20	Commissioned
Madya Pradesh	India	15	Commissioned
Neyveli	India	65	Under Commissioning
Telangana	India	15	Under Commissioning
NTPC Mandsaud	India	50	Under Commissioning
Total		215	

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C. Trip Report: Sponsor Visit to BHEL (December 2018)

Note: A complete version of this trip report that includes photographs and quality documentation has been uploaded to the FirmEx virtual data room.

1. Introduction

BHEL solar PV manufacturing facilities (cells and panels) as well as the design engineering and product support teams are located in Bangalore. The project engineering team is located in Delhi.

BHEL's renewable energy strategy is to become a leading EPC and O&M contractor both in the domestic Indian market and internationally. As a PV plant operator, BHEL has experience in hot and sandy/dusty environments including desert locations in Rajasthan and Gujarat in northwest India.

The PV cell and module manufacturing strategy is based on supplying BHEL's EPC contract needs only. In other words, BHEL does not aim to become a global module supplier pitted against Chinese companies.

BHEL has been involved in solar PV cell technology for past 30 years and has more recently set up a manufacturing plant in Bangalore which satisfies part of the internal module manufacturing needs.

2. Solar Cell Capability

Current cell manufacturing is undertaken using imported polycrystalline silicon wafers from China and Vietnam. BHEL cell manufacturing capability is able to supply ~85% of BHEL in-house module manufacturing needs.

Sponsors visited the semi-automated¹⁸ cell manufacturing area that has an annual output capacity of 175MW. The production room is a clean area under positive pressure. Access is well controlled, with all personnel entering the manufacturing area required to wear overshoes and enter via an air wash. Cells are 4.52W with dimensions of 156mm. Manufacturing yield is 97% and average efficiency 17.5%. Basic steps in the cell manufacturing include:

- Raw slurry-cut wafers are chemically treated using acid and alkaline fluids to texture the surface and thus increase absorption capacity and reduce reflection losses.
- Phosphorous diffusion takes place in a high-temperature furnace which forms the p-n junction.
- Wafers are then moved through an etching process to remove unwanted phosphorous from the edges and rear surface to improve low-light performance.
- Wafers are placed in a furnace where an anti-reflective coating (ARC) is applied to further reduce reflectivity losses.
- The metallisation process includes printing and firing of aluminium and silver strips on the cell front and back surfaces to form the positive and negative electrodes.
- Cells are then tested using a solar simulator and sorted by power rating into batches that will then be used to produce modules of the desired power output.

¹⁸ Whilst modern machines are fully automated, stacks of wafers are moved manually from one fully-automated processing line to another.

From a quality perspective, cells are traced using batch numbers allowing isolation of serial defects. Sponsors recommend that a desktop review of this quality plan is carried out by the LTA and any observations communicated to BHEL.

3. Module Manufacturing Capability

Sponsors visited the semi-automated module manufacturing area as well as the packaging area and warehouse. The compound was originally built in 1932 to produce porcelain insulators. It was bought by BHEL in 1980 and more recently converted into a solar PV cell manufacturing line which entered commercial production in February 2017.

The current annual manufacturing capacity is 200MW although this is going to be increased to 300MW in the near future. Basic steps in module assembly include:

- EVA (Ethyl Vinyl Acetate) is automatically unrolled and placed on top of a sheet of tempered glass.
- A group of 12 cells are robotically aligned and electrically interconnected using metallic strips, and these interconnected strings are then positioned on the sheet.
- Strings are then electrically connected using manual soldering, and then another layer of EVA is laid on top thus encapsulating the cells.
- X-rays are taken allowing close inspection for defects including micro cracks, chips, mis-alignment in the bus bars, etc.
- The glass, solar cells, backing panel, and EVA sheets are then heated under vacuum thus creating an encapsulated, hermetically sealed PV module.
- Edges of the module are then machine trimmed and an aluminium frame is added for protection.
- An epoxy sealant is used around the edges of the frame and the electrical junction box is also attached to the back of the module using epoxy.
- Each module then undergoes an EL (electroluminescence) test to determine power output which is recorded in a database, and modules are then sorted according to output wattage in pre-defined bands.
- Modules are then sent to a fully-automated packaging area where they are assembled into stacks of 28 and then placed into reinforced cardboard crates with pallet bases.
- Each crate has a datasheet pasted on the exterior which includes the key specifications (peak voltage, peak current, max output) for each module.
- Crates are then transported to a warehouse.

Each module weighs 22.5kgs. Modules have a unique serial number allowing complete traceability back to the manufacturing process and test results.

BHEL believes in involving its staff in formulating its manufacturing process to achieve the best quality and lowest module rejection rate. Since this programme was initiated approximately 12 months ago, module rejection rate has reduced from 2.5% to 0.88%.

BHEL has a strong supplier network located in the immediate vicinity. This includes for example cable and cable connector suppliers.

Modules come in standard power outputs and are interchangeable from one manufacturer to another and there is no issue combining mono-crystalline and poly-crystalline modules in one PV plant.

Sponsors recommended that a desktop review of this quality plan is carried out by the LTA and any observations communicated to BHEL.

4. Electronic Controls

Sponsors visited the main electronic controls production facility. BHEL manufactures an extensive array of electronic controls for most of its various business lines including power, transportation, and defence. With respect to power, the company can design, manufacture, assemble, and commission control panels and the related circuit boards, switches, relays, cards, and racks for plants up to 800MW.

In particular, BHEL design and manufacture a full range of power electronics with applications ranging from HVDC to Static VAR compensators, and rail traction electronics.

5. 50MW Solar PV Plant

NTPC (National Thermal Power Corporation) awarded a 250MW EPC contract to Tata Power, Lanco Solar Energy, BHEL, and Sterling and Wilson. This EUR 250 million Anantapur project was awarded through tender in which only Indian companies were allowed to participate. As the lowest bidder, Tata was allocated 100MW. The other three companies were each awarded 50MW. NTPC signed power purchase agreements (PPA) with two domestic utilities: Andhra Pradesh Eastern Power Distribution Co and Andhra Pradesh Southern Power Distribution Co.

The site is approximately 4 hours' drive northeast of Bangalore. Sponsors visited BHEL's 50MW plant, including the control room where we were able to view the operating SCADA system, the main inverters and transformers, and also walk around the site to view the panels and galvanised steel mounting structures which are fixed but adjustable to two tilt settings: 32.5° winter and 5° summer. For this project, some of the modules were manufactured by BHEL and others were sourced externally due to in-house manufacturing constraints. Photos are provided in Annex D.

Installation of the outdoor equipment (panels, string boxes, and cabling) was good and no obvious visual defects were noted.

The control building contained inverters, HV switchgear, SCADA equipment and HMI screens. The equipment itself seemed to be of good quality but the installation was inadequate in some areas. For example:

- Vermin were observed running freely throughout the control building (danger of eating cables).
- Inadequate labelling of switchgear (operations staff had added their own printed labels).
- Finish (painting) of the interior was of lesser quality.
- Installation of the fire protection panel seemed to be an afterthought and cabling to the panel was not fire resistant.
- Cleanliness in the switchgear room (dust and dirt) was inadequate.

All of the above issues can be easily controlled during construction and it is recommended that the Owner's construction management team pays particular attention to the quality of finish of the control and switchgear building and is adequately staffed to be able to do this effectively.

6. HVDC Station

Although not directly related to the Djermaya Solar PV project, BHEL was proud to host a visit to the 800kV 6.0GW four-pole HVDC terminal station located in the vicinity of Agra known affectionately as “the Taj Mahal of BHEL” which it built for the Power Grid Corporation of India. The 1,728 km HVDC interconnector facilitates bidirectional flow of power between the north-eastern region where there is abundant but seasonal hydropower, and the Uttar Pradesh state.

This rather impressive facility was built together with ABB between 2012 and 2017 and serves as a good example of the company’s significant technological achievements. The DC part of the facility is enclosed in a huge structure where positive inside pressure prevents dust accumulation caused by static electricity. The rectifier and inverter comprise 72-pulse water-cooled thyristor bridges that are also housed inside large structures. The remaining AC part of the system (capacitance and inductance banks, switches, circuit breakers, and switching station) is built in the open air. Photos are provided in Annex E.

7. Battery Storage Systems

Sponsors met with Ms. Renuka Gera, General Manager of the Energy Storage Systems Group (ESSG) based in Delhi. BHEL has an R&D team working on battery-backup and storage systems and currently is currently building a battery storage system as part of a 25MW solar PV plant located on an island.

BHEL aims to manufacture modular battery storage systems in 8MWh and 12MWh sizes. BHEL does not supply the batteries but does manufacture most of the remaining hardware¹⁹. These modules will be self-contained sealed units able to operate in dusty environments.

Sponsors discussed Djermaya system requirements, emphasising that the aim was system stability and not peak storage, and that we did not envision any need to factor in battery replacement cost. BHEL will begin working on a design and we should plan on a conference call early in 2019.

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¹⁹ BHEL manufactures lithium-ion batteries used in India’s satellite programme (but these would not be appropriate for solar PV plants)

IX. Construction and Operations

An overview of the Djermaya contractual structure and a more detailed description of the underlying project documents (the Framework Agreement and the PPA) were provided in Section V. In this section, the EPC and O&M contracts are reviewed in detail.

A. EPC Contract

1. Introduction

The parties to the EPC Contract (the “Contract”) include Bharat Heavy Electricals Limited (BHEL, or the “Contractor”), and Djermaya CDEN Energy Sarl (the “Owner”), a company registered under the laws of Chad.

The objective of the Contract is for BHEL to build the Djermaya 32MWp plant, as well as the transmission line and an extension of the Lamadji substation, on a fixed price, date certain turnkey basis with guarantees of plant performance.

Prior to the Taking-Over Date, BHEL will have to demonstrate the plant’s required performance. Liquidated damages will be payable for failure to achieve the performance guarantees. If performance fails to achieve a minimum level, the Owner will be entitled to reject the plant. The Time for Completion is 365 days with possible extensions in limited circumstances.

BHEL will be the EPC Contractor as well as the Operator under the O&M Contract. However, BHEL as Contractor shall manage the interface with the Operator and should co-operate with the latter in providing any required information. BHEL as the Operator will be present during testing and commissioning of the plant (i.e. during the Early Services Period which begins on the Effective Date of the Contract and ends at Taking-Over Date).

The contract is designed to be fully back-to-back with the Power Purchase Contract (PPA). In addition, penalties under the PPA are incorporated into the liquidated damages to be paid under the EPC Contract. All the Owner’s responsibilities in the PPA, if related to the EPC Contract, are addressed in the Contract.

2. Commercial Terms and Security Mechanism

The Contract Price is EUR 30,721,452. The Contractor will receive payments for the executed Works as per agreed milestones under Schedule 21, If and when the Owner agrees that performed Works are in accordance with the Contract.

An Advance Payment of 30% of the Contract Price will be made to the Contractor 10-days before Notice to Proceed (NTP). If the Contractor does not receive the Advance Payment within these 10 days they will be entitled to a day-for-day extension to the Time for Completion until the Advance Payment is received. The Contractor will issue an on-demand Advance Payment Bond of the same amount for the benefit of the Owner thus securing the Advance Payment. This bond will be issued prior to NTP by a Qualified Bank. The amount of the Advance Payment Bond will decrease each month and will expire when the Advance Payment has been fully repaid.

To secure the Contractor's performance of its obligations, the Contractor will furnish an on-demand Performance Bond, prior to the Commencement Date in favour of the Owner for an amount equal to 20% of the Contract Price valid for this amount up to the Taking-Over Date. Its amount will decrease upon the agreed milestones to 0% of the Contract Price at the expiry date of all the Defects Notifications.

Both Bonds shall be issued by an institution having an international long-term credit rating of not less than BBB+ from S&P. If the rating falls below this Minimum Credit Rating, the Contractor shall provide a substitute performance bond from another satisfactory financial institution.

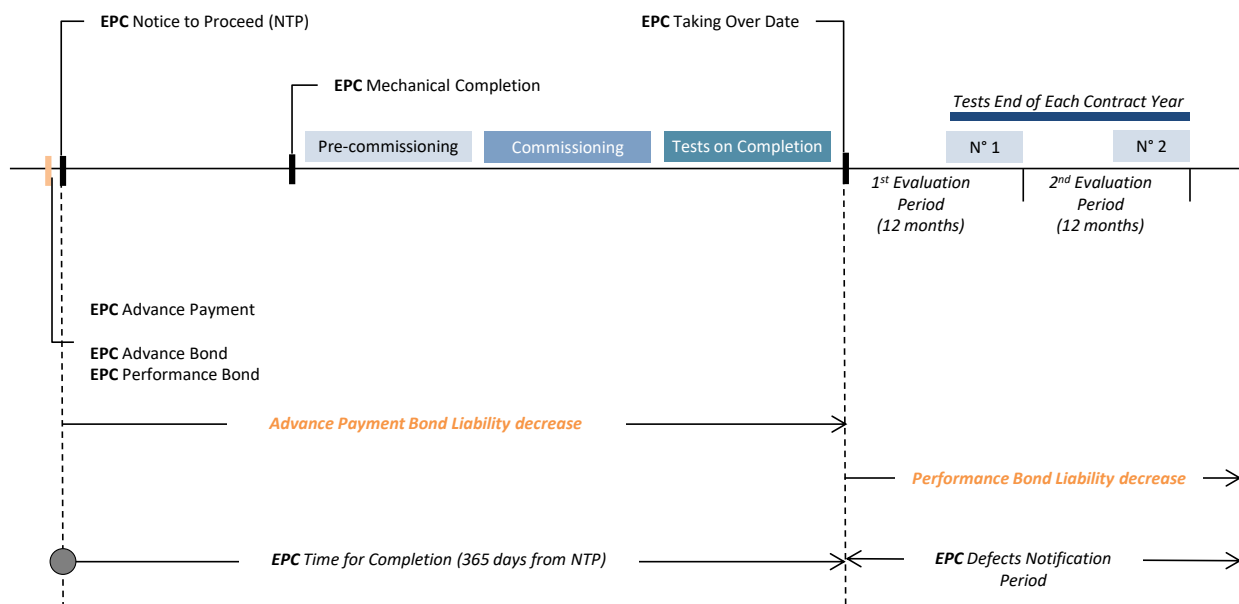
3. Contract Timeline

The Works will start on the Commencement Date the Contractor which will occur when the following conditions are satisfied

- Contract's Signature Date is effective;
- all conditions precedent have been satisfied (including for example that the Contractor has provided Performance and Advance Payment Bonds, and each party has issued their respective insurances;
- the Owner has issued a Notice to Proceed; and
- this date does not exceed the Longstop Commencement Date which is stated to be 250 days after the Signature Date.

The Time for Completion is 365 days from the Commencement Date. The Owner will be Taking Over the plant when *inter alia*: (i) all Works have been completed, within the Time for Completion period; (ii) Tests on Completion have passed; (iii) *Certificat de l'Ingénieur Independent* and *Certificat de Mise en Exploitation Commerciale de la Centrale* have been issued to the relevant party; (iv) all required documents have been provided to the Owner.

Figure 35: EPC Timeline



Under the Contract, the Contractor can ask for an Extension of Time for Completion when *inter-alia*: (i) a Variation has been agreed; (ii) delay attributable to the Owner; (iii) changes in law or tax, occur.

4. Logistics and Construction

The intended transport route for Goods to the Site is shipment to Douala's port in Cameroon, then transported by truck through the country and imported into Chad via one of the agreed CEMAC import roads. The Owner will be the importer of record.

The Contractor will determine and be responsible for its choice of importation route. It will work closely with the Owner to ensure that the process is done in respect of any other agreements and to minimize costs, especially in regards to applicable taxes. The Contractor will pay all incurred taxes in regards to their business, and any taxes or duties do authorities in the country of first import into the African continent (e.g. Cameroon). The Owner is to pay all legal import and custom duties to enter Chad and is to withhold any taxes that would apply on payments to the Contractor.

The Contractor may sub-contract some of the required Works. In accordance with the PPA and the *Convention d'Etablissement*, the Contractor will undertake to give preference to local Chad subcontractors so to promote the local economy. Prior to signing the Contract, the Owner will receive the approved list of Subcontractors.

5. Delays and Performance Liquidated Damages

Delay Damages

Under the Contract, the Contractor is to finish all Works prior to the expiry of the Time for Completion. In the event additional days are needed for the Contractor to achieve Taking Over, the Contractor will have to pay for Delay Damages at a daily rate, as described in Schedule 7, to account for all of the Owner's incurred and spent costs (interest payments, O&M contract, PPA penalties etc.) and loss of revenue. Delay Damages are capped at 15% of the Contract Price.

Under the PPA, the Owner will pay LDs to the SNE for each day the plant has not been operating past the Electricity Supply Target Date (FCFA 2.0m, capped at 1% of the Contract Price of the EPC Contract). This cost is factored in the daily Delay Damages calculation amount.

Performance Damages

The Contractor also provides a Guaranteed Performance Ratio of 80.0%. Performance Ratio (PR)²⁰ is the ratio between the actual energy delivered at the measuring point with the theoretical energy that would be generated if modules converted irradiation into useful energy according to rated capacity. Capacity Tests will be performed prior to Taking-Over Date. If the Actual Performance Ratio is less than the Guaranteed Performance Ratio, the Contractor will be liable for Performance Liquidated Damages as set out in with Schedule 5²¹.

²⁰ PR is a measure of the quality of a PV plant that is independent of location, and it therefore often considered to be a measure of quality. PR is stated as percent and describes the relationship between the actual and theoretical energy outputs of the plant. It thus shows the proportion of the energy that is actually available for export to the grid after deduction of energy losses (e.g. due to thermal loss and conduction loss) and of energy consumption for operation.

²¹ The calculation is carried out in two steps. Firstly, the difference in energy produced = Guaranteed PR less Actual PR multiplied by the energy that would have been produced according to irradiance levels. Secondly, this difference is multiplied by the average revenues per kWh over the Contract Year.

If the Contractor is not able to achieve the Minimum Performance Ratio (95% of the Guaranteed Performance Ratio) the Owner will be entitled to reject the plant. The liability of the Contractor to the Owner for liquidated damages in aggregate (Delay Damages Performance Damages) shall not exceed 20% of the Contract Price.

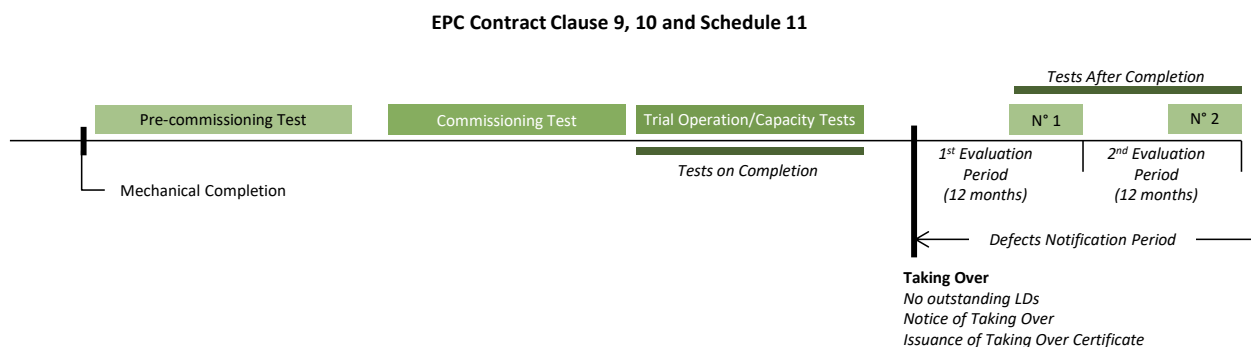
6. Testing and Performance

The Contractor is to inform the Owner of Mechanical Completion which will be confirmed by an inspection. From there, the Contractor is to perform the Tests on Completion for it to obtain the Taking-Over Certificate. Tests on Completion comprises: (i) Pre-commissioning test, which consists of a visual inspection of the plant and testing of individual plant components; (ii) commissioning tests to verify the proper operation of the plant (the commission procedure for each item is described in schedule 11 of the Contract); (iii) performance tests demonstrating the plant's reliability and that it meets the Contract requirements, in particular its Guaranteed Performance Ratio.

Upon completion of this phase, the Contractor will send reports to the Owner demonstrating correct operation of the plant. The Independent Engineer will issue the *Certificat de l'Ingénieur Indépendant* certifying the plant has successfully passed all those tests and granting the Owner the *Certificat de Mise en Exploitation* under the PPA. The *Certificat de l'Ingénieur Indépendant* is one of the conditions for the Contractor to be granted the Taking-Over Certificate.

During the first two years from Taking Over, the performance ratio will be measured twice and certificates will be delivered accordingly: Performance Ratio 12 (24) months Certificate. In the event that the measured PR is below the Guaranteed Performance Ratio the Contractor will have to pay Performance Liquidated Damages. In case the latter does not meet the Minimum Performance Ratio, then the Owner has the right to reject the Works and the Contractor should reimburse the Owner for any past and future costs incurred.

Figure 36: EPC Testing and Commissioning



7. Spare Parts

Spare Parts means spare or replacement parts for the operation and use of the completed Works. An extended list is available in schedule 14. The Contractor is to provide a stock of Spare Parts from start of operations until expiry of the Defects Notification Period

Spare Parts are the Owner's property when delivered on site. Therefore, they are the Contractor's responsibility to import. Spare Part price is binding under the Contract.

8. Defects

The Defects Notification Period is equipment-specific (up to ten years depending on the equipment) and which is extended by two years from the date the part is put into service. The Contractor agrees to provide the following module performance guarantee: At least 90% of initial peak capacity after 10 years and at least 80% of initial peak capacity after 25 years.

The Contractor benefits from Manufacturer's Guarantees, detailed in schedule 24 of the Contract up to the end of the Defects Notification Period. If guarantees last longer than the Defect Notification Period, then the Contractor will transfer them (and all related documents) to the Owner.

The Contractor guarantees and warrants that all works are free from Defects. If any Defects occur under the Defects Notification Period, the Contractor shall procure the replacement and associated works at his own cost. If the Contractor fails to remedy the Defect, then the Owner is entitled to: (i) to carry out the work and have the Contractor pay the Owner for any extra costs ;(ii) come to an agreement with the Contractor so to reduce the overall Contract Price; (iii) if the Defect or damage deprives the Owner of the whole or substantially the whole benefit of the Works, terminate the Contract if no solution can be found, and all costs are to be repaid.

9. Variations

Prior to Taking Over, the Owner may request changes to the Works – or Variations – which the Contractor is bound to execute. In particular, the Owner might ask for:

- Additional tests to be carried the cost of which should be borne by the Contractor if the results show that the Plant is not in accordance with the Contract (otherwise if the tests are conclusive the Owner is to pay).
- Changes in some of the Works, as per the Contractor's suggestion, so to lower the total value of the Works and/or accelerate Time for Completion.

10. Insurance

Both parties must contract insurance with an institution rated no less than A- from S&P and be able to provide evidence of such to the other Party. Parties' obligations under the agreement must not be impacted by any discussions or payments made under the assurance policies. As a general rule, the Contractor and Owner would cooperate so to recover insurance claims. Insured items, responsibilities and costs are described in Schedule 27:

- The Owner shall insure the Works for a value no less than the overall cost of reinstatement, including any demolition and removal costs. This will be applicable from Signature Date until 12 months after the Taking Over Date. Moreover, the Owner should obtain, as a minimum, marine cargo, construction all risks, terrorism and third-party insurances.
- The Contractor must insure his own equipment for their full replacement value, including their delivery to the site. It should also obtain employee liability insurance.

11. Force Majeure

A Force Majeure Event includes Political Force Majeure (which only covers Chad) and Natural Force Majeure. If an event occurs, the affected party will give notice to the other Party as soon as reasonably practicable and try use its best endeavours to mitigate the effect of the Force Majeure Event.

The claiming party shall be relieved from the duty to perform its obligations under the Contract and any liability for failure to perform. None of the parties will be responsible for delays caused by a Force Majeure Event, and periods or dates stated in this Contract shall be extended on a day-for-day basis.

If a Force Majeure Event lasts for a period of 180 consecutive days, or 180 days in total over a 12-month period, then either party shall have the right to terminate the Contract, unless: (i) there are ongoing discussions under the PPA between the Owner and the Buyer concerning the event; or (ii) the relevant performance of the obligation has been completely restored.

If the Contract is terminated under Force majeure, the Owner will pay the Contractor all outstanding liabilities, works already carried out, all removal costs and the Contractor's staff repatriation costs.

12. Termination

The Owner may terminate the Contract when there is *inter-alia*: (i) a breach by the Contractor which has a material impact and is not cured within 21 days, or if it cannot be cured so quickly, has at least commenced an action to cure the problem; (ii) abandonment for a continuous period of 30 days or more; (iii) any act or omission of the Contractor which causes the termination of a Project or Finance Document; (iv) the Contractor default in making undisputed payments within 21 days after notice; (v) the Contractor's failure to meet the Contract's schedule, in particular if Works haven't started at Longstop Commencement Date; (vi) unapproved subcontracted works; (vii) Performance Delay or overall LD cap is exceeded; (viii) expiration of the Contractor's insurance without it being replaced within 5 days; (ix) PPA termination as a result of expropriation or Force Majeure.

If the Contract is terminated by the Owner, the Contractor will:

- Withdraw from the site, assign or novate to the Owner its subcontracts, provide the Owner with all technical, design and other information, deliver required goods, insure protections of life and works, novate at best subcontracts to the Owner, assign manufacture's guarantees to the Owner hence assist with affecting the transfer of authorisations, remove materials and equipment, return any spare parts in custody, and otherwise;
- Pay all costs, losses, and damages incurred, and also pay the Owner any additional cost of engaging a replacement contractor; and
- any additional amounts payable to such contractors and liabilities of the Owner to any Lender

The Contractor may terminate the Contract *inter-alia*: (i) should the Owner default in making undisputed payments when due within 42 days after notice; (ii) should the Owner fail to meet his obligations; (iii) should the Owner's prolonged suspension of works for more than 12 months.

If terminated by the Contractor, the Owner shall:

- Pay all works carried out, costs incurred for ordered materials, all of the Contractor's outstanding liabilities, all of removal costs and all of repatriation costs;
- Return any bank guarantees issued in favour of the Contractor.

13. Contractor's Obligations

The Contractor shall design and execute the Works in accordance with the Owner's Requirements, any consents, Applicable Laws and ESHS Requirements. The fully Contractor shall be responsible for supplying the Owner with a complete, new photovoltaic plant in accordance with the requirements of the Contract.

The Contractor represents that it has obtained all necessary data regarding the condition of the Site. The Owner shall supply any data in its possession regarding the condition of the site, however, the Contractor shall be responsible for verifying such data. Owner shall have no responsibility regarding the accuracy or completeness of any data which it supplies to Contractor. There will be no adjustment to either the Contract Price or the Time for Completion as a result of site conditions.

There will be no adjustment to the Contract Price or the Time for Completion as a result of unforeseen difficulties.

The Contractor represents that it has satisfied itself of:

- the adequacy of the Site and any Real Estate Rights provided by the Owner, for the purposes of constructing the Permanent Works and any necessary Temporary Works, and
- the adequacy of any access routes to the Site.

The Contractor shall be responsible for all activities associated with transport of goods to the Site, including the management of customs and import processes. Owner shall supply any necessary signatures and shall pay any import or customs duties required for the import of goods into Chad.

The Contractor shall be responsible for:

- providing all utilities necessary for construction and testing of the Works (including providing connections for any necessary power, water or other utilities);
- protection of the surrounding environment, and limit damage or nuisance to people or property;
- security of the Site;
- waste disposal from the Site; and
- providing monthly progress reports.

14. Design

The Contractor shall be responsible for the design of the Works and the accuracy of the Owner's Requirements. The Owner shall not be responsible for any error or omission in the Owner's Requirements or any data or information supplied by the Owner. Any information supplied by the Owner shall not relieve the Contractor of its responsibility for the correct design or execution of the Works.

Contractor shall submit certain documents to Owner for review. The review period is 21 days. Within 14 days of Contractor submitting documents to Owner for review, Owner shall inform Contractor if such documents are not in accordance with the Contract. Contractor may proceed with any work during such review period at its own risk. No review or comment by Owner shall relieve Contractor of its responsibility to perform its obligations under the Contract.

D. O&M Contract

1. Introduction

The parties to the Operations & Maintenance Agreement (the “Agreement”) include Bharat Heavy Electricals Limited (BHEL, or the “Operator”), and Djermaya CDEN Energy Sarl (the “Owner”), a company registered under the laws of Chad.

The objective of the Agreement is for BHEL to operate and maintain the Djermaya 32MWp plant on a long-term basis. The Term of the Agreement is 18 years with an Owner’s option to extend this for an additional 7-year Supplemental Period. BHEL will be EPC contractor as well as Operator, but the contractual roles are to remain separate. For example, BHEL as Operator is obligated to “liaise with, cooperate, review, and provide comments and necessary input to the EPC Contractor in developing the O&M Manual and the spare parts list”. BHEL as Operator will be present during testing and commissioning of the plant (i.e. during the Early Services Period which begins on the Effective Date of the Agreement and ends at Taking Over Date).

The contract is designed to be fully back-to-back with the PPA, meaning that penalties under the PPA can be fully passed through to the O&M Operator so that the Owner is not exposed to risk. The terms of the Agreement are confidential.

2. Commercial Terms and Security Mechanisms

During the Early Services Period (ref. timeline in the Annex) the Operator will receive payment in accordance with Schedule 7. During the Operating Period (which begins at Taking Over) the Operator will receive:

- A Fixed Fee for an amount of EUR 96,683 per year, escalated at 2% per annum beginning on the second year after the Effective Date and payable on a monthly basis. This fee is inclusive of all taxes including withholding tax (WHT) and VAT (i.e. if these taxes are imposed by the authorities the amounts will be deducted from the Fee prior to payment).²²
- Payment of Reimbursable Costs, invoiced on a monthly basis either in EUR or CFA depending on the currency in which costs were incurred.

The O&M Services Total Yearly Fee is therefore the sum of the Fixed Fee plus the amount of Reimbursable Costs.

To secure the Operator’s performance of its obligations, the Operator will furnish an on-demand Performance Bond in favour of the Owner for an amount equal to 100% of the O&M Services Total Yearly Fee valid for at least the remainder of the Contract Year. The Performance Bond shall be issued by an institution having an international long-term credit rating of not less than BBB+ from S&P. If the rating falls below this Minimum Credit Rating, the Operator shall provide a substitute performance bond from another satisfactory financial institution.

²² The definition in Clause 1.1 is very clear that withholding tax (WHT) is included in the term “Tax”. Clause 15.5(a) states that payments to the Operator by the Owner are inclusive of all taxes. This being said, the word “taxes” is not capitalised so WHT cannot be assumed. Clause 15.5(b) states that taxes on... the Operator’s profits... or deemed profits...(including any income tax, corporation tax, capital gains tax or other similar taxes) shall be borne by the Operator. Assuming that WHT is a form of tax on profits (i.e. that escape the Government) then it would appear again that the Operator is responsible for paying it.

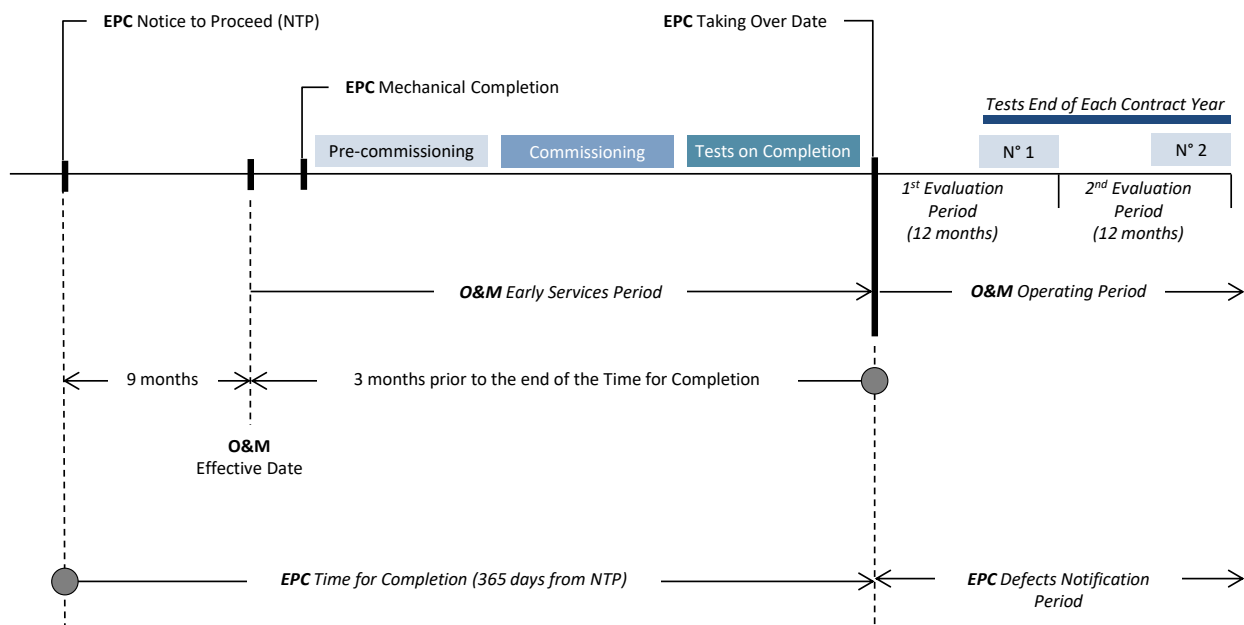
3. Contract Timeline

The Effective Date of the Agreement occurs when *inter alia*: (i) all conditions precedent have been satisfied (including for example that the Operator has provided the Performance Bond); and (ii) the project is 3 months prior to the Time for Completion as per the EPC Contract.²³

The Agreement will enter force on the Effective Date, and then remain in force for a period of 18 years beginning on the Taking Over Date. The Owner has the right to extend the Term for a Supplementary Period of 7 years, i.e. to the end of the PPA period.

If the conditions precedent have not been satisfied within 24 months of the EPC Notice to Proceed (NTP), or if the EPC contract has been terminated, then the Agreement will be terminated.

Figure 37: O&M Timeline



4. Human Resources and EHS

The Operator will employ appropriately skilled and experienced staff to perform its obligations. In view of supporting the local economy, the Operator is to make use of and train local employment as often as possible, including for subcontracting purposes. The Operator will in consultation with the Owner appoint Key Personnel which may not be transferred or changed without approval by the Owner.

²³ Drafting of this definition will change because Time to Completion is not a date, but an interval of time. Corrected wording should be “3 months prior to the expiry of the Time for Completion as per the EPC Contract.” Alternatively, this definition could be “9 months after the Notice to Proceed under the EPC Contract. (N.B. the EPC definition of Time for Completion is “365 days from the Commencement Date”).

The Operator must develop its own EHS management plan that is in accordance with the ESMP, and also comply with all laws in this respect. The Operator shall take all reasonable steps to protect the environment, both on and off the Site, and to limit damage and nuisance to others.

5. Sub-contracting

The Operator may sub-contract any part of the Services with prior consent of the Owner. The Operator will undertake to give preference to local Chad subcontractors.

6. Coordination

The Owner is responsible for obtaining and maintaining the Owner Authorisations as listed in Schedule 12. The Operator will be responsible for procuring, obtaining, and maintaining all other Authorisations which may be required. The Operator agrees to attend all meetings of the Coordination Committee (*Comité de Coordination*), which comprises representatives of the Owner and the Buyer (Société Nationale d'Electricité - SNE) and is in charge of all issues in connection with the operation and maintenance of the Plant, connection equipment, and the Grid.

7. Spare Parts

The Owner must provide the Operator the inventory of spare parts supplied by the EPC Contractor under the EPC Contract. The Operator must maintain and update the inventory showing those items that have been used, purchased, and those requiring replacement.

The Operator is responsible for sourcing of spare parts and for the transport, storage, and security of Spare Parts. During the Defect Notifications Period (as defined in the EPC Contract, generally 5 years from Taking Over Date, except for inverters, modules, trackers, and civil works which is 10 years), the Operator must replace spare parts with new parts at its own cost. After the Defect Notifications Period, the Operator must replace any spares with parts that are free of material defects and at least equivalent to the parts that have to be replaced.

8. Standards of Performance and Warranty

The Operator is responsible for enforcing the EPC Contract during the Defects Notification Period under the EPC Contract and will keep the Owner fully informed of any claims under the contract.

The Operator will at all times operate and maintain the Plant so that it will achieve the Guaranteed Availability and the Guaranteed Performance Ratio. Key to maintaining plant performance is module cleaning. The contractor will use high pressure water cleaning at a frequency of two cleanings per month throughout the year.

The Operator will perform its obligations (i) according to Good Industry Practices, and (ii) in keeping with Applicable Laws, Authorisations, OEM Recommendations, the Approved O&M Plan, the Environmental and Social Management Plan (ESMP), and Lender Standards. The Operator warrants that work that is carried out and spare parts that are supplied will comply with these standards, which together constitute the O&M Warranty.

The O&M Warrant expires 2 years after specific services are rendered, or in the case of spare parts, 2 years after installation. If the O&M Warranty is breached, the Operator will notify the Owner and correct any defects at its own cost including the supply of spare parts if necessary. Re-performed work and replacement parts will also benefit from 2 years of O&M Warranty (although the total O&M Warranty will never exceed 4 years).

If the Operator fails to perform the Services and/or operate and maintain the Plant in accordance with this Agreement, the Owner may, at its sole discretion and at the Operator's risk and cost, rectify such failure through the engagement of suitably qualified third party. The Operator agrees that the rectification of such failure shall not affect any of the Performance Guarantees nor the Operator's obligation to correct any defects under the O&M Warranty.

9. Maintenance, Inspection, and Testing

The Operator will inform the Owner at least 15 days before the beginning of a new Contract Year of any Scheduled Maintenance Outages. These will be undertaken in consultation with the Buyer in order to minimise any impact to the Grid (e.g. preferably during periods of low irradiance). With respect to Unscheduled Outages, the Operator shall advise the Owner and Buyer of the need for and the circumstances surrounding such outages, the proposed commencement date, and estimated duration of the work to be carried out.

The Operator must provide at least 10 days prior written notice of each test and/or inspection, and the Owner, Buyer, and Lenders are entitled to attend any test and/or inspection. The Operator will provide the Owner with a certified report of any test and/or inspection within 5 days after completion. If the Plant fails to pass any test and/or inspection the Operator must rectify and repeat the process.

10. Variations

A variation is a modification or improvement to the Plant or the nature of any Services. Either the Operator or the Owner may propose a Variation. Following a procedural exchange of information, if a decision is made to proceed based on mutual agreement the owner will prepare a detailed technical description with a list of impacts on shutdowns, costs, and on Performance Guarantees, as well as a calculation of the impact on the Fixed Fee resulting from such change. If the variation is carried out by the Operator, the Operator will be entitled to reimbursement of costs (on an open book basis) plus a fixed profit margin of 5%.

11. Dispatching

For each Contract Year the Operator will provide the Owner with a provisional indicative schedule setting out the anticipated Energy Effectively Delivered on a monthly and daily basis as per Schedule 18 of the PPA. The Operator will comply with the procedures of programme and dispatch (*Procédures de programmation et de dispatching*) included in Annex 6 of the PPA as established by the Buyer. Non-compliance with these procedures may result in the Buyer stopping the delivery of power due to breach.

Either the Buyer or the Owner has the right to provide the Operator with an Outage Notification if the dispatching conditions are not carried out as per Annex 6 of the PPA, and the Operator must comply with this Notification within 10 minutes of receipt.

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12. Performance and Liquidated Damages

The Guaranteed Availability is 98.8%. If the Actual Availability is less than the Guaranteed Availability in any Contract Year, the Operator shall pay the Owner Availability Liquidated Damages (LDs) calculated in accordance with Schedule 5²⁴. The Operator's liability for liquidated damages is capped at 100% of the Fixed Fee.

The Operator also provides a Guaranteed Performance Ratio of 80.0%²⁵. If the Actual Performance Ratio is less than (i) the Guaranteed Performance Ratio, or (ii) the performance ratio achieved after the Test on Completion, the Operator will have to pay Performance Liquidated Damages calculated in accordance with Schedule 6²⁶. The Operator's liability for liquidated damages shall be capped at 100% of the Fixed Fee.

The Operator's aggregate liability for liquidated damages for failure to achieve the Guaranteed Availability and failure to achieve the Guaranteed Performance Ratio will be limited to an amount equal to 100% of the O&M Services Total Yearly Fee.

Figure 38: O&M Liquidated Damages

Criterion	Guaranteed Value	Liability Cap
Availability	98.8%	100% of the Fixed Fee
Performance	80.0%	100% of the Fixed Fee
Aggregate Liability		100% of the O&M Total Yearly Fee

13. O&M Plans and Budgets

The proposed plan and budget for the Early Services Period will include the same information as requested for the O&M Plan. In addition, the Owner must also include an estimate for the total amount of Reimbursable Costs budgeted for the Early Services Period.

For the Operating Period, each O&M Plan will: (i) detail the operations of maintenance relating to the components and equipment of the Plant; (ii) set out the works (including major works) to be carried out on the Plant so that it complies with all applicable technical and operational requirements; (iii) include the maintenance program; and (iv) present a proposed Budget for the Contract Year on a monthly basis as shown in Schedule 3. Following a review, the Owner will within 30 days either approve the O&M Plan and Budget or otherwise propose amendments.

The Reconciliation Report will explain the difference between the total amount vs. budgeted of actual Reimbursable Costs and the reasons for any differences.

The 6-Year Plan covers major maintenance and capital repair and recommended improvements.

²⁴ The calculation is carried out in two steps. Firstly, the difference in energy produced = Guaranteed Availability less Effective Availability multiplied by the energy that would have been produced according to irradiance levels. Secondly, this difference is multiplied by the average revenues per kWh over the Contract Year.

²⁵ The formula for Performance ratio is the theoretical value for that year adjusted to the module degradation, hence maintaining a constant PR.

²⁶ The calculation is carried out in two steps. Firstly, the difference in energy produced = Guaranteed PR less Actual PR multiplied by the energy that would have been produced according to irradiance levels. Secondly, this difference is multiplied by the average revenues per kWh over the Contract Year.

Figure 39: O&M Plans and Budgets

Reporting Requirement	Applicable Period	When Submitted
Plan and Budget	Early Services Period	No later than 30 days after the Effective Date ²⁷
First O&M Plan (including Budget)	First Contract Year	120 days before scheduled Taking Over Date ²⁸
O&M Plan (including Budget)	Each Contract Year	120 days before the beginning of each Contract Year
Reconciliation Report	The Completed Contract Year	AS soon as practicable
6-Year Plan	6-year period	At the same time as the O&M Plan and Budget

14. Reporting

The Operator shall provide Monthly Reports as per Schedule 22 no later than 7 days after the end of each Month. These reports will contain inter alia: (i) test results; (ii) a record of significant events; (iii) details on the implementation of the O&M Plan for the current year (in particular with respect to major works to be carried out); and (iv) non-binding estimates of capacity availability and for Scheduled Maintenance Outages for the following 12 months.

The Owner shall provide an Annual Report for each Contract Year within 6 weeks following the expiry of the year, the content of which includes:

- *To the Owner and the Buyer:* (i) a complete operation and maintenance report for the previous Contract Year; (ii) the Operator's preliminary operations and maintenance plans for the then-current year; (iii) the Operator's preliminary operations and maintenance plans for the subsequent years.
- *To the Owner only:* a complete operation and maintenance report for the previous Contract Year.

15. Force Majeure

Force Majeure Event includes Political Force Majeure (which only covers Chad) and Natural Force Majeure. If an event occurs, the party affected shall give Notice to the other party as soon as reasonably practicable. The claiming party shall be relieved from the duty to perform obligations under the Agreement and any liability for failure to perform. Neither party will be responsible for delays caused by a Force Majeure Event, and periods allowed or dates required shall be extended on a day-for-day basis.

If a force Majeure Event lasts for a period of 180 consecutive days, or 180 days total over a 12 month period, then either party shall have the right to terminate the Agreement, unless: (i) there are ongoing discussions under the PPA between the Owner and the Buyer concerning the event; or (ii) the relevant performance of the obligation has been completely restored.

²⁷ This is not going to work. Here we are asking for a report covering the Early Services Period when this period has already begun.

²⁸ This is not going to work. According to Clause 2.1, the Effective Date begins 3 months prior to the End of Completion. Here we ask the Operator to prepare and submit a report before the Agreement is even effective.

16. Change in Law or Tax

Any change that results in an increase in the cost that is greater in the aggregate of the Tax & Law Threshold Amount of 5%, then the Operator shall be entitled to receive compensation from the Owner. Likewise, a change that results in a decrease in the aggregate of the threshold amount will entitle the Owner to reduce the Fixed Fee. The aim is to maintain economic equilibrium for both parties.

17. Dispute Resolution

If the Owner or Operator disputes any invoice (whether in respect of the payment of the Fixed Fee or any Reimbursable Costs), the Owner or Operator, as the case may be, will pay the part of the invoice that is not disputed and then seek to resolve the problem. If resolved, the amount due shall be paid together with interest at the Interest Rate (2.5% per annum).

Dispute resolution follows a three-step process. Firstly, parties can settle disputes in good faith through Informal Dispute Resolution. If this fails to resolve the problem, and if the disputed amount is less than EUR 500,000, the parties can refer the matter to an impartial expert (Expert Determination). If this second step is unsuccessful, or if the disputed amount is greater than EUR 500,000, the parties can submit the matter to the International Chamber of Commerce (ICC). In such a case the seat of arbitration will be London, the Agreement interpreted according to English Law, and proceedings conducted in the English language.

18. Termination

The Owner may terminate the Agreement when there is *inter alia*: (i) a breach by the Operator which has a material impact and is not cured within 30 days, or if it cannot be cured so quickly, has at least commenced an action to cure the problem; (ii) abandonment for a continuous period of 20 days or more; (iii) termination of a Project Document or a Finance Document; (iv) failure of the Operator to maintain any Authorisation; (v) Operator default in making undisputed payments within 30 days after notice. If the Agreement is terminated by the Owner, the Operator will:

- Withdraw from the Site, assign or novate to the Owner its subcontracts, provide the Owner with all technical and other information, remove materials and equipment, return any spare parts in custody, and otherwise assist with effecting the transfer of authorisations;
- Pay all costs, losses, and damages incurred, and also pay to the Owner the additional cost of engaging a Replacement Operator (being the amount by which the costs under the Replacement Operator's contract exceed the amount that would have been payable to the Operator under this Agreement).

The Operator may terminate the Agreement *inter alia*: (i) should the Owner default in making undisputed payments when due within 45 days after notice; (ii) as a result of the PPA being terminated by reason of an occurrence of a Force Majeure Event. If terminated by the Operator, the Owner shall pay the:

- Reimbursable Costs up to the date of termination;
- Pro rata amount of the Fixed Fee, and the Operator's reasonable costs related to demobilisation.

X. Environmental & Social Considerations

A. Project Description

The project is located in the region of Hadjer Lamis, in the department of Haraze Al Biar and the sub-prefecture of N'Djamena Fara, 30km north of the capital N'Djamena. The project site comprises of 100 hectares (one hectare being approximately the size of an international rugby field) situated approximately 300m from the main highway linking N'Djamena to Lake Chad and Massaguet.

The project is surrounded by rural communities grouped in villages and hamlets that are managed in a customary way, under the tutelage of a hereditary village chief. The original right of way of the site partially covered a wetland in the form of a seasonal lake called Dalakaina. The site footprint has subsequently been shifted east so as to avoid this wetland area. Within the proposed site, the project's direct zone of impact, there are no villages or dwellings except for nomadic Fulani settlements that are temporarily established in the area for very short periods of time. Those living and utilising the area within 5km of the project site, which corresponds to the indirect impact zone, are:

- Djermaya, a medium-sized town, located approximately 4km north-east of the project area near the Djermaya refinery.
- Five small villages: Am Soukar, the closest hamlet located less than 1km from the project area; Douguinaga, located less than 2km east of the project area; Am Koundjo, approximately 3km north of the project area (on the outskirts of Djermaya); Délékéna, approximately 2km southwest of the area; and Kilmé, approximately 4km south of the area.
- Several seasonal camps are also located in this area, frequented in the dry season by various nomadic Chadian groups.

Although the site is uninhabited, the land is predominantly tied to the village of Am Soukar, except for the north-west quarter of the site which is managed by the village of Am Koundjo. It is used by local community members for seasonal growing of crops and grazing of animals.

B. Red Flag Report

Egis Environment completed a Red flag environmental and social (E&S) report in September 2014. This document provided described the context, summarised the regulatory requirements in Chad, set out what were believed to be the main environmental and social issues, and presented a high-level summary of the potential risks and recommendations to follow. For example, the need to:

- Follow a development and implement a protocol that fully respects local populations impacted by the construction of the solar PV plant;
- Identify areas used for grazing and identify the correct ways of properly compensating impacted peoples;
- Carry out a public consultation in view of the possible modification to paths taken by migratory peoples and change in use of grazing lands;
- Conduct a full impact assessment;

- Complete topographical and hydrological studies in order to determine water runoff and collection areas and establish the boundaries of the adjacent pond in dry and wet seasons;
- Carry out a geotechnical study to determine appropriate design for module mounting and buildings.

The solar plant project was classified as a Category ‘A’ project, in line with Chadian regulations, which means the project is to be subject to the completion of an impact study. Consequently, the Sponsors applied to, and secured grant funding from, the PIDG Technical Assistance Facility (TAF)²⁹ requesting their support to undertake fully fledged E&S studies for the proposed solar project.

C. Environmental and Social Impact Assessment

The Sponsors conducted a competitive tender to select a consultant to undertake the project Environmental and Social Impact Assessment (ESIA), including the overarching Environmental and Social Management Plan (ESMP). In 2016, Artelia, an international consultancy and engineering company based in France, with extensive experience of E&S work globally on power and interconnection projects (including in Chad) was selected. Artelia teamed up with CIRA-SA, which is accredited by the Ministry of Environment and Agriculture in Chad for carrying out impact assessments.

The study has been undertaken in line with national Chadian environmental, social, health and safety standards, together with the guidelines of the World Bank Group, including the International Finance Corporation’s (IFC) Performance Standards, and AfDB’s standards, with a view of meeting minimum EU standards. E&S fieldwork on site was carried out in 2016 and 2017 in order to capture seasonal variations.

Given the characteristics of the site and the intrinsic qualities of PV power generation, E&S challenges associated with the site and project are limited. Furthermore, given that the amount of land secured is considerably more than the footprint of the first phase of the project, the Sponsors have some flexibility to adapt to E&S constraints.

During the assessment, impacts to the local community and wildlife were identified. The primary issue flagged related to the lake on the project site, namely in terms of the impact on the people who use the area seasonally to grow crops on the banks, and animals that may access the lake shore. Based on these observations, it was decided to shift the site east, away from the waterbody, so as to limit the impacts on the local community and fauna.

Although the impact on the local community has attempted to be minimised by moving the site, the remaining social impact of the project will be mitigated by putting in place a livelihood restoration programme, as well as compensating community members according. A Livelihood Restoration Plan has been drafted and is to be finalised following the Lenders’ completion of their due diligence.

A summary of the potential impacts identified are outlined below (extracted from the solar plant ESIA):

²⁹ TAF is a pool of funding within the PIDG Trust to assist PIDG companies with the aim to support capacity building and to help scope out potential investment opportunities.

Figure 40: Potential Impacts

	Impact
Construction	Construction activities and right of way to the project site could infringe on land use and have environmental consequences and impact local economic activity.
	During construction up to 400 persons will need to be recruited for unskilled and skilled positions.
	Civil engineering works will result in land clearing and earth digging and this temporary exposure of subsoil could result in erosion.
	The transportation and unloading of equipment and material will have an impact on highway traffic.
	Civil engineering work will require raw materials such as metal and sand and other products such as diesel, oil, etc.
	Civil engineering work could lead to the spilling of various effluents such as engine washing liquids, or toilet waste.
	Civil engineering work including digging and filling and land clearing will generate inert waste products as will construction work.
	Heavy equipment and vehicle traffic will generate atmospheric emissions (greenhouse gases, dust) and also noise and light.
	Poor management of the construction site could result in dumping of dangerous waste or cause fires.
Operations	The installation of PV modules and the fences will impact the landscape and also biodiversity and human activity.
	The project will generate electricity from solar energy for distribution via the electricity network.
	Normal operations of the plant could generate waste including those in the category of dangerous industrial waste (electric and electronic, maintenance oils, other fluids). The presence of 12 staff on the site could also result in domestic waste and sanitary waste water.
	The operation of a PV plant will generate little to no noise.
	Inadequate operation of the plant could lead to negative impacts on persons and the environment (e.g. accidental spillage, fire).

Specific mitigation for these impacts is detailed in the solar project ESIA. The Sponsors will ensure the findings of all the E&S studies are carried through to the construction team by putting in place contractual obligations and working with the EPC contractors on their E&S related management plans.

The ESIA report was submitted to the Ministry of Environment in December 2018 for review and public consultation. Once formally approved, an environmental licence for the project site will be issued.

D. Transmission Line ESIA

A further impact assessment will be undertaken for the transmission line aspect of the project. The Sponsors are currently working with SNE, Ministry of Urbanism and Artelia to analyse the different routing options. A crucial component is E&S issues, with the focus on minimising displacement, if any. It is anticipated that the transmission line will be classified as a Category B project (as per decree 630/PR/PM/MERH/2010) and hence be subjected to an impact notice, not a full ESIA. Similarly, if the analysis effectively manages to reduce the level of social impact, the transmission line project can be considered a Class 2 project based on AfDB's standards.

E. Security During Construction and Operation

In 2018 the Sponsors commissioned Drum Cussac Group Limited (Drum Cussac) to carry out an assessment of the security risks facing the Project Company as well as the contractor. Drum Cussac has provided two reports which have been distributed to all stakeholders: Threat Assessment dated 6th May 2018, and Security Review dated 10th December 2018. Perhaps the most pertinent statement is that “All risks are manageable but the security risks are fluid requiring repeated re-assessment.” Other important points as follows:

- Overall, Chad is considered to have a high security risk predominantly because of Boko Haram attacks along the western border near Lake Chad and of Al Qaeda affiliated attacks emanating from the north.
- N’Djamena, and its environs – including the project site located near Djermaya – are assessed as lower risk than the rest of the country, although the threats of terrorism and ordinary crime do exist and require mitigation.
- Whilst terrorist attack is considered, the highest risks are assessed as security forces human rights abuses and community or labour disputes resulting in a stoppage of work.
- In addition to security risks, safety risks especially road traffic accidents and fire and health risks warrant special attention.
- Commuting to and from the site could present a relatively predictable and attractive target for terrorists or armed robbers, as well as increasing the likelihood of road traffic accidents.
- The project site location has few existing security threats but this might change during construction and operation.
- In order for the risks to be kept ‘as low as reasonably possible’ the contractor will require robust security management, with regular and stringent oversight by the Project Company.
- Because international standard medical facilities are limited in N’Djamena and no suitable facilities exist in the area of the project site, it is strongly recommended that the camp has an equipped clinic and a doctor (companies bidding for the EPC contract should be able to demonstrate that their insurance will provide access to an approved clinic / hospital and international medical evacuation).
- The recommended security strategy is for the Project Company to specify standards and to carry out audits to ensure they are being met.

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XI. Project Development and Sponsors

A. The Project Company

1. Development History

The Djermaya project was initiated in 2012 by Citelum, a French lighting company. Shortly after the initiation of the project, Citelum was acquired by Electricité de France (EDF), which subsequently was unable to continue with its renewable development activities in Africa and in 2014 sold its rights to CDEN, a French renewable energy developer. CDEN established Djermaya CDEN Energy S.a.r.l. (DCE) as a company registered in Chad and holding the rights to the Project.

In 2015, JCM Capital, a Canadian registered principal investment firm with renewable energy development activities in Africa, and InfraCo Africa executed a Joint Development Agreement (JDA) with CDEN. In August 2016, CDEN was acquired by Smart Energies with full transfer of the project's rights and responsibilities. Also in 2016, ownership of DCE was transferred to Djermaya Holdings Limited (DHL), a newly formed UK holding company co-owned by the three Project Sponsors. The relationship between the Project Sponsors is governed by a Shareholder Agreement.

As part of a consolidation effort, JCM Capital has decided to focus its efforts on other projects and will not become a long-term investor in the Djermaya project. The Shareholder Agreement is being renegotiated to take into consideration this change.

2. Legal Structure and Shareholding

Djermaya CDEN Energy S.a.r.l. ("DCE" or "the Company") is registered in Chad having its corporate offices in N'Djamena. Issued share capital is XAF 1,000,000 divided into 100 shares all of which are held by Djermaya Holdings Limited (DHL), a UK registered company.

On 24th May 2018 the director, Mr. Guillaume Louis Mompeysson, was replaced by two new directors: Mr. Crispin Holiday, (Aldwych), and Mr. Hugues-Antoine Guinoiseau (Smart Energies). On 13th November 2018 Ms Fatou Gaye replaced Mr. Crispin Holiday.

DHL is a UK registered holding incorporated on 24th May 2016. DHL owns 100% of the share capital of DCE. Each of its three shareholders owns 33 ⅓ % of DHL: Smart Energies Africa S.A. holds 10 class A shares. InfraCo Africa Limited holds 10 class B shares. JCM Clean Power Development Fund holds 10 class C shares. Prior to financial close JCM Clean Power Development Fund will exit the shareholding.

DHL current has three directors: Mr. Hugues-Antoine Guinoiseau (Smart Energies), Ms. Fatou Gaye (InfraCo Africa), and Mr. Thomas Heintzman (JCM). Mr. Heintzman will be replaced when JCM Clean Power Development Fund exits the shareholding.

3. Organisation During Construction and Operations

Figures 41 and 42 on the following page provide two organisational charts for the project company, first during the construction phase, and then during the operations phase. Based on current planning, the project company will not have to employ more than three persons directly and will rely on service contracts with local firms for those tasks which are best outsourced.

Key to the operation will be the Managing Director (MD), who will also act as Chief Financial Officer (CFO). This person will be the sole director (“mandataire social”) of Djermaya CDEN Energy S.a.r.l. and will be appointed by the majority shareholder. Tasks will include: (i) running the business on behalf of the Owner and be responsible for preparing reports for Lenders; (ii) preparing drawdown requests for Lenders and certify Contractor’s invoices; (iii) being the “face” of the project and thus having to interface with press, local chiefs and government officials. The MD will need to be bilingual English/French and may need to be an expatriate.

During construction the project company will also employ a senior engineer, who will likely be an expatriate hired on a fixed term basis. This person will be responsible for ensuring quality of the Contractor’s work, and senior enough to be able to “persuade” the contractor into complying with the EPC contract and international standards to produce a quality final end product. The engineer will also be responsible for ensuring safety on the site. Unsafe working practises, whether the norm in Africa or not, will not be tolerated and this person must have the personality to enforce safety. The engineer will also be the first point of contact with the local community to explain what is going on and to deal with complaints.

The MD and the Engineer will be supported by an administrative assistant who will manage travel, calendars, and correspondence. This person should be able to set up a document management system for the hundreds of documents the contractor will supply. The project company will also have the services of a driver, although it is not yet certain if the driver will be a full-time employee of the company.

As a single shareholder limited liability company, Djermaya CDEN Energy S.a.r.l. does not need to have a Board of Directors. However, Sponsors believe that it will be advantageous to have an Advisory Committee comprising two influential persons, one Chadian that is well connected locally, and foreign national, probably having previously held a high-level position with a multilateral organisation. These two individuals will have access to the highest levels of the Government and SNE and lobby in favour of the project company. These positions will be remunerated commensurate with the required engagement.

All other activities necessary for running a successful business, including accounting, tax, and payroll, will be outsourced to competent firms selected through competitive tender. It is also expected that most of the reporting to Lenders will be carried out through a service agreement by the Shareholders.

As the project company transitions from the construction phase to the operations phase, the fixed term employment contract of the Engineer will be terminated. A new person will be hired on a full-time, permanent basis. This Asset Manager will assist the MD with all of the reporting requirements, liaising with local authorities, assisting with accounting and tax matters, and ensuring that all environmental and social issues are being handled effectively.

Figure 41: SPV Organisation During Construction

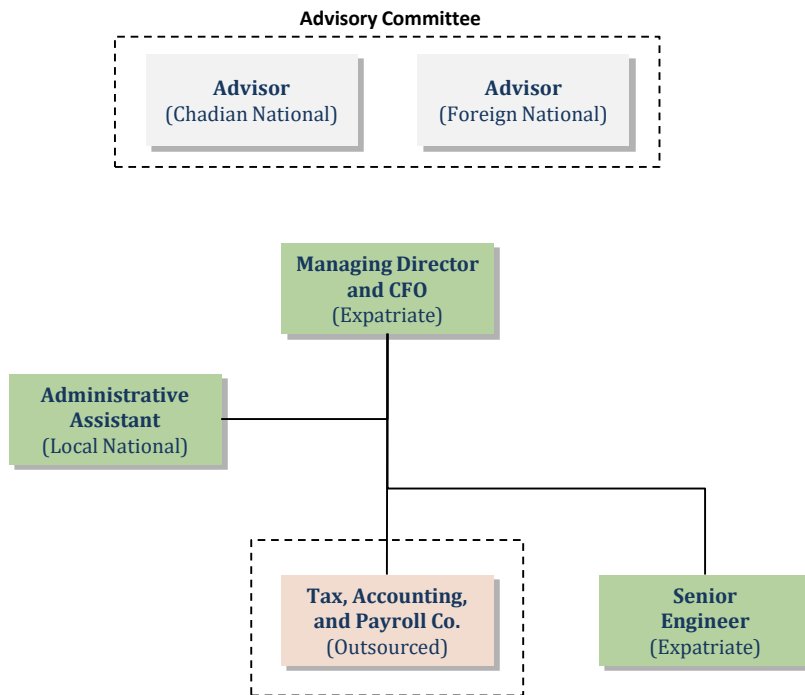
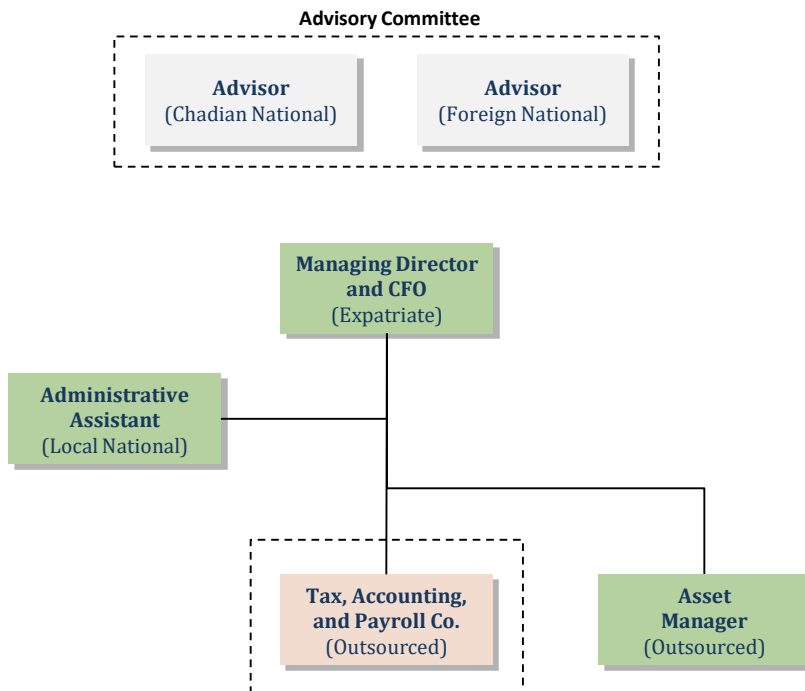


Figure 42: SPV Organisation During Operations



B. Smart Energies S.A.

1. Corporate Profile

Smart Energies S.A. is a privately held company with an 80-year track record in energy (oil and gas) and real estate investment. Smart Energies began investing in renewable energy and clean technologies in 2012. By December 2018 the company had more than 300 solar PV sites under construction or in operation for a total capacity of more than 100MW and an annual production of more than 130GWh.

Smart Energies is an independent power producer (IPP) with active involvement across the value chain, including early stage development, financial structuring, investment, construction management, asset management, and external growth through acquisition of brownfield assets.

The renewable energy portfolio is experiencing solid growth in Europe and abroad. After carrying out significant investments both in brownfield acquisitions and greenfield development from 2014 to 2017, Smart Energies plans to continue building on this strategy during the 2018-2020 period.

Solar PV and renewable energy solutions include not only photovoltaic plants but also batteries, hybrid technologies, pumped-storage, and other storage technologies. Clients range from commercial and industrial customers which cannot rely on a regular electricity supply from the grid, to national utility companies.

Union Générale de l'Énergie S.A. ("UGE") is the subsidiary holding all energy related assets. Smart Energies International ("SEI") is the subsidiary that develops and invests projects outside of France, with a particular focus on Africa, Eastern Europe, and the Middle-East. The shareholding structure for Djermaya CDEN Energy Sarl is provided in Annex A.

2. Renewable Energy Strategy

When Vianney de l'Estang was brought in as CEO in 2012 he changed the investment strategy from oil and gas to renewable energy, and Smart Energies rapidly became one of the leading players in decentralized renewable energy production and energy transition in Europe. For example, in France and in Italy the company now has more than 300 solar or hydroelectric sites in operation or under construction, of which solar PV accounts for a total capacity of more than 100 MWp.

In 2016 Smart Energies decided to expand internationally and acquired CDEN, a start-up renewable energy developer specialising in emerging markets. The company is putting more resources into developing projects in emerging market countries. For example, projects under development include:

- *South Africa* – A private off-take agreement with Power X (a broker holding the exclusive national license to trade renewable power in RSA) for 10 MW solar capacity. Together with a local partner, SEI is also developing a portfolio of <100 kWp projects for a total capacity of 10 to 30MW.
- *Chad* – The 60MW Djermaya Solar project near the capital N'Djamena.
- *Other sub-Saharan Africa* – In Ghana, Benin, DR Congo, Rwanda, Kenya, and Madagascar, SEI is co-developing a portfolio of decentralised plants for commercial and industrial clients as well as smaller scale hydro and hybrid mini-grid projects.

- *Tunisia* – In May 2018 an SEI partnership won a bid launched by the Ministry of Energy, Mines and Renewable Energy, and within this context SEI is currently developing a 10 MW solar PV plant in the Sidi Bouzid region. Construction is to start in Q1 2019 and expected commercial operations to occur in Q2 2019.
- *Eastern Europe* – SEI is expanding into Eastern Europe and aims to own a 100MW portfolio in the region by 2020. The company is currently finalizing the acquisition of a 50MW solar portfolio in Ukraine.

Figure 43: Selective List of Key Projects

Project Name	Location	MW	Technology	Project Phase	COD
Ever 4	Italy	0.6	Solar	Acquisition	Q1 2019
Port Elizabeth	South Africa	10.0	Solar	Development	March 2019
Vynohradove	Ukraine	5.0	Solar	Development / FC	April 2019
Bir el Haffay	Tunisia	10.0	Solar	Development / FC	Dec 2019
V.	Ukraine	32.0	Solar	Development / Acquisition	Q2 2019
Kherson, O.	Ukraine	5.0	Solar	Development / Acquisition	Q2 2019
SSU	France	2.0	Solar	Development	2020
Sub-total		63.6			
Smart solar	France	3.0		Construction	Q2 2019
Carmco	France	11.0	Solar	Construction	2019
Sub-total		14.0			
Evergrid	Italie	1.0	Solar	Operation	2011
Cestas	France	2 x 12.0	Solar	Operation	2015
SolarMed 1	France	1.3	Solar	Operation	2016
Yellow	France	3.5	Solar	Operation	2018
Sub-total		29.8			
Grand Total		107.4			

Note: Table sorted by project phase.

Key competitors in domestic markets (France and Italy) include Tenege, Cap Sud, Cap Vert, and EDF-Energies Nouvelles. Smart Energies competes effectively by focusing on portfolios of multiple small and mid-sized assets. Internationally, main competitors include companies such as Akuo, Neoen, Fotowatio, Volitalia, Veolia, Total-Eren, and Engie. Smart Energies aims to build on its successful business model and is able to mobilise investment in new markets quickly. The company is present throughout the value chain – from early-stage development through financing, investment, construction, and operations.

3. Financial Review

The accounts provided in Appendix B are for Union Générale de l'Energie S.A. ("UGE"), which is the Smart Energies subsidiary holding all energy related assets. Annex C includes a table of key ratios.

In 2017 UGE registered a turnover of €13.6 million, of which c. 20% was from asset management services billed to project companies, and c. 80% from revenues generated from project companies (i.e. electricity sales). Considering the EUR 4.4 million turnover in 2014, the cumulative annual growth rate (CAGR) has been 33%, although the growth from 2016 to 2017 was a reasonable 11%.

Nearly 90% of the operating costs consist of staff salaries³⁰. From 2014 to 2017, operating cost CAGR was 17%, which explains the improvement in EBITDA margin from 54% in 2015 to 67% in 2017. EBIT margin (21% in 2017) is lower due to high amortisation charges.

Since 2014 UGE has not only been involved in greenfield development, but has also been actively engaged in an external growth strategy. With reference to Annex D, it can be seen that over this period 28 projects have been acquired for a total amount of EUR 10.1 million: EUR 2.4 million (2014), EUR 1.2 million (2015); EUR 3.8 million (2016); EUR 2.7 million (2017).

The aforementioned acquisitions have been funded partly through internal resources and partly through external borrowing. In 2014 shareholders' equity was EUR 48.7 million, and this has declined to EUR 37.9 million in 2017 (a reduction of EUR 10.7 million).

With reference to the abbreviated statements in Figure 44 on the following page, it can be seen that net operating cash flow has strengthened since 2014, reaching EUR 5.6 million in 2017. This has been used both to fund acquisitions and also to pay shareholder dividends.

Whilst UGE has a significant amount of debt on the balance sheet (EUR 80.8 million), most of this results from the consolidation of long-term non-recourse debt within the capital structure of individual project companies. It includes to a lesser extent borrowings at the corporate level. Interest expense in 2017 was EUR 3.2 million, which indicates an average interest rate of 3.9% for the total amount of outstanding debt.

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³⁰ All staff are employed by Smart Energies. There is a services contract between UGE and Smart Energies, and this is the "external charges" line item in the Profit and Loss Account (€4.9 million in 2017).

Figure 44: UGE Abbreviated Financials

All values kEUR		2015	2016	2017
Abbreviated Profit and Loss Statement				
A	Turnover	7,228	12,283	13,650
B	EBITDA	4,059	8,268	9,183
C	EBIT	665	2,686	2,922
D	Net Interest	-1,038	-2,565	-2,955
	Net profit from consolidated companies	-2,843	-87	-198
	Share of companies using the equity method	-1,359	-1,910	-1,772
	Amortisation of goodwill	-1,146	-63	-123
E	Net Profit	-2,843	-2,060	-2,092
Abbreviated Cash Flow Statement				
	Gross Cash Flow	2,962	5,616	5,698
	Change in Working Capital	1	-1	-3
	Net (Operating) Cash Flow	2,963	5,615	5,695
	Cash Flow from Investing Activities ⁽¹⁾	1,204	3,836	2,666
	Cash Flow from Financing Activities	1,021	217	2,705
	Dividends Paid	-1,013	-1,292	-1,689
F	Other Sources of Financing ⁽²⁾	2,034	1,996	5,734
	Net Increase (Decrease) in Cash	2,780	1,996	5,734
	Cash Beginning of Year	4,429	7,209	9,205
	Cash End Year	7,209	9,205	14,939
Abbreviated Balance Sheet				
G	Current Assets	15,213	16,945	24,779
	of which Cash	7,209	9,205	14,939
	Fixed Assets	107,420	116,419	113,027
	Total Assets	122,633	133,364	137,806
H	Current Liabilities	20,490	20,442	19,041
I	Long-Term Debt	57,125	71,203	80,778
J	Shareholders' Equity	45,104	41,714	37,950
	Regulatory Capital	1,800	1,800	1,800
	Retained Earnings	16,322	12,932	9,168
	Revaluation Surplus	26,982	26,982	26,982
	Total Liabilities & Shareholders' Equity	122,633	133,364	137,806
K	Number of Employees ⁽³⁾	9	12	14

(1) It is assumed that total investments equal the sum of project acquisitions shown in the accounts.

(2) Cash flow statements are not available and the figures provided are calculated based on (i) net operating cash flow, (ii) cash flow from investing activities, and (iii) net increase in cash. There were no capital increases during the period 2014 – 2017, so it can only be assumed that the additional source of funding came from debt financing.

(3) Employees are registered under Smart Energies (the holding owning UGE).

Figure 45: UGE Key Ratios

		2015	2016	2017
A / J	Turnover per Employee	803	1,023	975
A	Growth in Turnover	64%	70%	11%
B / A	EBITA Margin	56%	67%	67%
C / A	EBIT Margin	9%	22%	21%
G / H	Current Ratio	74%	83%	130%
D / C	Interest Cover Ratio	1.6x	1.0x	1.0x
I / B	Leverage Ratio	14.1x	8.6x	8.8x
I / J	Gearing Ratio	1.3x	1.7x	2.1x

Figure 46: UGE Acquisitions (2013 – 2018)

Project	Acquisition Date	Price EUR k
Ecuries de Lansargues	21/08/2013	2
Loubieres	31/01/2014	300
Romo	31/01/2014	115
Capiscot	16/10/2014	105
Alpes Energie	23/12/2014	354
Agrisun	16/10/2014	8
CSC14	04/11/2014	752
CSC15	04/11/2014	752
La Baume	01/01/2015	217
Erable	01/01/2015	20
Peupliers	01/01/2015	20
Samfisol1	01/01/2015	70
Weymouth	01/01/2015	20
Sunmed2	30/03/2015	235
Sunmed3	30/03/2015	57
Ileo2	31/12/2015	256
Ileo6	31/12/2015	297
Eleo8	31/12/2015	12
Solarmed1	23/03/2016	2,479
Fauvettes	15/02/2016	259
Champgrand	15/02/2016	650
Samfisol2	18/02/2016	-
Ileoinvest13	13/07/2016	448
Soleil et rendement	21/06/2017	784
TSA09	12/01/2017	352
TSG09	12/01/2017	464
TSS09	12/01/2017	650
Strategeco	15/06/2017	386
Flexol Alpha	05/10/2017	30
Total Acquisitions		10,094

4. Governance

There are two types of limited liability joint stock company (*société anonyme*) in France if regulatory capital is less than EUR 150,000. Smart Energies has opted for the configuration that consists of an Advisory Board (*conseil de surveillance*), and an Executive Committee (*le directoire*).

The Supervisory Board is comprised of three members: Gregory Van der Not d'Assche (President), Frederick Dieterle (Vice-president), and Frédéric Fournier.

The Executive Committee has just one member: Vianney de l'Etang, President of the Committee and holding the title of Chief Executive Officer (*Directeur générale*). Vianney graduated from two top French engineering schools and has been working in renewable energy for more than 10 years. He managed renewable energy projects at Engie and Direct Energies before becoming Smart Energies CEO in 2012.

Mr. de l'Etang heads the Executive Management Team which meets every three months. During these meetings Smart Energies Europe and Smart Energies International report on progress made and present investment opportunities that have been identified in respective markets. All Investment decisions must be approved by the Smart Energies Board of Directors. Details on each of the members of the Executive Management Team are provided in the table on the following page.

Figure 47: Executive Management Committee

Name	Position	Biography
Hugues-Antoine Guisoiseau	Director of Emerging Markets	Hugues-Antoine is the founder of CDEN. He is graduated from HEC Paris and has developed and financed renewables projects in France, Africa, Middle East, and Eastern Europe over the past ten years.
Olivier Bzowski	Director of Development in Europe	After several experiences in private equity and investment banking, Olivier worked at EY for four years before joining Smart Energies in 2015.
Laurent Simon	Legal Director	Laurent developed the Business Law and Tax Department in two Parisian law firms over a 13 year period. He joined Smart Energies group in 2011 as General Counsel to handle the group's corporate activity and acquisitions.

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C. InfraCo Africa Ltd.

1. Corporate Profile

InfraCo Africa seeks to alleviate poverty by mobilizing investment into sub-Saharan infrastructure projects. It does this by funding teams of experienced developers or by investing directly into projects which need the financial commitment and leverage that InfraCo Africa can bring. Its involvement reduces the risks and costs associated with early stage project development and ensures projects are developed to the highest standards, from concept to financeable investment opportunity and beyond.

InfraCo Africa is a UK company and part of the Private Infrastructure Development Group ('PIDG'). Established in 2004, it is managed as a private company although funded by the governments of the UK (DFID), the Netherlands (DGIS) and Switzerland (SECO). InfraCo Africa's projects have mobilized over USD 2 billion of investment and provided new infrastructure for approximately 13 million people, improving living standards and powering economic growth in sub-Saharan Africa. InfraCo Africa currently has 21 employees.

Following a competitive tender in 2014, Aldwych was selected as a developer to act on InfraCo Africa's behalf by providing some of the resource and experience needed to originate and develop projects in sub-Saharan Africa. A Developers Service Agreement was executed between both parties in November 2014 which also provided for Aldwych Africa Development Limited (AADL) to be the nominated Aldwych subsidiary for the provision of these services. Under the Developers Service Agreement, AADL sources projects which meet established InfraCo Africa criteria at all times while working under the PIDG Operating Principles and Procedures (OPP). The Djermaya Project falls within this framework.

InfraCo Africa has recently secured a further tranche of funding and will be looking to double its existing pipeline of projects over the next five years.

2. Business Overview and Strategy

InfraCo Africa provides funding and expertise to infrastructure projects at their earliest stage, enabling them to grow from an initial concept to a bankable investment opportunity. It can work with projects directly where they already have an experienced lead developer, or it can provide on-the-ground project development expertise through one of the four developer teams that work on its behalf (as AADL does). InfraCo Africa can also fund the construction of pioneering projects by retaining or topping up its equity investment at financial close, or help finance innovative infrastructure businesses that need to scale-up and demonstrate commercial viability to attract further investment.

InfraCo Africa balances commercial viability and development impact, seeking to support those projects that the private sector considers to be too high risk but which have the potential to be bankable. It applies its own Operating Policies and Procedures as well as the IFC Performance Standards to ensure that projects are developed to international standards, have a positive developmental impact and become high quality, sustainable investment opportunities.

InfraCo Africa has a broad remit and can work across most infrastructure types with a few exceptions (listed in its OPPs), however its current focus is:

- *Energy:* InfraCo Africa can support power generation, transmission and distribution projects, including rural electrification. It is currently focused on developing of renewable power and storage solutions, as well as furthering the mini-grid / off-grid market in sub-Saharan Africa.
- *Transport:* InfraCo Africa can support projects that develop transport infrastructure including rail systems and services, ports and harbours, ferries, roads, bridges and tunnels. It prioritizes projects that create regional trade corridors and is currently focused on marine transport and logistics infrastructure
- *Water:* InfraCo Africa can support projects that provide water and water-related services to urban or rural populations. It is currently focused on the treatment and supply of clean drinking water and the development of irrigation systems for farms and market gardens.

InfraCo Africa can work across sub-Saharan Africa but prioritizes Least Developed Countries (LDCs), Other Low-Income Countries (OLICs) and fragile or conflict-affected states (as defined by the OECD). It is currently developing projects in: Chad, Ethiopia, Malawi, Mozambique, Nigeria, Guinea, Tanzania, Uganda and Zambia. It has also worked in Cape Verde, Ghana and Kenya.

3. Project Selection Process

InfraCo Africa has a very strict policy for selecting its projects, which the Djermaya project had and has to meet.

In order to apply for funding or additional expertise from InfraCo Africa, projects must meet the following criteria:

- be within sub-Saharan Africa;
- be within power, water or transport sectors;
- have the potential to be commercially viable (with InfraCo Africa's support);
- have a positive developmental impact on the communities they serve;
- need InfraCo Africa to bring something additional, something that no other party can bring to the project; and
- be developed in accordance with its Operating Policies and Procedures.

InfraCo Africa's mandate is to mobilise and not displace private sector investment. Therefore, it will not get involved in a project that could attract all of the financing it requires from other private or DFI investors.

If the project meets the criteria, there is a simple two-step process to go through:

Step One: Initial Approval

This part usually takes up to 3 months and it entails the following:

- getting a clear understanding of the project (technology, business model, financing and developmental impact);
- reviewing the market, business and regulatory environments;
- understanding the specifics of InfraCo Africa's involvement
- presenting the project to InfraCo Africa's Investment Committee; and
- if appropriate, signing a Memorandum of Understanding

Step Two: Due Diligence and Contracting

Once the Investment Committee has approved the project and allocated an initial budget the second part of process will begin. This typically takes 3 to 6 months during which InfraCo Africa and the organization applying for funding will work together to achieve the following outcome:

- confirm host government commitment and resourcing (if they are the customer / off taker);
- complete Know Your Customer (KYC) checks;
- complete technical, legal, financial, environmental and social due diligence;
- undertake market studies (if required);
- present the project to our Investment Committee; and
- if appropriate, sign a Joint Development Agreement or equivalent legally binding contract.

4. Project Development

Below are listed the main power projects to which InfraCo Africa is or has been contributing to.

Figure 48: Main Power Projects

Project Name	Location	(MW)	Technology	Project Phase
Djermaya Solar	Chad	60	Solar PV	Development
Khoumaguéli Solar	Guinea	40	Solar PV	Development
Abiba Solar	Nigeria	50	Solar PV	Development
Corbetti Geothermal	Ethiopia	70	Geothermal	Development
Virunga	Tanzania	2.4	Hydro	Development
Salima Solar	Malawi	20-40	Solar PV	Development
Pavua	Mozambique	160	Hydro	Development
Western Power	Zambia	180	Hydro	Development
Golomoti Solar	Malawi	26	Solar PV	Development
Elan Solar	Cameroon	50	Solar PV	Development
Total Development		~650		
Cenpower	Ghana	350	Combined Cycle gas turbine	Construction
Aba IPP	Nigeria	141	Gas-fired plant	Construction
Total Construction		~500		
Cabeolica	Cape Verde	25.5	Wind	Operational
Standard Microgrid	Zambia		Off-grid Solar	Operational
Kalangala Infrastructure Service	Uganda	1.6	Mixed Utility: Ferry, Water, Solar diesel	Operational
Redavia	Tanzania		Off-grid Solar	Operational
Total Operation		~30		
Total all Phases		<1000		

InfraCo Africa has six Solar PV projects currently under-development and three operating assets:

- *Djermaya Solar*: A 60MWp solar PV plant in Djermaya, 30km north of Chad's capital, N'Djamena. Development of Djermaya Solar will be phased to gradually integrate renewable power into Chad's national grid. The first 32MWp phase is intended to be fully financed in 2019 and operational in 2020. This will be followed by a second 28MWp phase. This project is being jointly developed by AADL (on behalf of InfraCo Africa) and Smart Energies.
- *Khoumaguéli Solar*: The Khoumaguéli plant will be the first grid-connected solar power plant in Guinea and will deliver 40MW of clean power to Guinea's national grid. Using existing grid infrastructure, Khoumaguéli will also be well-positioned to enable a planned West African Power Pool project linking Guinea with its neighbours. This project is being jointly developed by AADL (on behalf of InfraCo Africa) and Solvéo Energie.

- *Abiba Solar*: a 50MWp ground-mounted solar PV facility to integrate with existing grid infrastructure. The plant will be built on a 150ha site at Manchok in Nigeria's Kaduna State. Development of this project is being led by Access Power alongside InfraCo Africa.
- *Salima Solar*: An initial development of 20-40MWp of solar power for Malawi's national grid, situated 75km east of Malawi's capital, Lilongwe. This will be Malawi's first commercial Solar IPP. Development of this project is being led by JCM Capital and Matswani Capital PTY alongside InfraCo Africa. Development of this project is being led by JCM and Matswani, alongside InfraCo Africa.
- *Golomoti Solar*: a 26MWp solar photovoltaic project planned for development at Dedza, approximately 100km south east of Malawi's capital, Lilongwe. An 132kV transmission line will connect the plant to the nearby Golomoti substation, facilitating delivery of much-needed power to Malawi's national grid. Development of this project is being led by JCM alongside InfraCo Africa.
- *Standard Microgrid*: An innovative business providing distributed renewable energy services in Zambia: a 12kW micro power utility that integrates proprietary grid management tools with solar photovoltaic (PV) technology and batteries, to provide clean, reliable energy to approximately 150 customers (including households, small businesses, rural schools and clinics). InfraCo Africa is funding the scale-up of Standard Microgrid to deploy more than 150 x c10kWp micro solar PV power utilities across Zambia.
- *REDAVIA*: An innovative solar PV rental business: with the PV panels, fittings, electrical components and control systems required to provide up to 90KWp of installed capacity, all delivered in a standardized shipping container. These solar units can be supplied with batteries or can hybridise with diesel generators to deliver 24/7 power. InfraCo Africa is funding the scale-up of REDAVIA to provide containerised off-grid solar PV power to businesses and communities across Tanzania.
- *Kalangala Infrastructure Services*: A pioneering mixed utility on Bugala island which operates two roll-on roll-off ferries, has upgraded 66km of road, is distributing clean water and has developed a 1.6MW hybrid solar-diesel power plant.

5. Financial Overview

InfraCo is an investment entity and does not consolidate investment entities it controls. Instead, interests (equity and loans) in subsidiaries are classified as fair value through profit or loss, and measured at fair value.

InfraCo enters into joint development agreements in which the company takes on early stage development costs and risks. The company is compensated by a number of means. Development fees and reimbursement of costs incurred are receivable should funds be available from the disposal of such ventures to third parties, or by securing third party debt finance.

Revenue from the sale of projects is only recognised when the company has transferred the significant risks and rewards of ownership to the buyer and it is probable that the company will receive the agreed payments. The company also receives income in the form of grants to cover expenditures relating to its projects. Such grants are treated as deferred revenue and only credited to the statement of comprehensive income as the related expenditure is incurred. Grant receipts not yet utilised are included in creditors.

During the development phase of the project lifecycle, there is usually no identifiable market price for investments. The company's valuation policy is to write off the costs incurred prior to signing a joint development agreement (JDA). Projects prior to JDA are classified as business opportunities and fully expensed through profit and loss.

At the signing of a JDA and/or other key document (Framework Agreement, PPA) a value based on an appropriate valuation methodology will be attributed to the project. During post-development phase, a value based on an appropriate valuation methodology in accordance with Private Equity and Venture Capital (IPEV) valuation guidelines will be attributed to the project.

For 2017, most (85%) of income came from grants. With respect to operating expenses, two-thirds came from project development fees, one quarter from administrative expenses, and the remainder losses on foreign exchange.

Figure 49: InfraCo Consolidated Financial Statements

Abbreviated Income Statement				
<i>All values GBP millions</i>		2015	2016	2017
A	Turnover	506	1,275	1,330
B	EBITDA	(-14,533)	(-8,194)	(-10,727)
C	Net Interest Expense	225	432	1,043
D	EBIT	(-14,308)	(-7,762)	(-9,683)
	Net Profit	(-14,308)	(-7,762)	(-9,698)
Abbreviated Cash Flow Statement				
	Gross Cash Flow	(-14,308)	(-7,762)	(-9,698)
	Change in Working Capital	10,489	1,397	(-423)
	Net (Operating) Cash Flow	(-3,056)	(-11,020)	(-14,667)
	Cash Flow from Investing Activities	(-3)	(-11)	(-100)
	Cash Flow from Financing Activities	9,861	21,969	13,799
	Net Increase / Decrease in Cash	6,802	10,937	(-967)
	Cash Beginning of Year	13,278	20,079	31,017
	Cash End Year	20,079	31,017	30,048
Abbreviated Balance Sheet				
E	Current Assets	24,065	34,183	32,842
	<i>of which Cash</i>	20,079	31,017	30,048
	Fixed Assets	7,105	11,771	16,350
	Total Assets	31,170	45,954	49,192
F	Current Liabilities	3,918	4,495	3,116
G	<i>of which Current Portion of Long-Term Debt</i>	0	0	0
H	Long-Term Debt	0	0	0
I	Shareholders' Equity	27,252	41,459	46,075
J	Total Liabilities & Shareholders' Equity	31,170	45,954	49,192
K	Number of Employees (accounts)	10	12	16
Key Financial Ratios				
A / K	Turnover per Employee	50.6	106.3	83.1
B / A	EBITA Margin	(-28.7)	(-6.4)	(-8.1)
C / A	EBIT Margin	(-28.3)	(-6.1)	(-7.3)
E / F	Current Ratio	6.1	7.6	10.5
D / C	Interest Cover Ratio	(-63.6)	(-18.0)	(-9.3)
H / B	Leverage Ratio	-	-	-
(F + G) / J	Gearing Ratio	-	-	-

5. Governance

Figure 50: Management

Name	Position	Biography
Peter Hutchinson	CEO	Peter is Chief Executive Officer of InfraCo Africa and a member of the PIDG Ltd. Africa Investment, and HSES Committees. After graduating from Oxford University in Chemistry, he qualified as a Chartered Accountant with Deloitte. Post-qualification, he worked in a variety of engineering companies in the electrical sector, specialising in project and export finance in the developing world. In 1998, he joined Balfour Beatty as Finance Director of their railway engineering group in the UK, taking responsibility for a major acquisition programme developing a substantial presence in mainland Europe. He subsequently acted as Managing Director of both the UK and later, the German rail businesses. Most recently he was Executive Director of Green Africa Power, a PIDG facility providing intermediate capital to renewable energy generation projects in sub-Saharan Africa.
Claire Jarratt	CIO	Claire joined the InfraCo Africa's business development team in 2012. Before joining InfraCo Africa, Claire was an Assistant Director in the Corporate and Project Finance team at International Power PLC and worked at Price Waterhouse Cooper as an Assistant Manager where she gained her ACA from the Institute of Chartered Accountants in England and Wales.
Richard Chubb	CFO	Richard joined InfraCo Africa in 2018. Prior to this spent almost seven years as Regional CFO at International SOS, and seven years working in the power sector with RRPV and various Globeleq companies.
Elizabeth Hipwell	COO	Elizabeth joined InfraCo Africa in 2014 and is also responsible for setting the company's strategy, business plan and monitoring developmental impact. Prior to this, Elizabeth spent 8 years in Accenture's Strategy practice predominantly working within the Financial Services sector, designing HR operating models and implementing transformation programmes.
Alex Childs	General Counsel	Alex joined InfraCo Africa in 2013. He is a project finance and corporate/M&A lawyer, having previously worked at Shearman & Sterling (advising on energy projects across the MENA region) and Fulbright & Jaworski.

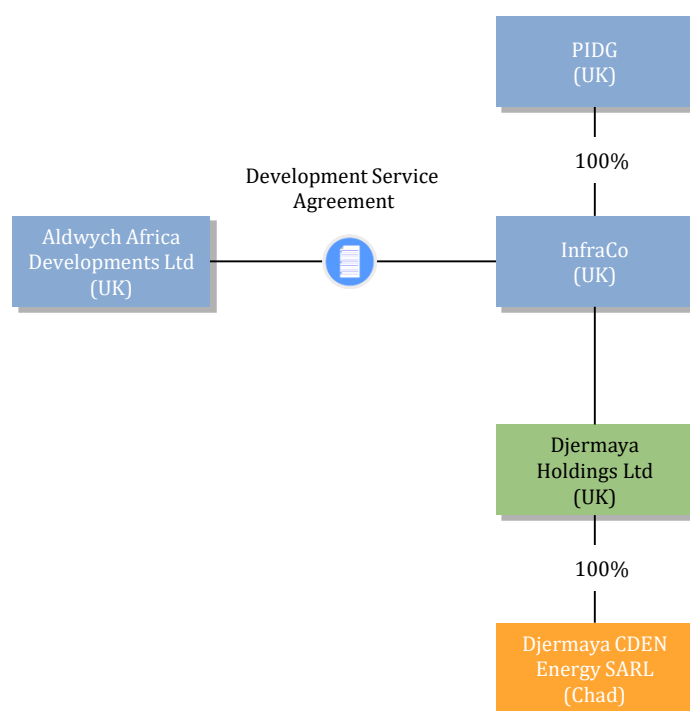
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Figure 51: Directors

Name	Position	Biography
Philippe Valahu	Chairman	Philippe became Chairman in March 2018. He is also CEO of the Private Infrastructure Development Group (PIDG) and has over 25 years of experience in infrastructure in developing economies, working with both the public and private sector.
Mike Chilton	Non-Executive Director	Mike joined InfraCo Africa's Board in March 2018. He is also CFO for the Private Infrastructure Development Group (PIDG). Mike joined PIDG from Cenkos Securities where he was finance director.
Jim Cohen	Non-Executive Director	Jim is a member of the Investment Committees of DIF II and III, infrastructure funds run from the Netherlands, which invest in both renewable power and government PFI/PPP programmes in Europe and Canada.
Godfrey Mwindaaire	Non-Executive Director	Before joining the InfraCo Africa Board, Godfrey was the West Africa Director of Acumen Fund Inc. for six years. Prior to this role, Godfrey spent seven years at the African Development Bank and left as its Chief Investment Officer.
Tania Songini	Non-Executive Director	Tania brings over 20 years of financial management experience. Most recently she served as Director of Finance and Operations for the International Rescue Committee UK before focusing on a portfolio career.

7. Djermaya Shareholding

Figure 52: InfraCo Shareholding in Djermaya



D. Aldwych Africa Developments Ltd.

1. Corporate Profile

Aldwych Africa Development Limited (AADL) is a wholly owned subsidiary of Aldwych International Limited. Aldwych's business is to develop, construct (or acquire), manage, own, and operate electricity generation assets across sub-Saharan Africa. All projects developed by AADL benefit from long-term off-take agreements and are financed by long-term non-recourse debt.

In sub-Saharan Africa AADL develops projects on behalf of InfraCo Africa Limited ("InfraCo"). InfraCo provides funding for early stage development and may exit at or before financial close, or alternatively may choose to take a longer-term strategic investment on a case by case basis.

Since 2017, Aldwych International has been absorbed into the Anergi Holding Company, a corporate structure resulting from the merger of Aldwych's historical shareholder (PAIDF)³¹, managed by Harith General Partners) and Africa Finance Corporation (AFC)³².

2. Developers service agreement

Following a competitive tender in 2014, Aldwych was selected as a developer to assist InfraCo in fulfilling its mandate of developing private power projects in its target investment region of sub-Saharan Africa. A Developers Service Agreement was executed between both parties in November.

Under this agreement, AADL sources projects which meet established InfraCo criteria while working under the PIDG Operating Principles and Procedures ("OPP"). InfraCo provides the necessary resources to develop the projects, in effect mitigating the early risks associated with such projects.

3. Aldwych Corporate History

Aldwych was established in 2004 for the purpose of developing, owning and operating power generation, and transmission and distribution projects in emerging economies, primarily in sub-Saharan Africa. Between them, the Aldwych founders and senior management (mostly former executives of the AES Corporation) have an impressive history of international power industry experience and have successfully financed or re-financed 30 power projects worldwide totalling some 16GW in aggregate capacity, in both the developed and the developing world. These AES executives implemented and operated a number of successful IPPs in Africa (Tanzania, Nigeria, Cameroon and South Africa), and also Hungary, England, Northern Ireland, Netherlands, Former Soviet States, Dominican Republic, and the USA.

With an early grant from the Shell Foundation and significant funding from management, Aldwych secured corporate financing in 2005 from FMO and a group of private equity investors. Further financing was secured in early 2009 involving ABSA Capital, the Emerging Africa Infrastructure Fund ("EAIF", managed by Frontier Market Fund Managers), the Pan African Infrastructure Development Fund ("PAIDF", managed by Harith Fund Managers), and again FMO.

³¹ Pan African Infrastructure Development Fund

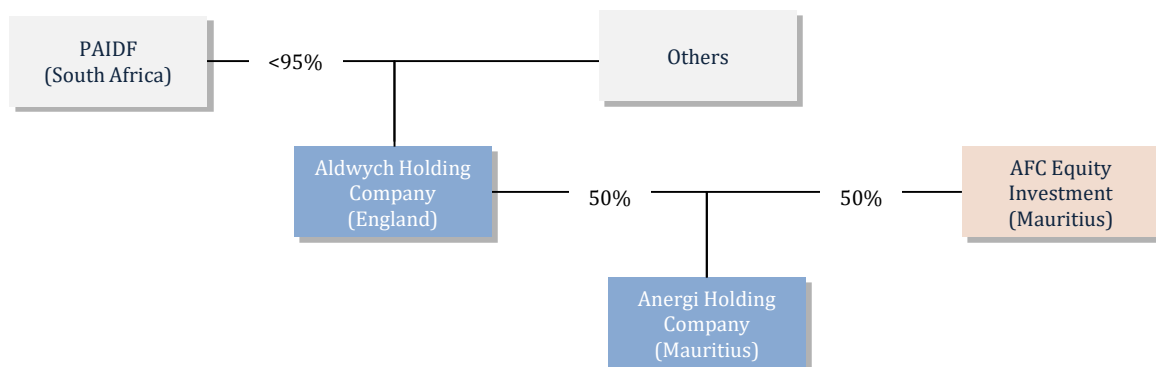
³² Africa Finance Corporation

4. Establishment of a pan-African Energy Company: Anergi

In 2017, Aldwych Companies were incorporated into the Anergī pan African power company (Aldwych International Limited (AIL) and AADL still keeping their existing corporate status within the Anergī group).

The Anergī Group is the result of a shared vision by Africa Finance Corporation (AFC) and Harith General Partners (“Harith”), which decided to establish a leading African power company to address the growing need for energy, and to accelerate the pace of energy delivery on a sustained basis. AFC and Harith with Aldwych Holding Limited joined forces in June 2016 to form the Anergī Group and, on 5 December 2017, Anergī was formally incorporated. Together, these two institutions ensure that the Anergī Group is perfectly positioned to be a major diversified energy player in the African energy sector. From a corporate perspective, Anergī is owned 50% by AFC and 50% by Aldwych Holding Limited which, in turn, is owned by Harith managed funds.

Figure 53: Anergī Corporate Structure



Aldwych International, and its employees, remains intact but is aligned with Anergī’s strategy, such as: developing new greenfield opportunities, supporting on the ground teams during construction and/or operations, and exploiting additional growth opportunities associated with the existing Anergī assets.

5. Business Strategy

Anergī is committed to supporting the economic and social development of the countries in which it invests by delivering the power they need in a sustainable and affordable manner, and in a spirit of partnership and cooperation.

Anergī maintains extensive relations with commercial, bilateral, and multilateral financing institutions and seeks to structure project financings in a robust and innovative manner. Experience of completing financings in developing markets is a cornerstone of the Anergī value proposition. Anergī considers its core business to include:

- Electricity generation projects based on traditional fuels or renewable technologies. They can be captive power plants for industrial customers or grid connected projects for wider distribution.
- Distribution and transmission projects which are benefit from pre-determined and contractually secured tariff adjustment mechanisms.
- Associated fuel related projects such as oil, gas or biomass production, transport and storage in conjunction with power generation projects.

Anergi's mid-term strategy is to become one of the main private investment energy actors in Africa by actively expanding its asset portfolio in size and in geographical scope, i.e. in countries where it sees strategic value. Anergi is committed to be a long-term investor and shareholder.

6. Joint Venture and Partnership credentials

Anergi holds a stake in the 600MW Kelvin coal-fired power plant in Johannesburg where it sits on the executive committee of the board of directors and is actively involved in providing commercial and technical advice and support to management. Through these interests, the Company is accustomed to working with co-shareholders in a spirit of constructive cooperation.

In partnership with BWSC of Denmark, Aldwych developed, constructed and Anergi is now operating, once again in partnership with BWSC, a 90MW diesel-engine power plant, burning heavy fuel oil, at Rabai near Mombasa in Kenya. Also in Kenya, Anergi has built the 310MW Lake Turkana wind farm together with a consortium of investors. This project was commissioned in Q4 of 2018.

Anergi is the main shareholder (<90%) in Cabeolica 25MW wind power plant, whose wind turbines are spread over 4 of Cape Verde's Islands.

In Nigeria, Anergi worked with a group of Nigerian financial investors to develop the 450MW Azura gas-fired power plant, in close cooperation with the relevant State government and the Federal Government. This project has reached COD in May 2018.

In Ghana, Anergi is similarly engaged with other partners and is building the 250MW Amandi gas-fired IPP CCGT project. Still in Ghana, thanks to the merger with AFC, Anergi now owns shares in the 350MW Cenpower CCGT plant. Both plants are expected to reach COD in 2019.

During construction, great emphasis is placed on training the workforce, both local and international, in best practice, especially with respect to high safety and environmental standards. Much time is also spent with the local community, to ensure that they feel part of the development and take ownership of it. In particular, the construction management team works with the local people to identify what areas of their lives the project could improve, as part of corporate social responsibility programmes.

On the operations side, Anergi is committed to skills transfer and to minimizing as far as possible the number of expatriate employees and the time they spend in the operating businesses. Expatriate employees are typically tasked with finding local replacements for themselves in short order.

At the Rabai plant in Kenya and at Kelvin for instance, Anergi has no expatriate employees and in Ghana there are currently only two foreseen. Anergi does, however, provide very hands-on support from head office both on a regular basis and if special needs arise, and continues to monitor safety and environmental performance very closely.

On the development side, Anergi is looking at late stage investments as pure equity player, brownfield projects in obtaining additional shares in the existing portfolio and also in greenfield projects in particular wind projects in Zambia and Tanzania. Finally, with InfraCo, AADL has now been involved for several years in supporting the development of the 40MW Koumaguéli Solar project in Guinea, alongside another sponsor whose expertise is in solar projects and technologies (Solveo).

7. Financial Overview

Aldwych International Ltd (AIL) financial statements are provided in the following table. Figures are only available up to 2016. AIL is a limited company, based in England which is fully owned by Anergi. Anergi being a new company has no financial statements available as yet.

Also, it should be noted that Anergi is currently in a debt raising process in view of financing the near-term project pipeline, and is therefore able to demonstrate a strong financial structure with regular cash flows coming from its current assets to meet Lenders requirements. Planned Investments in 2019 include additional acquisitions in existing operating assets, hence increasing cash flows at Anergi level as of 2019/2020.

Figure 54: Aldwych International Financial Statements

Abbreviated Income Statement

<i>All values Euro millions</i>		2014	2015	2016
A	Turnover	16,419	5,411	5,554
B	EBITDA	8,754	(1,920)	682
C	EBIT	8,754	(1,920)	682
D	Net Interest Expense	(1,448)	(1,387)	(1,854)
	Net Profit	10,658	(2,082)	(102)

Abbreviated Cash Flow Statement

	Gross Cash Flow	9,065	(1,863)	(766)
	Change in Working Capital	(11,194)	5,160	(1,353)
	Net (Operating) Cash Flow	(2,129)	3,297	(2,119)
	Cash Flow from Investing Activities	4,186	(143)	793
	Cash Flow from Financing Activities	(1,852)	(2,412)	1,385
	Net Increase / Decrease in Cash	205	742	59
	Cash Beginning of Year	1,655	1,789	2,571
	Cash End Year	1,789	2,571	2,441

Abbreviated Balance Sheet

E	Current Assets	19,197	13,492	16,209
	<i>of which Cash</i>	2,694	2,816	2,686
	Fixed Assets	1,448	2,744	3,976
	Total Assets	20,645	16,236	20,185
F	Current Liabilities	6,763	12,292	22,916
G	<i>of which Current Portion of Long-Term Debt</i>	2,579	9,300	19,038
H	Long-Term Debt	15,086	7,205	623
I	Shareholders' Equity	(1,204)	(3,261)	(3,354)
J	Total Liabilities & Shareholders' Equity	20,645	16,236	20,185
K	Number of Employees	19	20	19

Figure 55: Key Financial Ratios

A / K	Turnover per Employee	864	270	292
B / A	EBITA Margin	53%	(35%)	12%
C / A	EBIT Margin	53%	(35%)	12%
E / F	Current Ratio	2.8	1.1	0.7
D / C	Interest Cover Ratio	6.0	1.4	(0.4)
H / B	Leverage Ratio	2.0	(8.6)	29
(F + G) / J	Gearing Ratio	(14.7)	(5.1)	(5.9)

8. Project Development

Anergi owns 7 plants across Africa which are in operations or close to reaching COD. Anergi is also developing several projects. The list and details can be found in the table below:

Figure 56: Power Portfolio

Name of Project	Project Location	MWs	Technology Type	Phase	Commission Date
Lake Turkana	Kenya	310	Wind	Commissioning/ Operations	12/2018
Cenpower	Ghana	350	CCGT	Construction	03/2019
Amandi	Ghana	200	CCGT	Construction	08/2019
Total Construction		860			
Azura	Nigeria	461	OCGT	Operation	05/2018
Kelvin	SA	360	Coal	Operation	2001
Rabai	Nigeria	90	HFO	Operation	2009
Cabeolica	Cape Verde	26	Wind	Operation	2012
Total Operation		937			
Kelvin CCGT South Africa	South Africa	(450)	Gas	Development	Q2 2023
AADL Khoumagueli solar power	Guinea	30	Solar	Development	Q2 2021
AIL Fula Rapids hydro power	South Sudan	42	Hydro	Development	Q2 2024
Total Financial Close Equity		145			
AADL Djermaya solar power	Chad	32	Solar	Late Stage	Q3 2020
Total Late Stage Business Development		32			

9. Governance

The following persons are directors of both AADL and Aldwych and overseeing the Djermaya project.

Figure 57: Directors

Name	Position	Biography
Helen Tarnoy	Designated Person	Helen is one of the two co-founders of Aldwych and led its original financing. She has been directly or indirectly responsible for Aldwych's investments in CEC, the private Zambian transmission business, the 600MW Kelvin power plant in Johannesburg, the 90MW Rabai IPP in Kenya, the 310MW Lake Turkana wind project, also in Kenya, the 450MW Azura-Edo gas-fired power plant in Nigeria, the 192MW Amandi IPP in Ghana. Helen started with Aldwych as Commercial Director and became Managing Director in 2010. She was responsible for Aldwych's membership of the Power Africa Initiative, within which Aldwych has committed to develop almost 1200MW of new generation capacity in sub-Saharan Africa.
Neil Hopkins	CFO	Neil joined Aldwych in 2009 as Finance Director and is responsible for corporate financing, as well as financial reporting for the various Joint Ventures that Aldwych participates in. Neil is the Company Secretary for all of Aldwych's UK registered companies and is also the Compliance Officer, reporting in that role to the Audit & Risk Committee.
Bob Chestnutt	Head of BD	Bob joined Aldwych in May 2006 and successfully led the Rabai Project in Kenya to financial close in October 2008. Having been the Aldwych Project Director for the 310MW Lake Turkana Wind project located in Northern Kenya during its development phase Bob is now a Director of the project company, Lake Turkana Wind Power. This landmark project has completed its construction phase and is now in operations. Through the course of his career Bob has successfully participated in most aspects of development and financial structuring of complex projects in developing markets.

During the development period, Anergi (and more specifically people dedicated to the project within the AADL structure) work with InfraCo towards the successful completion of the Djermaya project. At Financial Close both InfraCo and Anergi have investment rights in the project. By this time, Anergi will have to make a recommendation to its board (members from Harith, AFC and 2 Mauritian Directors to fall within Mauritius law) for the long-term envisioned strategy, especially in regards to InfraCo's shares for which Anergi has the right to equal any offers made to InfraCo for buying their shares (most likely at Financial Close or at COD).

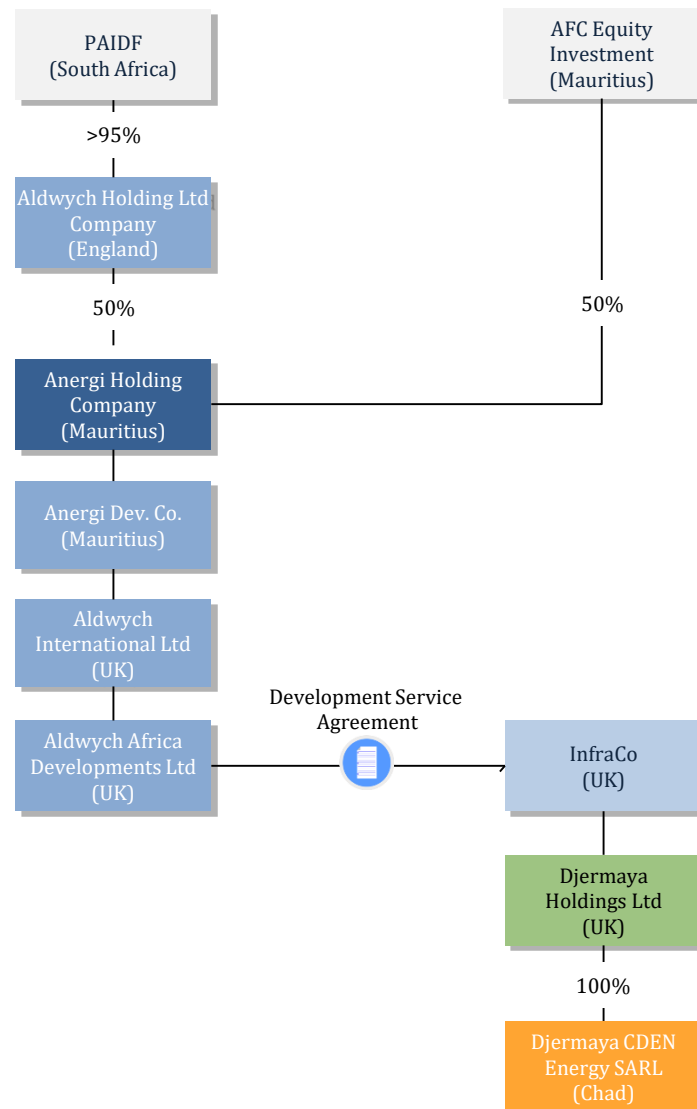
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10. Djermaya Shareholding

Anergi is actively participating in the development costs of the Project and is entitled to investment rights at financial close equal to 8% of the equity.

The current relationship of Aldwych / Anergi to the Djermaya project is outlined in the following shareholding diagram. This may change at Financial Close if Anergi decides to exercise its right to become a shareholder in the Project. Annex IV includes a complete diagram of the possible shareholding at financial close.

Figure 58: Aldwych Shareholding in Djermaya



XII. Project Bankability

The Partial Risk Guarantees (PRG) and the Put-Call Option Agreement (PCOA) or equivalent contractual mechanisms are instruments that are often used in power project financing in frontier markets, in particular where off-takers are structurally insolvent and host country governments are not considered to have the necessary credentials to back off-taker obligations. In this section both of these key credit enhancing instruments are examined in detail so that Lenders and other financing partners can better appreciate their role in supporting an overall bankable contractual structure for the Djermaya project.

It is important to highlight right from the start that instruments such as PRG and PCOA will only function within the context of a supportive Government. In the case of PRG, this means that the multilateral providing the risk guarantee will have to enter into a counter-indemnity agreement with the host country government. In the case of PCOA, the host country government will be a signatory of the agreement entered into directly with the project company.

Whilst these instruments do not have a direct impact on public finances in the immediate term, they can give rise to off-balance sheet obligations in the mid to long-term and which may need to be taken into consideration by government authorities. When countries such as Chad are under IMF supervision, implementation of such instruments must be fully disclosed to the IMF. Section II of this Memorandum discussed the role of the IMF and the efforts expended by Sponsors to obtain a “non-objection” with respect to PRG and PCOA in the context of the Djermaya project.

A. Payment Guarantee and PRG

1. Basic Mechanics of PRG

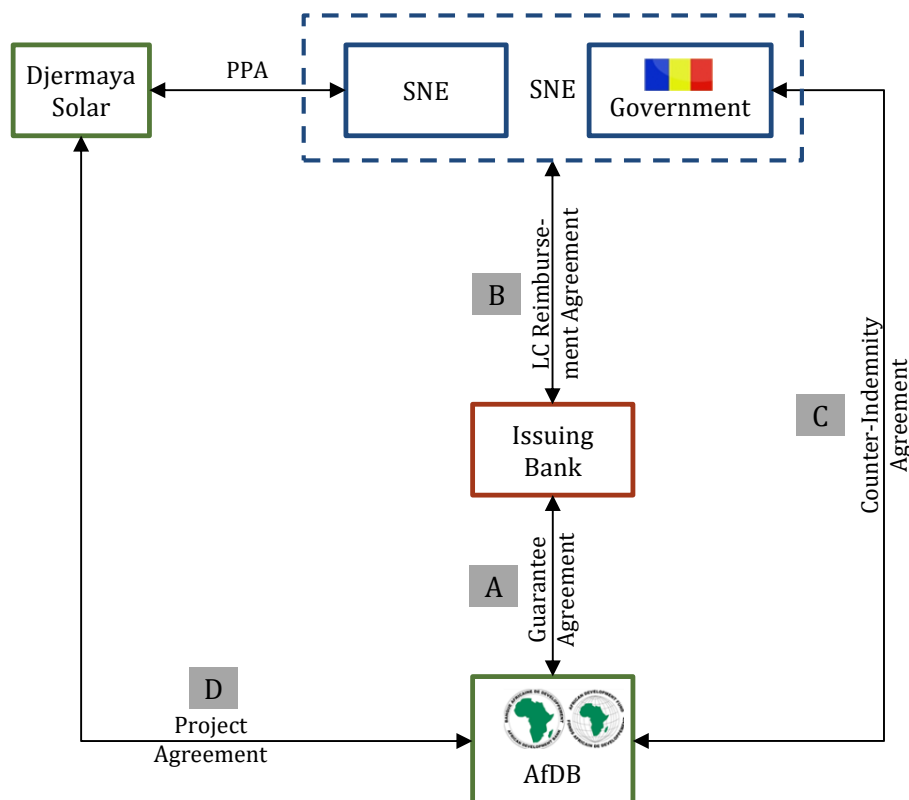
Whilst off-taker payment risk can be partially mitigated through project structuring (e.g. funded debt service reserve accounts), it must often be supplemented by an on-demand liquidity backup facility implemented by a commercial bank backed by guarantees from multilateral or development finance institutions.

The Partial Risk Guarantee (PRG) is such an instrument. It facilitates the implementation of an off-taker payment guarantee by transferring issuing bank counterparty risk to a solid ‘AAA’ entity. The basic mechanics of PRG are relatively simple, and with reference to Figure 59 on the following page functions as follows:

- The off-taker sets up a Letter of Credit (LC) with a commercial bank which is issued in favour of the project company [A].
- If the off-taker fails to make on-time payments for electricity purchases as per its contractual obligations under the PPA [B], the project company can call on the LC [C].
- If the LC is drawn, the commercial bank will request the off-taker to reimburse the amounts under the facility by replenishing the LC [D].
- If the off-taker fails to replenish, the commercial bank may after a certain elapsed time request the Government of Chad to reimburse the amounts drawn by the project company by replenishing the LC [E].
- If the Government fails to honour its obligations, the commercial bank can request the African Development Bank (AfDB) to reimburse the amounts drawn as per the PRG Indemnity Agreement by replenishing the LC [F].

- When the AfDB replenishes the commercial bank LC, the amounts drawn will be registered as a Government debt obligation under the African Development Fund (ADF) [G]. PRG Drawn amounts will decrease the remaining LC coverage.
- The Government of Chad will reimburse the ADF immediately or as may be indicated by the ADF any amounts that the ADF would have been required to pay under the PRG, including all interest thereon.

Figure 59: Partial Risk Guarantee



2. PPA Stipulations

Off-take risk (i.e. the risk of non-payment by SNE) is a key consideration for Investors and Lenders, which must have the assurance that Djermaya Solar will be receiving regular monthly payment thus enabling the Company to service its debt obligations over the 18-year senior loan amortisation period. With this in mind, the following structure has been contractually agreed and set out in the PPA signed on 9th August 2018.

- Prior to Financial Close, SNE and/or the Government of Chad must obtain from a commercial bank an on demand guarantee in the form of a non-revocable revolving letter of credit (LC) for an amount of EUR 7,931,000³³ issued in favour of Djermaya Solar.

³³ This amount represents an estimated 12-months of electricity invoicing by Djermaya Solar and may increase over the lifetime of the project up to EUR 9,700,000. However, there will never be an equivalent of 12 months of revenues drawn down under the LC. This is because the authorised amount must also include the interest which is due on these amounts (which is a floating rate of Euribor + margin as stipulated in the Credit Reimbursement Agreement). This means that the LC bank would probably run out of headroom around month 10 or 11.

- The LC can be called by Djermaya Solar within 5 days after a formal non-payment notification has been sent to SNE and when no further action by SNE has taken place. The Government cannot interfere and/or otherwise block the payment made by the issuing bank to Djermaya Solar under the LC agreement.
- SNE and/or the Government are contractually bound by the PPA to replenish the LC within 10 days of it being called by Djermaya Solar. The LC must be renewed throughout the duration of the project and remain in force for the entire 18-year period during which the senior project finance loans are amortised. The LC must be issued by a bank having a minimum international credit rating of BBB+.

3. PRG Contracts

The fundamental purpose of the PRG is to substitute SNE/Government risk for AfDB risk thereby lowering the cost of LC issuance and the cost impact on the electricity tariff. To be clear, within the context of a PRG, the ultimate counterparty for the LC issuing bank is not SNE/Government, but instead a 'AAA' rated multilateral organisation which guarantees to pay all outstanding amounts drawn under the LC – including accrued interest – at the end of the 12-month period up to an amount of EUR 10 million.

There are four key contracts that underpin the PRG, which are listed in the Figure 60.

Figure 60: PRG Contracts

	Contract Name	Contractual Party	Contractual Party
A	Guarantee Agreement	Issuing Bank	AfDB
B	Reimbursement and Credit Agreement	Issuing Bank	SNE and/or Government
C	Counter-Indemnity Agreement	Government	AfDB
D	Project Agreement	Djermaya Solar	AfDB

The *Guarantee Agreement* signed between the LC issuing bank and the AfDB sets out the terms by which the AfDB will reimburse amounts (and accrued interest) which might be drawn by Djermaya Solar up to an amount of EUR 10 million. The *Reimbursement and Credit Agreement* contractualises the SNE/Government obligation to replenish the LC within 12 months when amounts are drawn. The *Counter-Indemnity Agreement* provides the AfDB full recourse to the Government should obligations under the Reimbursement and Credit Agreement not be complied with. Lastly, the *Project Agreement* establishes the relationship between Djermaya Solar and the AfDB taking into consideration the entire project contractual structure.

There will be no cross default clauses between PRG and the underlying project documents. PRG will not be a party to the Inter-creditor Agreement.

PRG is not designed to be a revolving instrument and that once sums are drawn they will never be replenished. PRG is designed to be a very strong means of dissuasion and will most likely never be drawn. For example, if PRG is drawn and should the AfDB not decide to implement some form of restructuring of the repayment terms of the debt obligation, this could result in a cross-default situation with respect to all other international donors. As can be seen, the consequences of activating PRG are of such magnitude that it is extremely unlikely that the project will ever reach this point.

When reviewing PRG world-wide track record, history shows that it has only been drawn once, and that was in Argentina (World Bank). As for the AfDB, PRG has never been drawn. If it ever appears that a project is not functioning correctly, there will always be a high-level delegation sent to speak with the host country government long before PRG is ever called. In the case of Djermaya, the AfDB will use its strong influence to make sure the host country government complies with its obligations.

4. Liquidity Mechanism in Practice

Should SNE fail to make payment to the SPV within timeframes provided for under the PPA, the project is entitled to draw on the LC as per PPA Clause 8.4. Ten days following the draw of the LC, SNE and the Government have an obligation to replenish the LC.

Pursuant to clauses 10.2 (c) and 10.3 (b) of the PPA, a SNE or Government Event of Default respectively will materialise 6 months after the first draw on the LC should the LC not be replenished. From that time onwards and so long as the default continues, the project company has the option of serving a Notice of Termination. Such termination would become effective 30 days following notice.

If Sponsors choose to terminate the PPA, the senior loans would be accelerated and the option under the PCOA allowing the SPV to “put” the plant to the Government would be exercised, opening up a 150-day window by the end of which the Government will have the obligation to pay the purchase price. As a result, and pursuant to the PPA and PCOA, the path leads to the purchase price becoming due 360 days from the initial payment failure.

5. Pricing and Other Issuing Bank Matters

In its offer, the issuing bank will be expected to quote an initial LC set-up fee as well as an annual LC reissuance fee. The issuing bank will also need to provide the interest rate that will apply to drawn amounts under the Reimbursement and Credit Agreement. For example, assuming a bank refinancing rate of 12-month EURIBOR plus a margin reflecting the ‘AAA’ counterparty credit rating, the commercial terms of the offer might be summarised as follows:

- Initial set-up commission..... []%
- Annual re-issuance commission..... []%
- Interest on drawn amounts..... [12] month Euribor + []bps

Sponsors and the AfDB are aware that no commercial bank operating in Chad will meet a required minimum ‘BBB+’ credit rating. In addition, it is probably unlikely that any financial institutions in Chad will have a sufficiently strong balance sheet to issue an LC for the required EUR amount.

With this in mind, the issuing bank could be: (i) an international issuing bank meeting the required rating – Scenario 1; (ii) a local issuing bank which is a subsidiary of a large international banking group meeting the rating requirement – Scenario 2; or (iii) a large international banking group not meeting the minimum rating but able to work with a confirming bank having the required rating – Scenario 3.

With respect to scenario 2 above, the parent company would issue the LC and use the subsidiary for carrying out the XAF payment transactions.

With respect to Scenario 3, the parent company would issue the LC, which would be confirmed by an international correspondent bank meeting the minimum rating requirement, and the subsidiary in Chad would carry out the XAF payment transactions.

Banks working within the framework of PRG must be aware that incidents of corruption, fraud, or other reprehensible act will invalidate the agreements. With this in mind, it is very unlikely that a confirmation bank will assume operational risk of the issuing counter-party that is outside its control. Scenario 3 is therefore considered to be highly improbable.

B. Put-Call Option Agreement

1. PCOA Overview

Put-Call Option Agreement (PCOA) ensures equitable treatment for Lenders and Investors upon a Termination Event. The PCOA establishes a direct contractual obligation between the host country government and the project company/shareholders whereby the government must purchase the Facility upon Events of Default. The Purchase price is then used to indemnify Lenders and potentially investors.

The agreement provides a Put option for project shareholders allowing the sale of the assets to the government, and it provides a Call option to the government allowing a purchase of the assets of the project company. The PCOA defines the conditions under which these options can be exercised and the formulas for determining the Purchase Price (also known as the Termination Payment).

In the case of Buyer default, the Purchase Price will include outstanding project debt and also some element of expected return for shareholders. In the case of Seller default, the purchase price will be limited to just the outstanding project debt. In the case of termination for Force Majeure, the Purchase Price falls somewhere between.

2. Specific PCOA Terms

Note: Some of the information presented here is currently being negotiated and therefore subject to change.

The Put-Call Option Agreement (*Promesses de vente et d'achat* – PCOA) will be signed by Djermaya CDEN Energy S.a.r.l., the Ministry of Finance and Budget, and the Ministry of Petroleum and Energy (representing the “State”).

In the event of PPA termination, the project company can transfer and the State must buy the plant (Put Option), or the State can buy and the project company must transfer the plant (Call Option). Each of these rights can be exercised unilaterally and each party commits irrevocably and unconditionally to act on those rights.

In view of ensuring an efficient transfer of the asset, the project company will use best efforts to include in all contracts with third parties related to the construction and operation of the plant, to include step-in rights for the State and the obligation of the contracting party to continue executing their obligations for 12 months following the Termination Event.

Within one week of receipt of a Termination Notice, the parties agree to meet and to decide the conditions under which the construction or operation of the plant will continue being carried out until the Date of Transfer. Within 15 days of receipt of the Termination Notice, the parties will take all necessary measures to ensure an efficient transfer of responsibilities related to the construction and/or operation of the plant.

With respect to the price paid for the plant, this will depend on the reasons for the Termination as set out in Figure 61 below.

Figure 61: Price to be Paid at Termination

Termination Cause	Right to Put	Right to Call	Applicable Price
Default due to SNE (Buyer)	Yes	No	Price 3
Default due to the State	Yes	No	Price 3
Default due to the Project Company (Buyer)	Yes	Yes	Price 1
Delay in COD (due to Seller)	Yes	Yes	Price 1
Delay in COD (due to Buyer)	Yes	Yes	Price 3
Force Majeure (Natural)	Yes	Yes	Price 2
Force Majeure (Political)	Yes	No	Price 3
Change of Law	Yes	Yes	Price 3

Price 1 = D + IF + IA

Price 2 = D + FP + IF + CR – IA

Price 3 = D + FP + IF + CR + IC – IA

- D** Total amount of outstanding debt at the PPA Termination Date, including fees, interest, and penalties related to early redemption.
- FP** The total amount of shareholder capital committed to build the plant.
- IF** An amount to cover the tax impact on the project company, Lenders, and shareholders as well as any negative financial impact on the project company due to legal changes post termination.
- CR** An amount required to cover all taxes payable by the project company as a result of the termination and transfer and all sums due (including those relating to penalties) by the project company to its suppliers.
- IA** All sums received by the project company or the Lenders under insurance policies net of tax that have been used or committed to finance upgrade works on the plant.
- IC** Indemnity paid to shareholders equal to 5 years of return on equity at a rate of 18.5% calculated according to the following formula:

$$IC = FP \times 5 \text{ years} \times 18.5\%$$

The price paid by the State will be paid in whole and without netting off at the latest the End Date, which is defined as 150 days from receipt of the Notification, unless a technical audit reveals the existence of conditions that will result in a decrease in plant output due to a fault of the project company and not related to or caused by the State, SNE, or an event of Political Force Majeure. In such a case, the cost of repairing or restoring the plant will be deducted from the price, noting however that the amount can never be less than the amount calculated under “Price 1”.

Any questions related to the technical audit and price impact can be referred to an Independent Expert (the “Expert”). This Expert can be nominated from a list of 3 candidates that are registered with the International Chamber of Commerce and the State will choose one of the three candidates. Within 15 days of appointment, the parties will submit their questions, and the Expert will revert with the decision within 45 days from being appointed. The fees of the Expert will be paid by the project company.

The PCOA is subject to Chadian law. The State renounces sovereign immunity. Disputes will be referred to the International Chamber of Commerce in Paris for arbitration and all procedures will be undertaken in French.

C. Political Risk Insurance

Project Sponsors are seeking coverage to backstop political risks associated with the Project through Political Risk Insurance (PRI). In 2015, GOC entered into all required host government agreements enabling the Multilateral Investment Guarantee Agency (MIGA) to provide PRI to investors looking to invest in Chad. Project Sponsors have registered the project with MIGA and are seeking to secure the standard four point coverage:

- Expropriation
- Currency transfer restriction and inconvertibility
- War and civil disturbance (excluding short-term business interruption)
- Breach of contract

Sponsors received a non-binding indication of pricing on 24th January 2019 which is valid for 90 days. Percentage of coverage for equity is 90%, and that for such cover MIGA requires at the time of a claim payment either assignment of the equity shares rights free and clear of any encumbrances or limitations, or execution of the Claims Cooperation Agreement (CCA) with the project’s Lenders to which shares have been assigned.

Cost of PRI is included in the financing plan of the project.

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XIII. Financial Model

A. Financing Plan

Djermaya Solar will be financed following the principles of non-recourse project financing. The project is intended to be funded via a combination of grants, equity and debt as described below:

- *Senior Debt* – 69.9% of total project cost is expected to be provided by Senior Lenders.
- *Equity* – 18.4% of total project cost to be secured by the project sponsors.
- *Grants* – 11.6% of total project cost, consisting of a EUR 6.35 million repayable capital provided by the European Africa Infra-structure Trust Fund (“EU-AITF”) from their Sustainable Energy for All (“SE4All”) envelope.

The currency of the debt and equity is in Euros. To mitigate exposure to interest rate risks, a significant portion of the debt will be swapped to secure fixed interest rates.

With reference to Figure 62 below, total investment cost for the 32 MWp project and connecting infrastructure is EUR 52.9 million.

Figure 62: Sources and Uses

Sources of Funds			EUR m		
Senior Debt - AfDB	50.0%	18.5			
Senior Debt - Proparco	50.0%	18.5			
Senior Debt		37.0		69.9%	
Equity - Third Party	46.0%	4.4			
Equity - Anergi	8.0%	0.8			
Equity - InfraCo Africa	25.0%	2.4			
Equity - Smart Energies	21.0%	2.0			
Equity		9.5		18.4%	
Refundable Grant		6.4		11.6%	
Total Sources			52.9	100.0%	

Uses of Funds			EUR m		
EPC Costs		27.1		51.3%	
Interconnection Costs		3.6		6.8%	
Battery System		2.5		4.6%	
Contingency		1.7		3.1%	
Construction Management		1.0		1.9%	
Construction Insurance		0.7		1.3%	
WHT / Reg. / Import Duties		2.5		4.8%	
Construction Costs		39.0		73.8%	
Development Costs		7.5		14.1%	
Development Fee		1.3		2.4%	
Development Costs		8.7		16.5%	
Closing Costs		0.3		0.5%	
Pre-funded DSRA		2.1		4.0%	
Interest During Construction		1.7		3.2%	
Pre-funded Working Capital		0.4		0.8%	
Other Financing Costs		0.6		1.2%	
Closing and Financing Costs		5.1		9.7%	
Total Uses			52.9	100.0%	

Source: Financial model January 2019

Referring again to Figure 62 on the previous page, a description of the uses of funds is provided as follows:

- EPC costs represent the bulk of the construction cost and include all materials and equipment (panels, inverters, trackers, etc.) as well as civil works.
- Interconnection costs include all materials and civil works related to the transmission line and transformers and bays at the Lamadji substation.
- Battery system includes all costs related to BESS including batteries and also the necessary automation and controls an interfacing systems.
- A contingency provision is assumed to be used by the project in anticipating of any unbudgeted/unexpected capital expenditures and/or financing costs during construction. It is set at 5% of total EPC costs (3.1% of total project cost).
- Construction management costs cover fees paid to the company overseeing the construction on behalf of the Project Sponsors.
- Development costs includes all costs not covered by grants that were incurred over the 5-year development cycle, including (i) corporate set-up, land securing, technical studies, legal advisors, permits, etc. and (ii) internal developer costs.
- Development fee is the amount to be paid to sponsors in view of remunerating the at-risk capital committed to project development.
- Interest during construction includes interest accrued and paid during construction.
- Pre-funded Debt Service Reserve Account (DSRA) will be drawn from the proceeds of financing and represents an amount equivalent to 6 months of senior debt service.
- Pre-funded working capital will be drawn from the proceeds of financing at the end of the construction phase to cover on-going project company financial obligations.
- Other financing costs include Lenders' upfront and commitment fees.

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B. Financial Model Inputs

Figure 63: Key Model Inputs

Timeline	Input	Units	Comment
Financial Close	30/06/2019		
Start Construction	01/07/2019		
Commercial Operation Date (COD)	01/07/2020		12 month construction period
PPA End Date	30/06/2045		25-year PPA
Tariff			
Initial Tariff (local currency)	79.00	XAF	As per PPA
Initial Tariff (hard currency)	12.04	EUR Cents	Based on XAF/EUR = 655.96
Tariff Indexation	1.75	%	
Tariff Adjustment (1 st)	78	XAF	5 years after COD
Tariff Adjustment (2 nd)	76	XAF	10 years after COD
Tariff Adjustment (3 rd)	70	XAF	15 years after COD
Tariff Adjustment (4 th)	62	XAF	20 years after COD
Capacity / Irradiation / Output			
Peak Capacity (DC)	32.0	MW	
Peak Capacity (AC)	27.5	MW	
Irradiation (P50)	2,623	kWh/m2	
Performance Ratio	80.05	%	
Specific Yield	2,100	kWh/kWp/year	
Annual Production - Gross (P50)	67.2	MWh	
Annual Production – Net (P50)	66.1	MWh	After maintenance and grid unavailability
Module Degradation	0.45	%/year	More conservative compared to industry average
Financing Costs			
Upfront Fee	1.70	%	
Commitment Fee	0.80	%	
Monitoring Fee	20,000	EUR/year	
Agent Bank Fee	5,000	EUR/year	
Senior Debt			
Margin	550	Basis Points	
Door to Door	17	Years	
Grace Period	12	Months	
Amortisation Profile	Sculpted		
Sculpting DSCR (Min – P90)	1.30x		
Equity			
Dividend Distribution	Semi-Annual		
Legal Reserve	20	%	
Shareholder Loans	99.99	%	
Equity	0.01	%	

C. Financial Model Outputs

1. General

Financial statements (FS) comprising the construction and first 18 years are provided in Figures 66 and 67. Those are based on the latest model assumptions and are therefore subject to change. The Model will continue to be updated and adjusted as the transaction progresses, namely negotiation of the project documents and later on the finance documents. The Model and its assumptions are currently consistent with the current status of the Project.

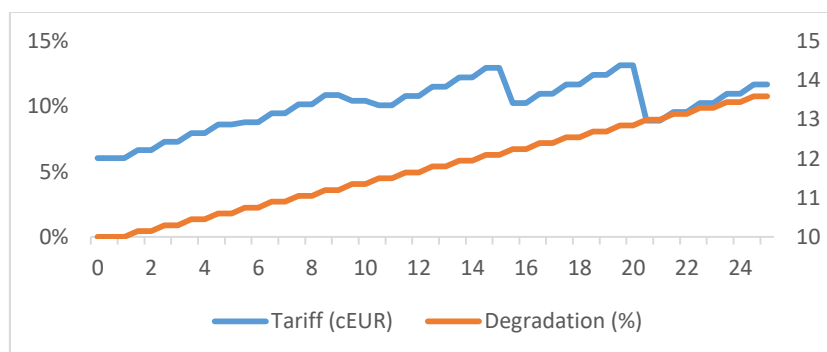
2. Timing

The annual FS are calculated on the basis of a P50 scenario and figures are EUR denominated. As explained above, while the PPA specifies that SNE's invoices are paid in local currency (XAF), those will be changed to EUR as soon as possible. Construction will last 12 months, this constituting the 1st year of the FS. From year 2, no CAPEX expenditures are assumed and the plant will be fully operational. Whilst 25 years of operation are modelled, only 18 are shown which corresponds to the senior debt amortisation period.

3. Revenues and technical assumptions

Overall, revenues remain constant throughout the PPA. This can be explained by both the tariff schedule and structure implemented in the PPA, and module performance degradation. While degradation increases over time (0.45% less of production year on year), tariff compensates for the technical losses. Figure 64 depicts the four distinct tariff periods with the reference price (*Prix de référence*) stepping down at set intervals.

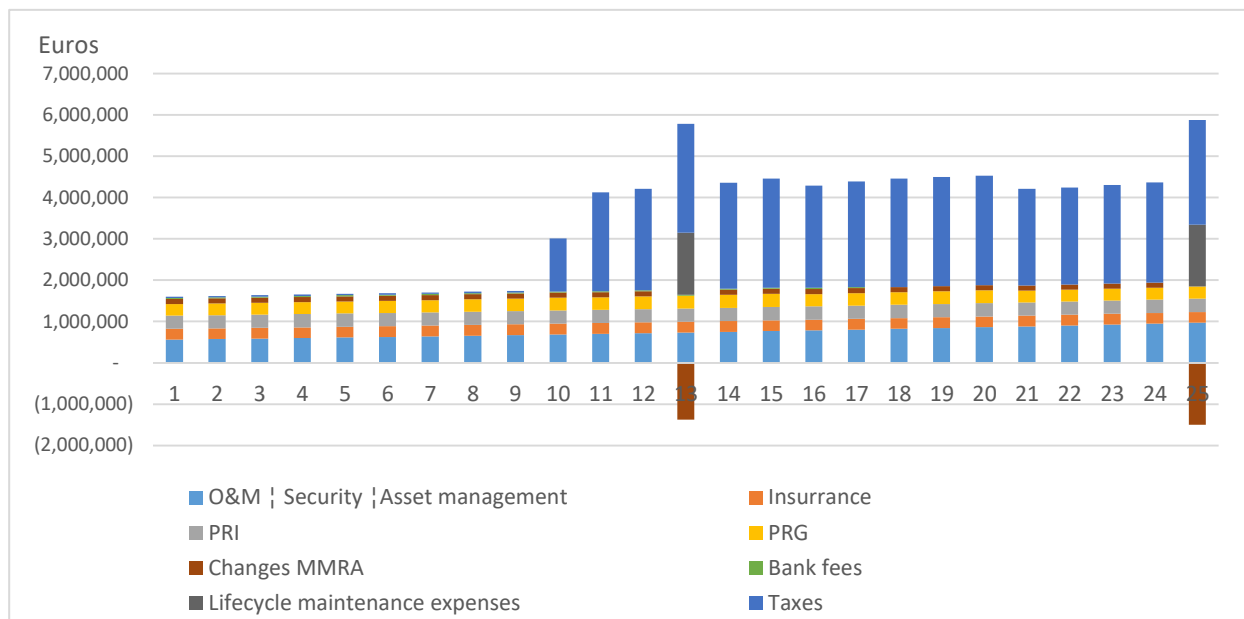
Figure 64: Tariff Schedule and Panel Degradation



4. Operating costs

A breakdown of the operating costs as well as the taxes is presented in Figure 65 on the next page. Whilst operating costs remain more or less constant over the duration of the PPA, the end of the corporate tax exemption as per the Framework Agreement (which we assume will be for 10 years³⁴) changes the whole dynamic of the model and has an impact on debt repayment and on returns.

³⁴ Based on the latest negotiations and discussions we had with the Minister of Commerce

Figure 65: Operating Costs Breakdown


The operating expenses line of the P&L shows that during the year when the plant is being built there are EUR 7.0 million of losses. This represents the substation and battery losses from an accounting point of view as those are transferred to the SNE at COD.

5. Financing assumptions

Regarding the senior debt financing assumptions, DSCR ratios are calculated as per the table in Figure 65. The FS shown are for a P50 scenario, but it should be noted that the debt is sculpted at P90. DSCR ratios are (i) to be above 1.30x at any given time, and (ii) the repayment profile of the debt is to be no slower than a mortgage-style repayment schedule. Corporate tax is being repaid from year 10 of operations (when the exemption under the Framework Agreement expires). DSCR is maintained at 1.30x for the second period to meet both (i) and (ii). Debt repayment is greater during the beginning years than at the latter part of the repayment period.

As for the EU-AITF grant, repayments are driven by equity IRR and available cash flows.

Figure 65: Tariff Schedule and Panel Degradation

Year (Operations)	1	2	3	4 - 5	6	7	8 - 16	17	Post
DSCR (P90)	1.53	1.48	1.48	1.48	1.30	1.30	1.30	1.30	N/A

Figure 66: Djermaya CDEN Energy S.a.r.l. Abbreviated Income Statement and Cash Flow Statement

Scenario	p-50																		
All figures '000 EUR	Construction	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations
Year Count	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Financial Year Start	30/06/2019	01/07/2020	01/07/2021	01/07/2022	01/07/2023	01/07/2024	01/07/2025	01/07/2026	01/07/2027	01/07/2028	01/07/2029	01/07/2030	01/07/2031	01/07/2032	01/07/2033	01/07/2034	01/07/2035	01/07/2036	01/07/2037
Financial Year End	30/06/2020	30/06/2021	30/06/2022	30/06/2023	30/06/2024	30/06/2025	30/06/2026	30/06/2027	30/06/2028	30/06/2029	30/06/2030	30/06/2031	30/06/2032	30/06/2033	30/06/2034	30/06/2035	30/06/2036	30/06/2037	30/06/2038
Income Statement	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Net Energy Output (GWh)	0	66	65	65	65	65	64	64	64	63	63	63	63	62	62	62	61	61	61
Revenues	0	7,904	8,006	8,110	8,214	8,319	8,319	8,426	8,533	8,642	8,511	8,398	8,504	8,612	8,721	8,831	8,236	8,340	8,444
Operating Expenses	-7,786	-1,478	-1,495	-1,512	-1,529	-1,547	-1,561	-1,580	-1,599	-1,618	-2,098	-2,083	-2,076	-2,274	-2,087	-2,079	-2,047	-2,040	-2,008
EBITDA	(7,786)	6,426	6,512	6,598	6,685	6,772	6,758	6,846	6,934	7,024	6,412	6,315	6,429	6,338	6,634	6,752	6,190	6,300	6,436
Margin	N/A	81.3%	81.3%	81.4%	81.4%	81.4%	81.2%	81.2%	81.3%	81.3%	75.3%	75.2%	75.6%	73.6%	76.1%	76.5%	75.2%	75.5%	76.2%
Depr. & Amortisation	0	-1,703	-1,703	-1,703	-1,703	-1,703	-1,703	-1,703	-1,703	-1,703	-1,703	-1,703	-1,703	-1,766	-1,828	-1,828	-1,828	-1,828	-1,828
EBIT	(7,786)	4722.82	4808.21	4894.36	4981.29	5068.97	5054.51	5142.40	5231.03	5320.41	4708.94	4611.62	4725.32	4572.17	4805.47	4923.18	4361.06	4471.50	4607.55
Margin	N/A	59.7%	60.1%	60.4%	60.6%	60.9%	60.8%	61.0%	61.3%	61.6%	55.3%	54.9%	55.6%	53.1%	55.1%	55.7%	53.0%	53.6%	54.6%
Interest Income	0	15	11	11	11	12	12	13	13	14	11	9	9	9	9	9	8	8	1
Interest Senior Loans	0	-2,817	-2,745	-2,642	-2,533	-2,397	-2,254	-2,089	-1,878	-1,635	-1,392	-1,204	-1,060	-903	-734	-547	-350	-155	0
EBT	(7,786)	1,921	2,074	2,264	2,460	2,684	2,812	3,066	3,366	3,698	3,328	3,416	3,674	3,678	4,080	4,385	4,019	4,325	4,608
Corporate Tax	0	0	0	0	0	0	0	0	0	0	-785	-1,921	-2,011	-2,009	-2,146	-2,253	-2,122	-2,229	-2,329
EAT	(7,786)	1,921	2,074	2,264	2,460	2,684	2,812	3,066	3,366	3,698	2,543	1,496	1,663	1,669	1,934	2,132	1,897	2,095	2,279
Cash Flow Statement	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Operating Cash Flow	0	6,442	6,523	6,609	6,696	6,784	6,770	6,859	6,948	7,037	6,423	6,324	6,438	6,347	6,643	6,760	6,198	6,308	6,437
Taxes Paid	0	0	0	0	0	0	0	0	0	0	-785	-1,921	-2,011	-2,009	-2,146	-2,253	-2,122	-2,229	-2,329
Change Working Capital	0	-670	-9	-9	-5	-13	0	-9	-5	-13	11	10	-5	-13	-9	-9	54	-13	-9
Change in MMRA	0	-125	-125	-125	-125	-125	-125	-125	-125	-125	-125	-125	-125	1,376	-125	-125	-125	-125	-125
Cash Flow before Capex	-	5,646	6,389	6,475	6,566	6,646	6,644	6,725	6,818	6,899	5,524	4,288	4,296	5,701	4,363	4,373	4,005	3,941	3,974
Capex	-50,370	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Release of MMRA	0	0	0	0	0	0	0	0	0	0	0	0	0	-1,502	0	0	0	0	0
CF available Debt Service	(50,370)	5,646	6,389	6,475	6,566	6,646	6,644	6,725	6,818	6,899	5,524	4,288	4,296	4,200	4,363	4,373	4,005	3,941	3,974
Equity Drawdown	9,495	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Drawdown EU-AITF	6,350	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Drawdown Senior Debt	37,050	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Senior Debt Service	0	-3,463	-4,070	-4,125	-4,183	-4,234	-4,233	-4,864	-4,932	-4,990	-4,046	-3,098	-3,102	-3,026	-3,148	-3,154	-2,882	-2,833	0
CF after Debt Service	2,525	2,184	2,319	2,350	2,383	2,412	2,412	1,861	1,886	1,909	1,478	1,190	1,194	1,174	1,214	1,219	1,122	1,108	3,974
EU-AITF Repayment	0	0	0	0	0	0	-272	-445	-463	-897	-718	-446	-448	-440	-455	-508	-373	-532	-352
CF after EU-AITF	2,525	2,184	2,319	2,350	2,383	2,412	2,140	1,415	1,423	1,012	760	744	746	734	759	711	749	575	3,621
DSRA	-2,125	29	0	0	-31	-26	-240	-79	-36	483	438	0	0	0	0	136	23	1,428	0
CF for Distribution	400	2,212	2,319	2,350	2,352	2,386	1,901	1,336	1,388	1,494	1,197	744	746	734	759	847	772	2,004	3,621

Figure 67: Djermaya CDEN Energy S.a.r.l. Abbreviated Balance Sheet and Key Ratios

Scenario **p-50**

All figures '000 EUR	Construction	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations	Operations
Year Count	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Financial Year Start	30/06/2019	01/07/2020	01/07/2021	01/07/2022	01/07/2023	01/07/2024	01/07/2025	01/07/2026	01/07/2027	01/07/2028	01/07/2029	01/07/2030	01/07/2031	01/07/2032	01/07/2033	01/07/2034	01/07/2035	01/07/2036	01/07/2037
Financial Year End	30/06/2020	30/06/2021	30/06/2022	30/06/2023	30/06/2024	30/06/2025	30/06/2026	30/06/2027	30/06/2028	30/06/2029	30/06/2030	30/06/2031	30/06/2032	30/06/2033	30/06/2034	30/06/2035	30/06/2036	30/06/2037	30/06/2038
Balance Sheet	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Cash	400	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150
Accounts Receivables	0	670	679	687	692	705	705	714	719	733	721	712	717	730	739	749	694	707	716
DSRA & MMRA	2,125	2,221	2,347	2,472	2,627	2,779	3,143	3,348	3,508	3,151	2,839	2,964	3,089	1,712	1,838	1,827	1,929	626	751
Current Assets	2,525	3,041	3,175	3,309	3,470	3,634	3,998	4,212	4,378	4,034	3,710	3,826	3,956	2,592	2,727	2,725	2,773	1,483	1,617
Total Fixed Assets	42,584	40,881	39,177	37,474	35,771	34,067	32,364	30,661	28,957	27,254	25,550	23,847	22,144	21,879	20,051	18,222	16,394	14,565	12,737
Total Assets	45,109	43,922	42,353	40,783	39,240	37,701	36,362	34,872	33,335	31,287	29,260	27,673	26,099	24,472	22,778	20,947	19,167	16,048	14,353
Current Liabilities	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Long term debt	37,050	36,404	35,079	33,595	31,944	30,107	28,128	25,353	22,300	18,945	16,291	14,397	12,355	10,232	7,818	5,211	2,679	-	-
L-T Liabilities	37,050	36,404	35,079	33,595	31,944	30,107	28,128	25,353	22,300	18,945	16,291	14,397	12,355	10,232	7,818	5,211	2,679	-	-
Shareholders' Equity	8,060	7,518	7,274	7,188	7,296	7,594	8,234	9,519	11,035	12,342	12,970	13,276	13,744	14,240	14,960	15,736	16,488	16,048	14,353
Total Equity and Liabilities	45,109	43,922	42,353	40,783	39,240	37,701	36,362	34,872	33,335	31,287	29,260	27,673	26,099	24,472	22,778	20,947	19,167	16,048	14,353

Key Ratios	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
EBITDA	(7,786)	6,426	6,512	6,598	6,685	6,772	6,758	6,846	6,934	7,024	6,412	6,315	6,429	6,338	6,634	6,752	6,190	6,300	6,436
CFADS	(50,370)	5,646	6,389	6,475	6,566	6,646	6,644	6,725	6,818	6,899	5,524	4,288	4,296	4,200	4,363	4,373	4,005	3,941	3,974
Senior Debt Service	-	(3,463)	(4,070)	(4,125)	(4,183)	(4,234)	(4,233)	(4,864)	(4,932)	(4,990)	(4,046)	(3,098)	(3,102)	(3,026)	(3,148)	(3,154)	(2,882)	(2,833)	-
Total Debt	37,050	36,404	35,079	33,595	31,944	30,107	28,128	25,353	22,300	18,945	16,291	14,397	12,355	10,232	7,818	5,211	2,679	-	-
Cash	400	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150
Net Debt	36,650	36,254	34,929	33,445	31,794	29,957	27,978	25,203	22,150	18,795	16,141	14,247	12,205	10,082	7,668	5,061	2,529	(150)	(150)
Equity	8,060	7,518	7,274	7,188	7,296	7,594	8,234	9,519	11,035	12,342	12,970	13,276	13,744	14,240	14,960	15,736	16,488	16,048	14,353
Balance Sheet Total	45,109	43,922	42,353	40,783	39,240	37,701	36,362	34,872	33,335	31,287	29,260	27,673	26,099	24,472	22,778	20,947	19,167	16,048	14,353
Current Assets	2,525	3,041	3,175	3,309	3,470	3,634	3,998	4,212	4,378	4,034	3,710	3,826	3,956	2,592	2,727	2,725	2,773	1,483	1,617
Current Liabilities	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Senior DSCR	NA	1.63	1.57	1.57	1.57	1.57	1.57	1.38	1.38	1.38	1.37	1.38	1.38	1.39	1.39	1.39	1.39	1.39	NA
Solvency (Equity / Total BS)	18%	17%	17%	18%	19%	20%	23%	27%	33%	39%	44%	48%	53%	58%	66%	75%	86%	100%	100%
Net Debt / EBITDA	-4.71	5.64	5.36	5.07	4.76	4.42	4.14	3.68	3.19	2.68	2.52	2.26	1.90	1.59	1.16	0.75	0.41	-0.02	-0.02

Annex

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Annex A-I

Sponsor and Lenders External Advisors

	Sponsors	Lenders
1. Early Development Studies		
(a) <i>Geotechnical</i>	Terrasol Setec	-----
(b) <i>Hydrology</i>	Setec Hydrotech	-----
(c) <i>Topographical</i>	Lagemot	-----
2. Technical	Azorom	3E
4. Network Stability	CAPSIM	-----
5. Resource Assessment	DNV GL	3E
6. Environmental & Social		
(a) <i>PV Plant</i>	Artelia	3E
(b) <i>Transmission Line</i>	Artelia	3E
(c) <i>Livelihood Restoration</i>	-----	
(d) <i>Indigenous Peoples / Traditional Law</i>	-----	
7. Legal	Norton Rose Fulbright	Linklaters
8. Insurance	JLT Speciality	
9. Tax	PWC	
10. Accounting	NConsulting	
11. Model Audit	-----	
12. Security	Drum Cussac Group	-----

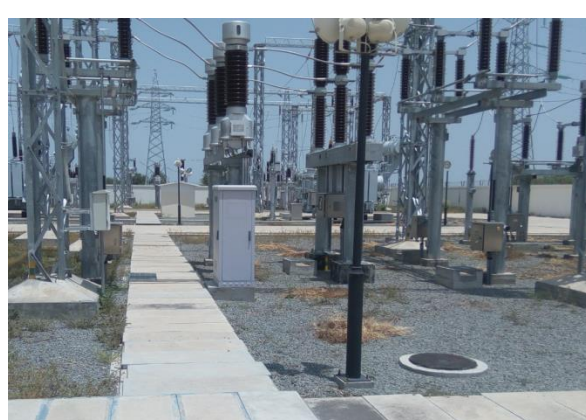
Annex A-II

Photographs of Project Site



Annex A-III

Photographs of Lamadji Substation



Annex A-IV

Potential Shareholding at Financial Close

Smart Energies S.A. Ultimate Beneficial Owners

	Surname	First Name	Percentage
1.	Van der Noot d'Assche	Anne	27.64%
2.	Donà dalle Rose	Francesco	11.11%
3.	Donà dalle Rose	Chiara	11.11%
4.	Gaia Imperiali	Gaia	11.11%
5.	Ladreit de Lacharrière	Nicolas	8.33%
6.	Ladreit de Lacharrière	Marc	8.33%
7.	Ladreit de Lacharrière	Paul	8.33%
8.	Juthier	Marie	8.33%
9.	Van der Noot d'Assche	Charles-Antoine	1.89%
10.	Van der Noot d'Assche	Gregory	1.89%
11.	Camu	Gaël	1.89%
12.	Van der Noot d'Assche	Antoine	0.02%

Note: The Ultimate Beneficial Owners (UBOs) of Smart Energies consist of 12 physical persons. Their names and other pertinent details must remain confidential.

Note: Anergi (parent company of Aldwych) is entitled to investment rights (8%) at financial close and may acquire the rights to the equity marketed by InfraCo, which if exercised would result in the following shareholding stakes:

- Anergi 54%
- InfraCo 25%
- Smart Energies 21%

